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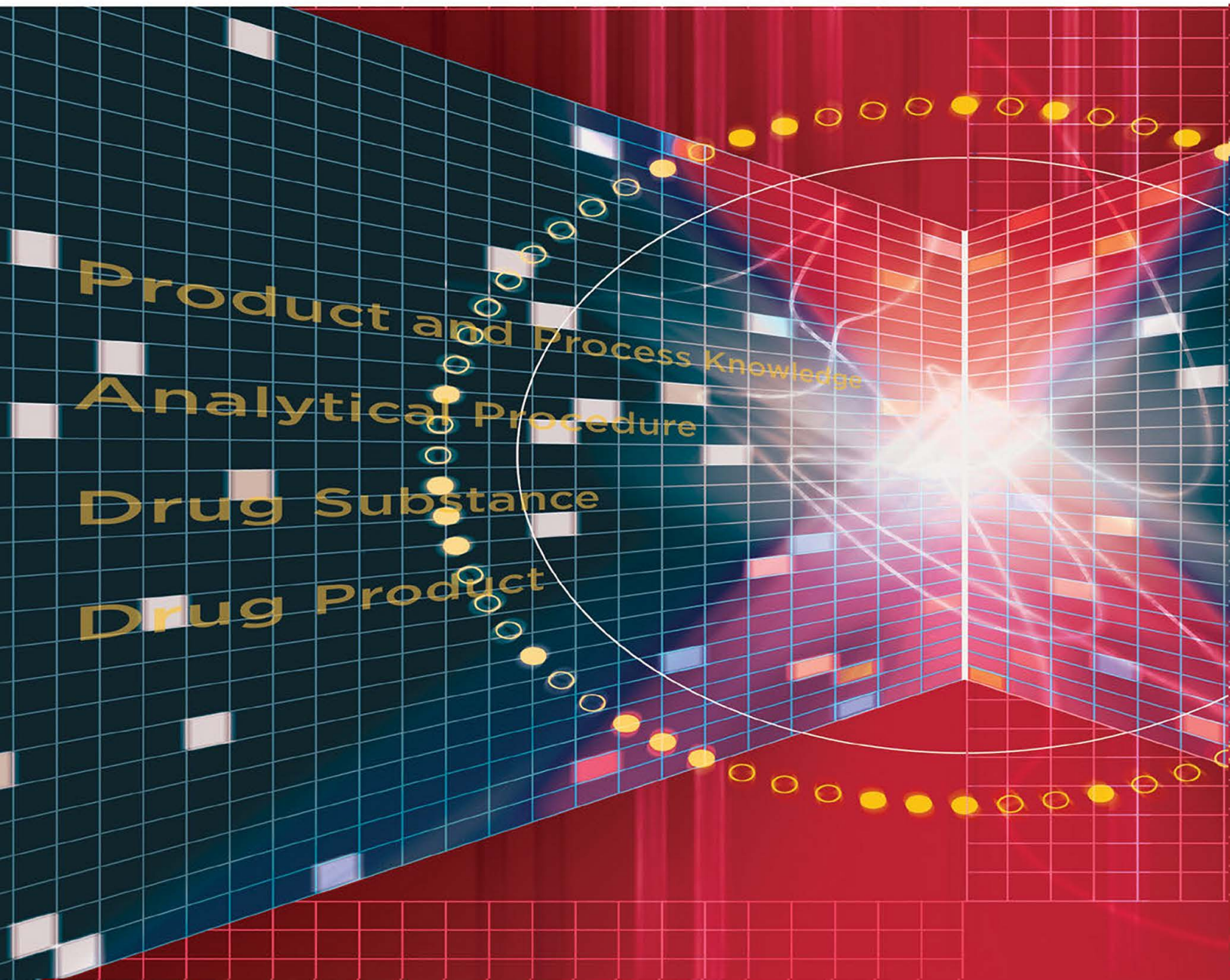
GOOD PRACTICE GUIDE:

Technology Transfer

Third Edition



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GOOD PRACTICE GUIDE:

Technology Transfer

Third Edition

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Preface

Transfer of manufacturing processes and analytical procedures between facilities or laboratories is a necessary part of pharmaceutical development and commercialization. Technology transfers take the outputs of process or method development activities and transfer the knowledge to a different location where a process or analytical procedure will be operated. This third edition of the ISPE Good Practice Guide: Technology Transfer was developed by an international team of authors from across the industry. This Guide presents a general approach and good practices for effective technology transfer with redacted case studies as examples. The intent is for the reader to utilize Chapters 1, 2, and 3 as a foundation and the subsequent chapters as applicable. The reader is encouraged to utilize the various lists, tables, figures, and templates for illustrative purposes.

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1 Introduction

1.1 Background and Purpose

Transfer of manufacturing processes and analytical procedures between facilities or laboratories is a necessary part of pharmaceutical development and commercialization. Technology transfers may utilize the outputs of process and method development activities and/or documentation of established processes and methods. Knowledge of the product and the manufacturing process is the basis for transfer to a different location (which could be a different site, country, or facility on the same site) where a process or analytical procedure will be operated.

This Guide focuses on how technology transfer can be achieved.

Technology transfer is defined in ICH Q10 [1] paragraph 3.1.2 as follows:

> *"The goal of technology transfer activities is to transfer product and process knowledge between development and manufacturing, and within or between manufacturing sites to achieve product realisation. This knowledge forms the basis for the manufacturing process, control strategy, process validation approach and ongoing continual improvement."*

Technology transfer projects may take place at various points during the product lifecycle. Successful transfers depend on robust project management processes combined with appropriate product and process understanding. They require partnership, cooperation, and coordination between the sending and receiving units to ensure successful and efficient completion, such that the receiving unit can manufacture, test, and release a safe, efficacious, and quality product comparable to that of the sending unit.

Technology transfer projects are dependent on the development and characterization of robust processes that allow consistent and predictable operation of these processes. Similarly, development of robust analytical methods enables timely transfer of methods. These are particularly important aspects; if processes are not well developed and sources of variation are not well known, e.g., due to incomplete or inadequate process/product knowledge, then technology transfer will not be robust.

This Guide has been designed to present industry good practices for successful and efficient execution of technology transfer projects and to achieve a balance between risk management and cost effectiveness while aligning with applicable regulatory expectations, as described in ICH Q7, Q8(R2), Q9, Q10, and Q11 [2, 3, 4, 1, 5] and other regulatory documents. It covers the principles of technology transfer and provides tools for its practical application.

The impetus for technology transfer varies and may be based on several factors, for example:

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Progression in a product development lifecycle, from development through scale-up to commercialization
The need for additional manufacturing capacity driven by increased demand or risk mitigation

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- The need to streamline capacity when volumes fall, e.g., at patent expiry, when products change commercial status from novel to generic, etc.
- Strategic requirements to relocate business units, e.g., regional economic advantages, regional requirements for local manufacture, global expansion in different regions of the world, etc.

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- Identification and qualification of local analytical facilities for testing of intermediates and finished products to meet regional regulatory requirements
- Dual sourcing strategies to mitigate demand variability or to reduce sourcing risks from manufacturing sites

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- Ensuring adequate turnaround times for analytical laboratories
- Utilization of virtual organizational structures and increasing use of third parties

1.2 Rationale for this Third Edition

The Guide has been updated to ensure alignment with relevant guidance publications and terminology used by industry and regulators. Practical case study examples have been added to this third edition; see Chapter 9 (Appendix 2) and Chapter 10 (Appendix 3).

This third edition of this Guide highlights the following:

- Alignment with science and risk-based (Quality by Design (QbD)) principles described in ICH Q8(R2) Pharmaceutical Development [3], ICH Q9 Quality Risk Management [4], ICH Q10 Pharmaceutical Quality System [1], and ICH Q11 Development and Manufacture of Drug Substance [5], including reference to Quality Target Product Profile (QTPP), Critical Quality Attributes (CQAs), Critical Process Parameters (CPPs), material attributes, design space (where used) and, in particular, control strategy
- Alignment with the process validation concepts and lifecycle approach as described in:
  - ICH Q10 and Q11 [1, 5]
  - FDA Guidance for Industry – Process Validation: General Principles and Practices (January 2011) [6]
  - EMA Guideline on process validation for finished products – information and data to be provided in regulatory submissions (November 2016) [7]
  - EU GMP Annex 15 Qualification & Validation [8] and similar PIC/S [9] guidances
  - EMA Guideline on process validation for the manufacture of biotechnology-derived active substances and data to be provided in the regulatory submission (April 2016) [10]
- Recognition that knowledge management is a critical component of effective technology transfer and that companies have employed various practices and technologies to identify and manage knowledge throughout the development and manufacturing lifecycle including technology transfer.
  - Knowledge includes product and process understanding and the science and risk-based approaches to gaining such understanding. Refer to the *ISPE Guide Series: Product Quality Lifecycle Implementation (PQLI®) from Concept to Continual Improvement* [11] for examples of practical application of science and risk-based approaches. See also the *ISPE Concept Paper: Implementing Knowledge Management in Bioprocess* [12].
  - Knowledge management also includes assessing what data will be collected to support a control strategy, how this will be managed, and identifying Key Performance Indicators (KPIs). These activities ensure that the full product lifecycle (from initial development to product discontinuation) and all elements of the technology transfer are optimized to avoid, for example, irrelevant data collection/analysis or making comparisons between the style or approach of the sending and receiving units.
  - Examples of knowledge to be transferred include:
    > Scientific and operational information
    > Product quality attributes (e.g., CQAs and material attributes)

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> Overview of process, including unit operations (e.g., reference to Process Flow Diagrams, CPPs, material attributes, etc.)

> Control strategy

> Continuous/continual improvement ideas and plans

> Tacit knowledge (i.e., knowledge from experience rather than explicit knowledge) and means to communicate this knowledge

> Health, Safety, and Environment (HSE) requirements

> Learning points from previous collaborative activities

> Implications for use of specialist tools and techniques such as Process Analytical Technology (PAT), multivariate data analysis, calibration of models, advanced data analytics, etc.

> Current understanding of potential risks to product quality

- Industry developments and potential regulatory impacts:
  - Technology transfers driven by acquisitions
  - Global expansion to developing countries
  - Facility consolidation within a company
  - Clinical to commercial transition within the same or different facilities
  - Supply chain driven events (e.g., economic drivers)
  - Utilization of Contract Manufacturing Organizations (CMOs) and contract development organizations, including virtual organizations
  - Increased utilization of platform and single-use systems
  - Introduction of generics and biosimilars
  - Cost efficiencies, either capital or revenue driven
  - Reduced cycle times
  - Local regulatory requirements (e.g., which may influence need to manufacture locally)
- Recognition that having a robust quality culture, for both sending and receiving units, is important for ensuring successful technology transfer. This is particularly important where cultures are very different between sending and receiving units, which may be due to internal organizational structures, management styles, country specifics, or just individual approaches; these differences should be recognized early and steps taken to ensure appropriate understanding and alignment during the technology transfer. Refer to the *ISPE Cultural Excellence Report* (April 2017) [13] for additional information on this topic.

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Helpful guidance is also given in the WHO Technical Report No. 961, Annex 7, WHO guidelines on transfer of technology in pharmaceutical manufacturing [14] and the Japan National Institute of Health Sciences, Guideline for Technology Transfer (Draft) [15]. The FDA Guidance for Industry – Contract Manufacturing Arrangements for Drugs: Quality Agreement, November 2016 [16] also mentions transfers.

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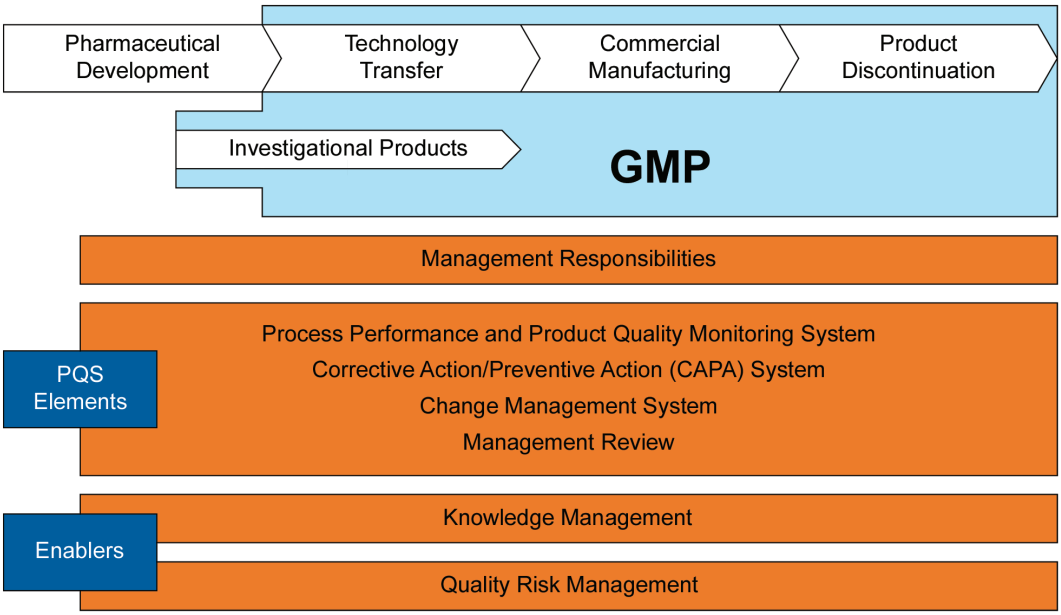
1.2.1 ICH Q10 Pharmaceutical Quality System [1] and Technology Transfer

ICH Q10 [1] illustrates the Pharmaceutical Quality System (PQS), as shown in Figure 1.1, and states:

> *"The PQS covers the entire lifecycle of a product including pharmaceutical development, technology transfer, commercial manufacturing, and product discontinuation as illustrated by the upper portion of the diagram."*

Figure 1.1: Diagram from ICH Q10 Pharmaceutical Quality System [1]



ICH Q10 [1] also covers the following topics and states the following about them in relation to technology transfer:

- **Product Lifecycle**

  The product lifecycle, in relation to technology transfers, includes the following activities:

  - *"New product transfers during Development through Manufacturing"*

  - *"Transfers within or between manufacturing and testing sites for marketed products"*

- **Knowledge Management**

  "Product and process knowledge should be managed from development through the commercial life of the product up to and including product discontinuation. For example, development activities using scientific approaches provide knowledge for product and process understanding. Knowledge management is a systematic approach to acquiring, analysing, storing and disseminating information related to products, manufacturing processes and components. Sources of knowledge include, but are not limited to prior knowledge (public domain or internally documented); pharmaceutical development studies; technology transfer activities; process validation studies over the product lifecycle; manufacturing experience; innovation; continual improvement; and change management activities."

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- **Management of Change in Product Ownership**

  "When product ownership changes, (e.g., through acquisitions) management should consider the complexity of this and ensure:

  (a) The ongoing responsibilities are defined for each company involved;

  (b) The necessary information is transferred."

- **Application of Process Performance and Product Quality Monitoring System throughout the Product Lifecycle**

  "Monitoring during scale-up activities can provide a preliminary indication of process performance and the successful integration into manufacturing. Knowledge obtained during transfer and scale up activities can be useful in further developing the control strategy."

- **Application of Corrective Action and Preventive Action (CAPA) System throughout the Product Lifecycle**

  "CAPA can be used as an effective system for feedback, feedforward and continual improvement."

- **Application of Change Management System throughout the Product Lifecycle**

  "The change management system should provide management and documentation of adjustments made to the process during technology transfer activities."

- **Application of Management Review of Process Performance and Product Quality throughout the Product Lifecycle**

  "Aspects of management review should be performed to ensure the developed product and process can be manufactured at commercial scale."

1.2.2 Product Lifecycle

Technology transfer expectations are different during the various phases of the product lifecycle. This Guide addresses the transfer of technology from a sending unit to a receiving unit, which may occur at any time during the lifecycle. To be a manageable and useful tool, this Guide assumes all work performed prior to initiating a transfer phase is adequate for enabling transfer of drug substance and/or drug product processes and analytical methods during the product lifecycle.

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Reference to the three stages of the process validation lifecycle described in FDA Guidance for Industry – Process Validation: General Principles and Practices [6] is helpful, particularly Stage 1 (Process Design) which relates to product and process understanding.

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The adequacy of product and process understanding is fundamental to successful technology transfer. It starts in design and development and captures knowledge in a development report (e.g., includes initial document review, summaries of experiments, QTPP, CQAs, material attributes, identification of sources of variation, and potential risks to product quality, etc.). For generics, historical data on product and process understanding should be documented based on current knowledge.

Typically, limited development occurs during the technology transfer phase. Exceptions during transfers can be handled by using risk-based approaches when, for example:

- There are incomplete characterization or process development

- Scale-up is needed

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- It is decided to start transfer very early (e.g., due to differing production and development site locations) and to make feasibility batches and carry out development work

Utilization of stage gates is an effective approach to initiating and managing a transfer when development and/or characterization activities are occurring in parallel to the transfer. See Chapters 2 and 3 for details of the phases. Refer to *ISPE Good Practice Guide: Project Management for the Pharmaceutical Industry* [17] for additional information on project management activities.

Investment in experience in the sending unit with the elements of the process or method being transferred can be used to accelerate transfers and close gaps in the technology transfer package. Examples include:

- **Solvent runs:** Running the process or selected operations of the process with process solvents only. This technique is useful to complete definition and test the effectiveness of site specific requirements in procedures, automation, and sampling strategies.

- **Live cell runs (in biological processes):** Can elucidate site specific sensitivities to process equipment and utilities by executing cell culture stages and monitoring performance. The cell culture phases are typically prioritized based on risk assessments which are informed by characterization data.

- **Simulation activities:** Can be used to complete the transfer package and test effectiveness without requiring either plant or laboratory time. These are often termed table tops, make-a-batch, simulation runs, virtual batch, or test-a-batch.

- **Engineering runs:** Executed prior to the initiation of process validation and typically without full GMP controls. Engineering runs (sometimes called dry runs) are valuable to obtain experience with processes which have significant differences from the sending unit knowledge base, e.g. changes in scale and equipment. In addition, they are useful for generating quantities of new sample types in support of enhanced control strategies. In some instances, they may be carried out after validation of the manufacturing process to characterize changes before making the choice of requalifying or revalidating equipment/processes (e.g., change of drug substance source).

The level of detail and depth of product and process knowledge (which generally will increase over time) should be adequate for the point in the lifecycle at which the transfer is taking place.

1.2.3 Intellectual Property and Confidentiality Agreements

Sending and receiving units would normally agree at an early stage what intellectual property can be shared between them or with third parties. From a commercial viewpoint, this is an important fundamental agreement that needs to be implemented early on before a transfer can be organized or its scope quantified. Intellectual property agreements and related confidentiality agreements are typically company and product specific.

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1.2.4 Quality Agreements

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A quality agreement is required when using a contract partner. It should be a separate document and includes the following aspects:

- Data integrity

- Data management and responsibility

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- Information technology requirements

- Continued Process Verification and its reporting

- Any continual improvement initiatives

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Refer to FDA Guidance for Industry – Contract Manufacturing Arrangements for Drugs: Quality Agreement [16], in which a quality agreement is defined as follows:

> *"A quality agreement is a comprehensive written agreement between parties involved in the contract manufacturing of drugs that defines and establishes each party's manufacturing activities in terms of how each will comply with CGMP."*

The FDA guidance [16] states that quality agreements should cover manufacturing activities and change control associated with these activities. Manufacturing activities include quality unit activities, facilities and equipment, materials management, product specific considerations, laboratory controls, and documentation.

1.2.5 Science and Risk-Based Approaches and Control Strategy

Technology transfer can be considered to have these main components:

- Business strategy
- Regulatory strategy
- Understanding and agreement of the knowledge to be transferred
- Understanding of the control strategy to be transferred
- Robust project management (including accounting for differences in culture and styles of working)

Using science and risk-based (or QbD) approaches enable understanding about a product and its process and the link to patient requirements, and hence enable effective technology transfer. For legacy products, historical information should be captured and documented, including, for example, agreed acceptance criteria and ranges within which product quality has been achieved and a qualitative assessment of product attributes, etc.

Technology transfer should ensure knowledge of product, process, and operational aspects are transferred effectively between sending and receiving units and that risks and mitigation plans during each of the phases are understood.

QTPP, CQAs, CPPs, material attributes, and control strategy, as described in ICH Q8(R2) [3], are elements to product and process understanding and should be available during technology transfer. For drug substance, refer to ICH Q7 [2] and ICH Q11 [5].

Close partnership between Research and Development (R&D), global manufacturing, quality, regulatory, and commercial functions is required to define, develop, and commercialize a drug, and therefore, enable successful technology transfer. Deliverables from early development should include initial assessment of unit operations at commercial scale, CQAs, CPPs, material attributes, and control strategy.

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Implementation of a robust control strategy is fundamental to a successful technology transfer. Establishing a robust control strategy requires a properly developed risk assessment (both for development and production), which depends on having adequate knowledge of the sources of variation.

As noted in Section 3.2.2, it is not the role of the technology to develop the control strategy. However, it is the responsibility of the technology transfer team to ensure the control strategy is effectively implemented; this will require specific inputs from the receiving unit (e.g., automation, alarm strategies, and process monitoring and sampling requirements).

An effective control strategy includes:

- Understanding the relationship between CQAs, CPPs, material attributes, and in-process controls

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- Controls on material attributes (including raw materials, starting materials, intermediates, reagents, primary packaging materials for the drug substance, etc.)
- Controls implicit in the design of the manufacturing process (e.g., sequence of purification steps (for biotechnological/biological drug substances) or order of addition of reagents (for chemical entities))
- In-process controls (including in-process tests and control of process parameters)
- Controls on drug substance
- Understanding of analytical methods
- Having appropriate data
- Alarm strategies
- Completed and qualified engineering modifications for equipment or utilities
- Where applicable, PAT systems, closed loop controls, and validated PAT methods and models

Implementation of a control strategy also depends on having other requirements in place, e.g., GMP controls, stability and release testing, ensuring requirements for drug substance, drug product and primary/secondary packaging are in place, etc. This is sometimes referred to as an integrated control strategy or holistic manufacturing control strategy. There should also be a means to capture and manage learnings for continual improvement, e.g., from deviations. It is important that the receiving unit has as much information as possible to understand why certain requirements are needed, as this helps minimize potential future errors.

Undertaking small scale batch observation runs can offer the following benefits (particularly when a CMO is being used for pilot and production manufacturing):

- Helps to get support and agreement from production teams
- Improve knowledge of the process, reduce errors, and build bill of materials lists
- Allows for the receiving unit team to view these at the sending unit's site and, for example, to make video recordings as appropriate
    - Gives the receiving unit team a chance to obtain an understanding of the documentation they may receive

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It can also include table top production batch walk throughs, using laboratory equipment, and models or drawings to mimic production and help build a scope of work.

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Technology transfer processes should enable the transfer of the knowledge to assure that all controls are in place, for all stages of a product lifecycle.

Ensuring operators understand what has to be implemented is important; some companies include a qualification and training program as part of operator training. Training procedures can include a review of the process/product understanding and reading Standard Operating Procedures (SOPs), batch record, and protocols to ensure the process knowledge is understood. Technical groups should explain to operators why the CPPs and controls have specific ranges and, in particular, those which will have a negative impact if out of range.

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1.2.6 Quality Risk Management

Quality Risk Management (QRM), as described in ICH Q9 [4], is an extremely important part of a technology transfer project. Successful technology transfer should ensure all high risk areas have been considered, appropriate mitigation has been put in place, and that any residual risks are assessed and agreed as acceptable between sites. The QRM process should be documented.

The sending and receiving units should follow QRM principles. SOPs related to QRM should be in place and followed in all aspects related to technology transfer.

The terminology, tools, and techniques from ICH Q9 [4] should be used appropriately. Refer to Section 3.3 for additional information. Refer to Chapter 12 (Appendix 5) and Chapter 13 (Appendix 6) for examples of Failure Mode and Effects Analysis (FMEA) and fishbone diagrams, respectively.

Risk assessments should ideally be carried out jointly by the sending and receiving units. There should be clear understanding of the purpose of each risk assessment. Any mitigations should be built into control strategies or other procedures at the receiving unit.

1.2.7 Communication

The success of technology transfer is critically dependent on good communication and relationships between personnel in technology transfer teams. At the start of a technology transfer, roles and responsibilities should be defined for transfer team members (from both sending and receiving units) including who are responsible and accountable for the components of the transfer.

Sending and receiving units should have the same level of understanding for all aspects of the transfer.

In addition, communication channels and methods (e.g., reports) that may affect the transfer of information should be defined. It is essential that good verbal and written communication channels are put in place, particularly when personnel are often not in the same room and therefore are unable to have face-to-face discussions, e.g., use of teleconferencing and video conferencing. The need to build good relationships and trust between the sending and receiving units is of great importance; capturing and documenting all appropriate knowledge is crucial so that it can be easily communicated between the sending and receiving units.

For transfers between companies, communication should be formalized by leveraging documents such as quality and supply agreements. A similar formality in principle should be put in place for intra-company transfers.

1.3 Scope

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This Guide is intended to be used as a generic guide to technology transfer and compiles information under three main topics:

- Technology transfer of analytical methods
- Technology transfer of drug substance (active pharmaceutical ingredients (APIs))
- Technology transfer of drug product (dosage forms manufacturing processes)

Each topic should be read in conjunction with Chapters 2 and 3 of this Guide.

It is recognized that transfers of analytical methods, drug substance, and drug product are closely linked. Drug substance/drug product transfers cannot usually be performed correctly if the corresponding analytical methods are not transferred to validate the transfer for quality.

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This Guide is intended to provide information for technology transfers between two parties (a sending unit and a receiving unit) for any applicable transfers in the product lifecycle, e.g.:

- Laboratory to laboratory scale
- Laboratory to development/pilot scale
- Development to clinical manufacturing
- Development to launch/commercial manufacturing
- Commercial to commercial manufacturing
- Drug substance to drug product manufacture
- Lead manufacturer to third party manufacturer

The sending and receiving units may be internal (within the same company) or external (between companies or to a third party such as a CMO). The transfer is normally from a sending unit that possesses the process knowledge, history, and operational experience to a receiving unit which needs to acquire that knowledge and leverage available experience to manufacture the drug.

Not all activities described in this Guide will apply to every transfer. Companies may select activities appropriate to the type and scale of technology transfer being performed.

This Guide does not intend to be a standard operating procedure for technology transfer. It reflects the understanding of industry as of the publication date.

This Guide is interpretive and ISPE cannot guarantee that a transfer performed in accordance with the recommendations in this Guide will be acceptable to regulatory authorities.

This Guide makes reference to, but does not cover, the following aspects:

- Regulatory guidance
- Qualification and process validation
- Product and process development

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As further described in Chapter 3, for the purpose of this Guide, a technology transfer is considered complete after the close-out phase. Therefore, the topics of Continued Process Verification (CPV) and Continued Quality Verification (CQV) are beyond the scope of this Guide. Refer to ICH Q8(R2) [3], FDA Guidance for Industry – Process Validation: General Principles and Practices [6], and ASTM (American Society for Testing and Materials) E2537-16 Standard Guide for Application of Continuous Process Verification to Pharmaceutical and Biopharmaceutical Manufacturing [18] for guidance on CPV and CQV.

However, process validation is an important step to ensure technology transfer has been completed and the links are summarized in Table 1.1 in simple terms.

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Table 1.1: Technology Transfer Links to Process Validation

Process Validation Stages [6]	Nomenclature (US)	Technology Transfer Links to Process Validation
1	Process Design	Technology transfer requires an understanding of what has to be transferred, e.g., CQAs, CPPs, material attributes, and control strategy.
2.1	Qualification of Utilities and Equipment	Utilities and equipment at the receiving unit (especially if it is different between sending and receiving units) is confirmed to operate over the ranges for the CPPs required.
2.2	Process Performance Qualification (PPQ)	Confirms the receiving unit can deliver the CQAs.
3	Continued Process Verification (CPV)	Ensures the receiving unit puts in place monitoring to ensure technology transfer has been successfully delivered.

1.4 Key Terms

This section introduces key terms as they are used in the context of this Guide. Refer to Chapter 15 for an expanded listing of definitions.

Control Strategy

As defined in ICH Q10 [1]:

> *"A planned set of controls, derived from current product and process understanding, that assures process performance and product quality. The controls can include parameters and attributes related to drug substance and drug product materials and components, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring and control."*

Receiving Unit

An organization, which includes core and auxiliary functions, where a designated product, process, or method is expected to be transferred to.

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Core functions normally include quality, manufacturing, and process engineering. Auxiliary functions may include regulatory, supply chain, warehousing, and HSE. The actual balance of functions would be project specific.

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Process Validation

- As defined in FDA Guidance for Industry – Process Validation: General Principles and Practices [6]:

  > *"Collection and evaluation of data, from the process design stage through commercial production, which establishes scientific evidence that a process is capable of consistently delivering quality product"*

Note: This is a lifecycle definition covering the three stages as described by FDA.

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- As defined in EMA Guideline on process validation for finished products – information and data to be provided in regulatory submissions [7]:

  "Documented evidence that the process, operated within established parameters, can perform effectively and reproducibly to produce a medicinal product meeting its predetermined specifications and quality attributes"

Note: This definition refers to a single stage – equivalent to Stage 2.2 of FDA Guidance [6])

Quality Target Product Profile (QTPP)

As defined in ICH Q8(R2) [3]:

"A prospective summary of the quality characteristics of a drug product that ideally will be achieved to ensure the desired quality, taking into account safety and efficacy of the drug product."

Sending Unit

An organization, which includes core and auxiliary functions, where a designated product, process, or method is expected to be transferred from.

Core functions normally include quality, manufacturing, and process engineering. Auxiliary functions may include regulatory, supply chain, warehousing, and HSE. The actual balance of functions would be project specific.

Stakeholder

Any individual, group, or organization that can affect, be affected by, or perceive itself to be affected by a transfer. Decision makers might also be stakeholders.

Technology Transfer Charter

The technology transfer charter captures the strategic intent and defines the overall scope of the technology transfer project. It should document the team members and their roles and responsibilities, explain the effort and time required for the technology transfer project, and identify significant assumptions and risks. The charter helps to ensure that management and team members from both sending and receiving units understand the project and agree upon deliverables.

Note: The technology transfer charter and proposal may be the same document, depending on the complexity of the technology transfer project. If the charter elements change, the project needs to be reevaluated, no matter what stage it is at in the technology transfer.

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Technology Transfer Proposal

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The technology transfer proposal defines the overall scope of the project, documents the team members and their roles and responsibilities, estimates the effort and time required for the project, establishes high level success criteria, and identifies significant assumptions and risks. This document is intended to ensure that stakeholders and team members from both sending and receiving units understand the project and agree upon the deliverables. The technology transfer proposal also identifies information needed for the technology transfer package. The technology transfer proposal should be reviewed and approved by the appropriate stakeholders from both sending units and receiving units. The proposal may be managed using formal change control procedures.

Note: The technology transfer charter and proposal may be the same document, depending on the complexity of the technology transfer project.

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Technology Transfer Plan

The technology transfer plan is based on the technology transfer proposal. It provides a more detailed description of the elements of the project (both technical and project management) that are to be completed by the team to achieve the overall goals. The technology transfer plan provides a means of ensuring alignment of the team members around the specific activities, as well as a means of tracking progress of the project. Results from risk identification/ gap analysis, risk assessment, and mitigation planning should be included. Any mitigation actions identified as part of the gap/risk assessment and expected deliverables should be included in the plan. Timeline, resources and budget, assumptions, regulatory strategy, and success criteria should be updated, as required. Major technology transfer execution activities should be listed. The plan should be managed using formal change control procedures, e.g., the receiving unit should follow their own change management system and related SOPs.

Technology Transfer Protocol

The technology transfer protocol is sometimes built into the proposal but may be a separate document. It defines the specific elements of the transfer activities, acceptance criteria (especially for analytical method transfers), and success criteria.

Technology Transfer Package

The technology transfer package is the collection of all (including local and process related) knowledge required to run the process and analyze the product. Depending on the stage of the product lifecycle, this may include, for example:

- Process definition documents containing tables of operating parameters and process development reports, e.g., for an early stage clinical process
- License requirements
- Validation studies
- Historical in-process data (can be information from many sources including annual product reviews and product quality reviews)
- Analytical method development history, including sample libraries
- Tacit knowledge and process experience, i.e., practical information that may or may not be initially documented

The technology transfer team should identify information (e.g., from product, process, equipment, raw materials facilities and suppliers) that needs to be collated, along with associated responsibilities, to help to ensure an effective knowledge transfer. A list of deliverables per discipline, including whether the sending or receiving unit is responsible, should be developed and formalized at the peer to peer meetings. This information gives direction and goals to both units, establishes who is the owner of the product lifecycle file, and ensures alignment of both units to an agreed plan (see Chapter 3).

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2 Technology Transfer – Planning Considerations and Success Criteria

2.1 Introduction

This chapter provides guidance on what to consider for a transfer, including establishing success criteria, prior to the formal formation of the full technology transfer team. Guidance provided is applicable to drug substance, drug product, and analytical procedure technology transfers.

A technology transfer typically starts with an agreement (referred to as a charter in Chapter 3) normally formalized or on a contractual basis between a sending unit and a receiving unit for the transfer of, e.g., a developed process or a commercial process or an analytical method.

The success of a technology transfer project will be largely dependent upon the skill and performance of individuals assigned to the project from both the sending unit and the receiving unit. It is, therefore, critical that a clear objective for any technology transfer project be developed. It is also critical that a project team comprised of individuals from both the sending unit and the receiving unit is established and that there is a precise understanding of each team member's role and responsibility prior to initiation of the technology transfer project.

Skill alone will not ensure a successful technology transfer project. Once a project has been identified and a team chosen, a clear and realistic project implementation plan is required to guide the project, manage expectations, and handle the inevitable deviations and changes that may present themselves during implementation.

Along with that plan comes the need to consider the relationship between the various tasks associated with a successful technology transfer. It should be clearly understood that each technology transfer assignment is unique; therefore, it is impossible to provide a generic technology transfer plan.

2.2 Technology Transfer Planning Considerations

In the early planning phase, a charter (as detailed in Section 3.1.2) may not be fully developed yet. In this case, the team should document, at a minimum, the major deliverables and assumptions not expected to change. Examples include:

- High level program deliverables such as delivery of material to clinical trials or early planning (e.g., for analytical methods)
- Agency submission strategies
- Site licensure milestones
- Budget
- Mechanism for handling changes, e.g., in scope or budget

As stated earlier, technology transfer can be considered a transfer of knowledge using well founded project management principles.

This section describes considerations for a successful transfer. For information on initial risk/feasibility assessments, see Section 5.2.

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2.2.1 Developing an Early Agreement Between the Sending and Receiving Units

An agreement (referred to as a charter in Chapter 3) is typically high level and developed with input from sending and receiving unit Subject Matter Experts (SMEs); input is rarely from the full technology transfer team. Securing an early agreement, sometimes referred to as site selection, may be necessary to release the full resources required for detailed planning and execution.

It is increasingly common that early stages with long lead times are initiated prior to formation of the entire team. It is also often necessary to make an early assessment of HSE aspects; an example is a facility that is not capable of handling particular compounds from an HSE standpoint.

By its nature, a technology transfer requires effective management of incomplete information; consideration of information and SME availability is required to determine appropriate planning timelines.

2.2.2 Sending Unit and Receiving Unit Working Closely

Technology transfer planning should be initiated as early as possible. Roles and responsibilities for sending and receiving units should be defined early and updated frequently in the early planning stages. Effective planning and diligent tracking of potential issues can help to expedite technology transfer projects and prevent delays.

A plan created by collaboration between sending and receiving units can help to unify the technology transfer team and to ensure that there is agreement on deliverables, success criteria, and ownership of the transfer at different stages. Project management tools such as the RACI[1] matrix can be useful to formalize roles.

The relationship within the technology transfer team should be actively maintained. This includes cultivating the skills needed to develop the team and building trust. Partnership between sending/receiving project managers and accountable functional area leads is key to actively managing the flow of known and emerging process, method, and site capability knowledge. When the sending and receiving units are within the same company, it is important not to overlook the need to have a formal plan which includes defining roles and responsibilities from individuals and departments.

It is important to understand the critical path timetable and ensure that the technology transfer planning and execution timelines are consistent with the critical path timing. Technology transfer timing should allow for the project manager to adjust the schedule, as needed, without moving major milestones: The timeline should be realistic. Communicating an unrealistic timeline to management could result in the team's acceptance of undesirable risks in order to meet their commitment. Management on both receiving and sending units should be apprised of the timeline and potential risks and their support should be obtained.

People to people interaction should be encouraged, especially between counterparts in the sending unit and receiving unit. This is especially important where different cultures exist between sending and receiving units. Implications of this may involve extra cost and therefore require management support (e.g., providing authorization for staff to be away from their normal jobs for travel).

The technology transfer plan should be reviewed and updated (as required) on an ongoing basis. Any changes that impact the budget, timeline for major milestones, or assumptions/risks should be incorporated and approved by stakeholders to ensure alignment.

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[1] **Understanding the RACI acronym [17]** – for every step of a planning project, the following should be defined:

R = Responsible	The person who is ultimately responsible for delivering the project and/or task successfully.
A = Accountable	The person who has ultimate accountability and authority; they are the person to whom "R" is accountable.
C = Consulted	Someone whose input adds value and/or buy-in is essential for ultimate implementation
I = Informed	The person or groups of individuals who need to be notified of results or actions taken but do not need to be involved in the decision-making process

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2.2.3 Clear Understanding of Roles and Responsibilities for Both the Sending and Receiving Units

A clear understanding of roles and responsibilities for team members, from both the sending and receiving units, serves to minimize redundancies and oversights. Knowing what is expected by other team members can increase accountability and help to minimize confusion and conflicts.

Additional recommendations include:

- Performing an assessment of the team capabilities relative to the skills required to complete tasks for which team members are accountable
- Defining decision rights (i.e., who will make the decision regarding particular aspects or differences of view)
- Defining clear communication routes both within the team and external to the team using, e.g., meetings, shared network or other project sites, meeting minutes
- Documenting roles and responsibilities with appropriate project management tools such as a RACI matrix – for additional information, refer to *ISPE Good Practice Guide: Project Management for the Pharmaceutical Industry* [17]

2.2.4 Effective Knowledge Transfer

It is very important that both sending and receiving units discuss and agree the knowledge to be transferred – scientific, regulatory, operational (including tacit knowledge), business, and cultural. Knowledge management is becoming an increasingly important topic.

Scientific aspects include, for example:

- Descriptions of the chemistry, physical chemistry, and biology involved in the design of the process
  - Understanding of process description/rationale/history, process flow diagrams, unit operations, scale, material transfers, etc. from dispensing through to packaging
  - Understanding what is important to control for both drug substance and drug product manufacture (see ICH Q8(R2) [3]) including, as appropriate, impact on packaging)
- Ensuring CQAs are understood by both sending and receiving units
- Ensuring analytical technology required to measure process performance is defined and enabled
  - For example: analytical system suitability and sample type qualification, receiving unit suitability/capability/applicability, and availability of technologies to be used for test methods transfers such as High Pressure Liquid Chromatography (HPLC) equipment, columns, etc.
- Understanding the requirements of the receiving unit, including facility fit, equipment constraints, and HSE (e.g., safe handling of the drug substance, drug product, and in-process intermediates)
- Transferring the control strategies from sending to receiving unit and operationalizing it
  - **Note:** even for legacy products, regulators are increasingly expecting a QTPP, CQAs and control strategies to be in place (e.g., FDA internal document MAPP 5016.1 [19])
- Being clear on CPPs and material attributes for each unit operation and understanding the impact of unit operations on CQAs, the expected and acceptable operating ranges and appropriate limits (including design space, if used)

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- Taking into account when multivariate approaches are used (e.g., a design space is developed, a model employed, PAT tools used for feed forward or feedback, real time release testing, etc.) and/or continuous manufacturing

- Potential sources of variation

Regulatory aspects include, for example:

- Compliance by both sending and receiving units with regulatory requirements in the country they operate in and the country where the product will be supplied to

  - Individual countries may have their own variation guidelines

  - For example, some countries may have specific requirements for conducting additional bioequivalence studies

- Ensuring validation is carried out appropriately including lifecycle approach

- Ensuring adequacy of the Quality Management System (QMS)/Pharmaceutical Quality System (see ICH Q10 [1])

- Taking into account that some regulations are slightly different in different countries (e.g., acceptability of three validation batches versus justifying the number of batches)

- Within both sending and receiving units, checking that there are no regulatory changes in progress that could later impact the implementation of the transfer

Operational aspects (including tacit knowledge) include, for example:

- Understanding the facility and utility requirements, e.g., facility fit, utility capacities, particular requirements that may be needed such as for aseptic facilities (classified areas, environmental controls, etc.)

- Understanding operator training and/or qualification requirements for the technology of the process, particularly if the receiving unit is less familiar with the technology or process

- Taking into account that equipment principles may differ (e.g., mixers or dryers may use different engineering principles to deliver the required homogeneity or dryness)

  - A check should be performed to ensure that there are no regulatory implications, e.g., impact on stability

  - Where needed, provide calculations, special instructions for scale-up, equipment equivalency reports, etc. if available

- Developing new manufacturing instructions and SOPs

  - The sending unit would normally concentrate on serious inaccuracies, omissions, or strategic problems with the content in the receiving unit's batch records or SOPs (e.g., not changing the layout)

- Understanding HSE requirements, e.g., need for containment or operator protection

- Putting maintenance procedures in place

- Understanding differences in operational principles, for example:

  - Number of shifts per day (e.g., three shifts versus day shift only), number of working days per week, whether shifts can be run consecutively, etc.

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- Automated versus manual operation
- Use of PAT tools
- Different scales
- Different batch sizes
- Different process flows
- Different sampling techniques
- Gravity flow versus horizontal transfers of powder

- Understanding tacit knowledge, i.e., knowledge that may not be explicit or documented but shows actual practices for doing certain tasks
  - Examples include scalability, the way a vessel is charged, or the way an analyst carries out a particular test
  - This is particularly important regarding such knowledge from the sending unit
  - It is also important that this knowledge is documented
- Assessing if long term planning is needed to obtain raw materials, e.g., due to import license restrictions
- Reviewing if operators from the sending unit can work on the receiving unit's site
  - This can be beneficial in enabling knowledge transfer and reducing errors on first batches
  - It is recommended to assess implications such as obtaining work visas, training, concerns by the receiving unit from Quality Assurance (QA), HSE, or regarding intellectual property

Business aspects include, for example:

- Importance of the product being transferred, which may differ from sending to receiving units, e.g. the profitability of one might be different to another, the transfer may be critically important to one company to release space for a more highly valued product, etc.
- Determining who pays for what (e.g., new equipment, training of personnel, etc.)
- Determining responsibility for capital; specifically, who pays for the investment itself, who owns the asset in terms of responsibility for depreciation, and for CMOs, whether that asset can be only be used by one client or is able to be used by multiple clients

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Cultural aspects are an important but difficult area to assess. Gaining an understanding of a company or country's culture can significantly contribute to the success of a technology transfer. Some cultures may be procedural, some may be hierarchical, some may be unstructured, some may be relationship dependent, etc. Language can be a factor; for example, while English may be the global language of a company, the local language for SOPs and operators may be different. Some cultures (even within a company) can be very different between units; for example, quality systems may have matured differently in practice, particularly where a company has recently taken over a site. Technology transfer teams should be sensitive to and aware of potential implications where culture may be significantly different and maintain an awareness to ensure similar understanding at both units. Refer to the *ISPE Cultural Excellence Report* [13] which provides a useful background.

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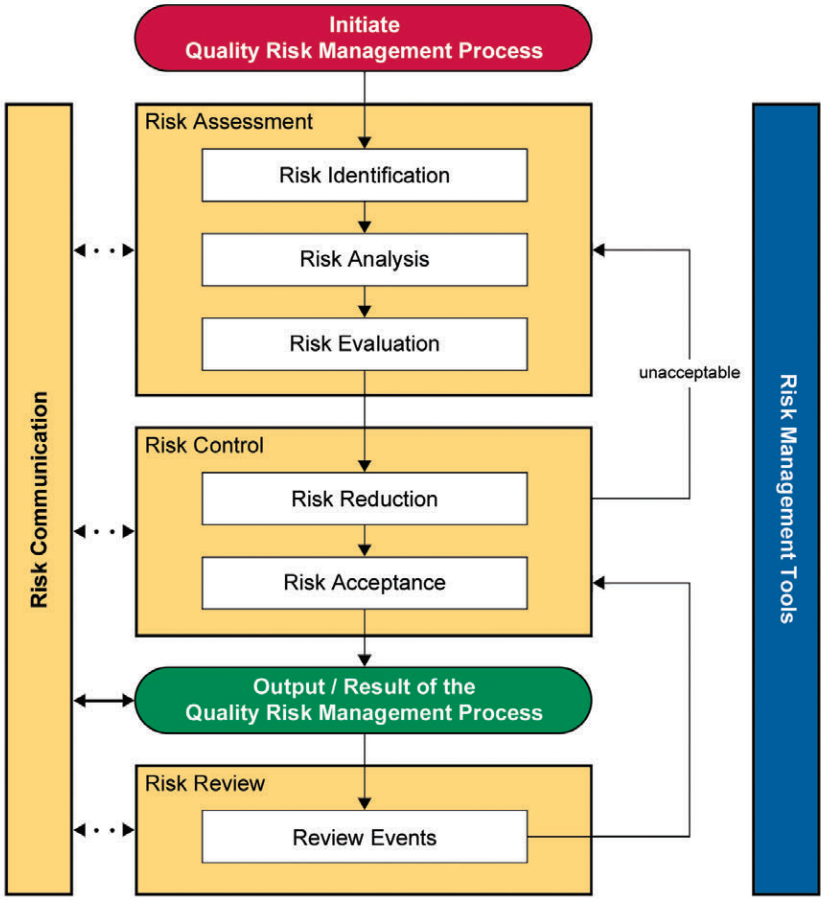
2.2.5 *Information Package*

A sufficiently complete technology transfer package of information should be developed to meet the requirements of the technology transfer. Refer to Chapter 3, Chapter 8 (Appendix 1), and Chapter 13 (Appendix 6) for additional information.

2.2.6 *Quality Risk Management*

QRM is integral to technology transfer and should follow ICH Q9 [4] principles (see Figure 2.1) or similar. It is important that teams are rigorous in their application of Q9 [4] and spend sufficient time during the initiation phase to ensure full understanding by both sending and receiving units of the purpose of the QRM exercise. Once the initiation phase is completed, a detailed risk assessment should be carried out (e.g., using tools such as cause and effect matrices, Ishikawa (fishbone) diagrams, gap analysis, and FMEA). After agreement on acceptable and unacceptable risks, then the unacceptable risks should be mitigated to an acceptable level and then controlled through a defined risk mitigation plan. This could involve experimentation and/or facility/equipment modifications and, when successfully implemented, will help to minimize risk and negative factors in the transfer process.

Figure 2.1: Flow Diagram from ICH Q9 Quality Risk Management [4]



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QRM is an iterative activity. All technology transfer projects require risk assessment. Existing risk assessments may need to be adapted to acknowledge changes in risk during the technology transfer. It is helpful to have a QRM plan if there will be multiple risk assessments. Note that there can be several types of risk assessment – for the project, the process, the product, the equipment, etc. The owners and focus of the risk assessments can be different. Normally, the project manager would own the project risk assessments and the technical team would own the technical risk assessments.

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2.2.7 Phases and Stage Gate Approach

The technology transfer project may be approached in phases (stages). Documentation of the project should include the important activities for each phase, e.g., reviews, deliverables, and projected timelines. Refer to Chapter 3 for the six phases of technology transfer.

Even in the early planning phase, there should be a mechanism for handling changes, e.g., in scope or budget. Throughout the project, stage gate reviews should be used to ensure that all critical items from the current project phase have been completed satisfactorily before moving on to subsequent phases of the project. This approach helps to ensure critical project items are completed on time and that later phases of technology transfer are not put at risk. Stage gate reviews should be attended by stakeholders and the technology transfer team.

For detailed information on the stage gate approach, refer to the *ISPE Good Practice Guide: Project Management for the Pharmaceutical Industry* [17].

2.3 Technology Transfer Success Criteria

Technology transfer is considered successful if the receiving unit can consistently produce material through the qualified process and demonstrate that quality attributes have been maintained through process monitoring and analysis of the material produced.

Simply stated, technology transfer can be considered successful if the receiving unit can routinely reproduce the transferred product, process, or method against a predefined set of specifications as agreed upon with the sending unit and/or development unit.

A comparability exercise may be completed as an assessment of sending versus receiving process/method performance. Interim success criteria are developed to track progress toward these goals. Success is ultimately demonstrated by regulatory agency licensure in the desired region(s). In addition, business performance parameters are established to balance risk versus cost, including resources.

Evaluation of the success of a technology transfer should be based on a pre-established set of criteria, which are evaluated as the project progresses. These criteria may be used to track progress of the transfer into subsequent phases and hence manage risk.

Industry R&D productivity, global product introductions, and increased generic competition (including biosimilars) are driving increased emphasis on establishing meaningful success criteria. These factors impart pressure by increasing the number of transfers, increasing global filing deliverable complexity, and reducing transfer duration. These pressures are driving innovation in the establishment of site criteria and measurement methods; hence, portions of the success criteria are transferred with the process or methods rather than established at the receiving unit. In addition, the need to increase the rapidity of measurements and subsequent decision cycles has created drivers for changes toward real time monitoring and on the floor measurement.

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Success criteria usually take into account the business situation and product requirements, for example:

- Stage within the product lifecycle
- Availability of product and process knowledge
- Experience of the sending units and receiving units
- Number of sites involved
- Whether it is a transfer between internal sites or to an external site

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From a business perspective, the success of a technology transfer is based on efficiently meeting the requirements of product quality, patient safety, and efficacy, and business requirements. Business requirements may include the following:

- Cost, including capital cost, expense and Cost of Goods Manufactured (COGM) targets
- Capacity/volume
- Equipment and facility capabilities
- Time frames
- Risk management
- Regulatory requirements or strategy, including meeting existing commitments
- Supply chain strategy

The relative importance of each of these factors should be established by the individual companies engaged in the technology transfer.

For example, when shared equipment is used in a multi-product facility, there may be additional requirements for risk assessments (e.g., cleaning validation assessments) before receiving regulatory approval for the newly transferred product to be manufactured at a receiving unit. For additional information, refer to the *ISPE Baseline® Pharmaceutical Engineering Guide, Volume 7 – Risk-Based Manufacture of Pharmaceutical Products (Second Edition)* [20].

Success criteria can take several forms and may be dependent on the technology being transferred and the phase of development in which the transfer takes place. The criteria should be measurable and achievable. Success criteria may include:

- Evaluation of the technology performance
- Adherence to the project timeline
- Adherence to the budget

Success criteria can be developed using data from the sending unit. The data could be quantitative (e.g., statistical analysis or other techniques) or non-qualitative (e.g., gel band shifts or mass spectra fingerprinting) and then transformed into quantitative values (e.g., assay endpoints). They can be based on the technology transfer team's understanding of the process, the risk identification/gap analysis and remediation, and other inputs such as regulatory and business requirements. The data set from the sending unit should be used to create a relevant set of success criteria for the receiving unit that allows the receiving unit to demonstrate a successful technology transfer. The data set to be used by the receiving unit should be identified during development of the success criteria. The data may be part or all of a package of comparability or validation analyses developed from the execution of the technology at the receiving unit as a part of development.

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Specific success criteria should be documented in the technology transfer plan. Main aspects may include:

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- Process performance – comparing in-process data (either traditional or PAT generated) with historical data
- Product quality that meets pre-established criteria:
  - Definition of data sets to be used for comparability analyses (e.g., earlier phase clinical data, manufacturing data from the sending unit)

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- Confirmation that the control strategy is executable
- Confirmation that the product control strategy from the sending unit has been successfully delivered by the receiving unit, including meeting requirements of product CQAs and specifications

  (**Note:** If a design space has been specified, this should be confirmed within agreed success criteria.)
- Process performance as expected based on historical data
  - For large molecule, examples include cell growth rates, titers, step yields, chromatogram profiles, and impurity clearance
  - For small molecule, examples include: impurity levels (e.g., solvents, elemental impurities), quality attributes (e.g., dissolution), material attributes (e.g., particle size distribution), process parameters (e.g., compression force settings to obtain required dissolution), process yield, and throughput (i.e., process efficiency)
- Analytical method transfer and equivalency
- Successful routine operation of the specified process over an agreed number of batches
- Successful routine operation of the facility and equipment with regard to operability, labor, and utilization plant occupancy times to improve Overall Equipment Efficiency (OEE) and to maximize overall operational efficiency
- Training of staff
- HSE objectives achieved
- Documentation fully and satisfactorily completed within the defined timescale
- Regulatory inspection outcomes and reviews, where appropriate
- Regulatory submission on time
- Facilitation and adjustment of the scope of process validation (e.g., by expanding prior knowledge of critical parameters)
- Improvement of process robustness and efficiency (exact 1:1 transfers are rare; there are normally opportunities for optimization which could not be realized initially at a sending unit)

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Consideration should also be made to establish longer term criteria (e.g., during process validation Stage 3 [6]) following closure of the technology transfer report. For example, technology transfer studies may fail to predict variance that is inherent in longer term evaluations of process stability. Also, there can be longer term variance from material or equipment that can be difficult to model with limited numbers of batches, even during a very robust technology transfer. Success criteria could include permitting natural material attribute variance or, if possible, intentional material attribute variance. These types of longer term criteria should be practical; consider on the one hand, historical variance if well understood by the sending unit, and on the other hand, criteria more robust than simply passing specifications. Longer term criteria should also consider unexpected excursions with scrutiny back to the closed technology transfer.

For additional information on evaluating transfer performance, see Section 5.2.5.

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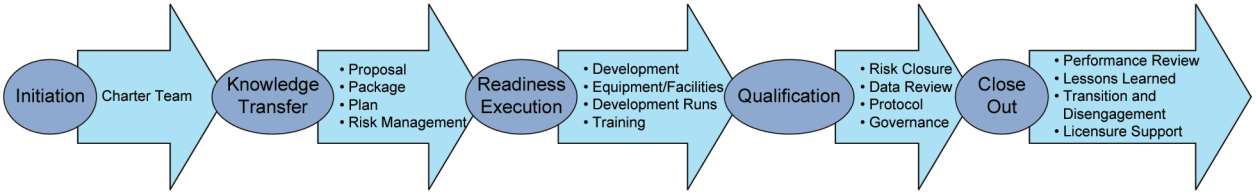
3 Technology Transfer Project Phases

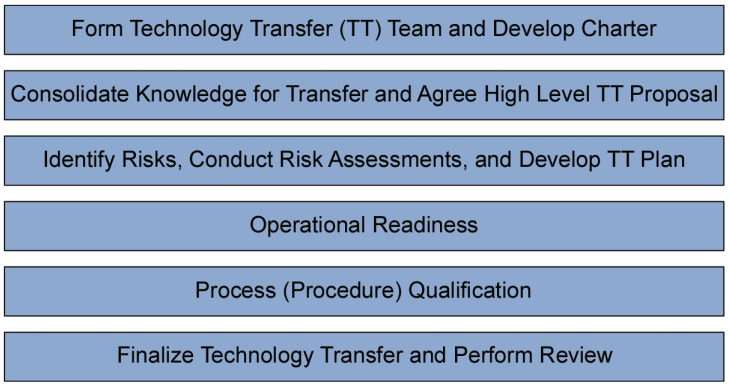
An overview of the chapter is provided in Figure 3.1.

Figure 3.1: Chapter Overview



This chapter provides a description for each of the six phases of a technology transfer. It is intended to be generic and describes general practices that apply to all transfers regardless of the technology being transferred. To note, at the time of publication, it is increasingly common for phases to overlap and/or be executed in parallel. The same principles used to manage and control a traditionally phased technology transfer can be leveraged for transfer with overlapping phases. The key principle in either case is to identify critical dependencies and control phase execution accordingly. Figure 3.2 shows the six phases of technology transfer.

Figure 3.2: Phases of Technology Transfer



3.1 Form Transfer Team and Develop Charter

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The initiation phase of technology transfer should establish the team, strategy, and goals of a project. Sending and receiving units should establish an integrated team of the required SMEs from the functional areas involved in the transfer. This team should develop a technology transfer charter to capture the goals, milestones, and roles and responsibilities for the project.

3.1.1 Establishing a Team

A formal technology transfer team should be established. The team structure, roles, and responsibilities should be identified. Team structure should include leadership from both sending units and receiving units, along with SMEs from the relevant functional areas. The team structure, roles for SMEs, and priorities will vary depending on whether the team is dedicated to:

- Analytical methods
- Drug substance

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- Drug product

At the start of the technology transfer program, key stakeholders, technical leads, and project managers should be identified. An overall project manager is usually responsible for delivery of the technology transfer and should coordinate activities such as finance, quality/regulatory, and materials management. Project managers may also be assigned within a functional area and will work closely with the overall project manager.

Technical leads are usually responsible for ensuring that project managers have the technical information to enable delivery of the technology transfer. These leads focus on the technical aspects of the technology transfer within their area of expertise and accountability. Depending upon the complexity of the project, a technical lead and project manager may be the same person. If the lead serves this dual role, it is important to ensure that staff who are trained in a technical discipline have the proper project management training and experience required to run the project. In addition, project management responsibilities cannot be neglected when competing priorities arise, for example, if the lead is needed for technical problem solving.

Team leads should assemble the technology transfer core team by including required SMEs. The composition of the core team is determined by the process being transferred and the required expertise to support the transfer. The core team should include representation from both the sending and receiving, as appropriate. Examples of functions represented in the core team include:

- Analytical sciences
- Analytical/Quality Control (QC)
- Engineering
- HSE and sustainability
- Late stage development
- Manufacturing operations
- Manufacturing technical
- Material management
- Planning
- Process development
- Project management
- QA
- R&D
- Regulatory affairs
- Supply chain

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Project stakeholders, such as senior management or budget managers, can provide oversight and guidance to the core team. Examples of functions represented by project stakeholders include:

- Finance

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- Manufacturing
- Process development
- Program management (commercial)
- Quality
- R&D
- Regulatory affairs
- Technical services

Technology transfer team leads and project stakeholders, across both sending and receiving units, should define the appropriate governance and controls for a project. Generally, this would cover the following:

- Scope of the project
- Timelines and key milestones
- Resources and budget
- Success criteria
- Reporting and escalation structure
- Key decision gates/approval process
- Change and risk management processes
- Quality management

Communication between the members of the team can be more effective when team members:

- From both units have a similar level of knowledge and experience of the techniques involved
- Have good team working skills
- Have the time to ensure that actions identified in the risk/gap assessment process are progressed
- Are familiar with the tools and techniques that support an effective knowledge transfer
- Define mutually accepted means of communication and collaboration – for example, geographically diverse teams may want to schedule face-to-face meetings for kickoffs and major milestone reviews, interspersed with regular web enabled teleconferences
- Acknowledge and manage cultural differences so as not to disrupt communications – team leads in international transfers will have an additional role of monitoring the effectiveness of team members at all levels and resolving communication impediments resulting from cultural differences

A leader who can promote an integrated team with the members from both the sending unit and the receiving unit can also help facilitate the process.

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3.1.2 Developing a Charter

The technology transfer team may develop a technology transfer charter. The charter can include:

- Assumptions and constraints
- Sites impacted (sending and receiving)
- Scope
  - Includes overview of the project and business drivers
  - Items that are out of scope are equally important to document in the charter
- Objectives and technical success criteria (see Chapter 2)
  - Examples include number of batches required, anticipated yield, successful registration and launch, etc.
- Roles and responsibilities
  - Clear understanding of expectations for each leadership role on the team and the deliverables for the respective functional area, including documentation, testing, disposition, and stability requirements
- Technology transfer team
  - Define who is within the team from each of the units, potential sub-team requirements, and understanding of key stakeholders
  - Should also include details of meeting frequency
- Governance – definition of decision rights and escalation path
- High level technology transfer plan[2]
  - Timescales of activities and resource requirements from each of the functional groups in across the sending unit and receiving unit
  - Communication strategy
  - Risk assessment review (including HSE)
  - Technology transfer documentation expectations aligned with individual responsibilities
  - Analytical transfer process and timelines
  - Commissioning and validation of any additional technology or infrastructure that is required
  - Manufacturing plan and sampling requirements
  - Data review plan, including deviation (non-conformance) reporting obligations between sending and receiving units

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[2] Development of a detailed execution plan is covered in Section 3.3.3; some elements of the plan may be noted as To Be Determined (TBD) during charter development and flagged for completion in the develop technology transfer plan phase.

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Note: Typically, deviations are reported back to the sending unit to facilitate support in review and for applicable development of corrective actions. However, if the sending unit is, for example, in production, utilizing methods transferred, under regulatory agency review, etc., then sending to receiving reporting will be equally important.

- Regulatory strategy
- Change control procedure
- Technology transfer completions of and formal transfer to clinical and/or commercial operations (release of sending technology transfer team members)

The technology transfer charter should be reviewed and approved by the appropriate stakeholders, executive sponsors, and senior management from both the sending and receiving units.

Note: The technology transfer charter and proposal may be either the same document or the charter may be versioned, depending on the complexity of the technology transfer project.

3.2 Consolidate Knowledge for Transfer and Develop Technology Transfer Proposal

3.2.1 Technology Transfer Proposal

A technology transfer proposal may be useful to define, facilitate review, and garner agreement of incomplete elements of the charter. In addition, prior to developing a detailed execution plan, a proposal can be used to develop detailed requirements and acceptance criteria for either key, or yet undefined, aspects of the plan.

The items listed below are recommended for inclusion in the technology transfer proposal; each team and project will likely need additional information specific to that project.

- HSE required assessments and known impacts
- Overview of project and business drivers
- List of team members and management/budget approvers (stakeholders)
- Broad objectives and deliverables
  - These may include projected clinical manufacture (successful number of runs), other GMP batches, any agreed upon process upgrades or improvements in yield, and successful registration and licensure
- High level timeline and resources needed
  - The project should be divided into phases, with a review and “approval to proceed” at the end of each phase (e.g., completion of technology transfer proposal, completion of engineering runs)
  - The timeline should be realistic, and not assume 100% success
  - Identify significant capital requirements and estimated Full Time Equivalent (FTE)
  - Identify which unit is responsible for funding elements of the project
- Issue resolution pathways and governance, including escalation

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- Validation strategy
- Regulatory strategy
  - Plans for product/process comparability, stability, and process validation
  - Meetings with regulatory agencies, filing dates, etc.
- Assumptions
  - List assumptions that are included in the proposed timeline
  - For example, availability of raw materials or process intermediates, successful transfer of analytics, the degree to which process changes will be acceptable, compatibility of data systems, process fit with minimal facility modifications, sufficient site utilities, no significant impact on site layout (e.g., material handling, personnel routes, safety)
- Potential risks to the project – high level risk assessment
  - For example, competition for FTE and resources from other programs, unfavorable clinical or regulatory outcomes, gaps in process understanding when changes and scale are within the in scope of a technology transfer, inability to mitigate significant gaps, inability to mitigate facility fit, limitations to available analytical support, product comparability issues
- Identification of information that is needed for the technology transfer package
- Identification of opportunities for members of the technology transfer team (e.g., receiving unit personnel) to observe the process at the sending unit to pick up any tacit knowledge (nuances or hands-on information which are not detailed in manufacturing instructions) and visual documentation (e.g., wet massing consistency, solution clarity)
- Requirements for disposition, storage, shipping, and forward processing, such as drug substance to drug product

Note: The technology transfer charter and proposal may be the same document, depending on the complexity of the technology transfer project. Where a drug substance or drug product has received a marketing authorization, the regulatory strategy may be a key contributory factor in the strategic intent of the project.

Both the sending unit and the receiving unit should review and agree on the technology transfer proposal. A summary of the success criteria, as described in Chapter 2, is useful to ensure alignment across organizations and at different levels of management.

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3.2.2 Technology Transfer Package

The technology transfer package should be organized efficiently to provide easy access to necessary information. It should also be comprehensive, so that a receiving unit can become be self-sufficient in running a process or analytical procedure.

Basic requirements for the technology transfer package include, e.g.:

- HSE related requirements
  - It is prudent to prioritize HSE impact assessments early in the transfer as these impacts may require long lead activities such as facility modification, regional licenses, and new training and monitoring programs

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- Business requirements

- Product specific requirements, particularly CQAs and control strategy and how these will be delivered for both drug product and drug substance at the receiving unit

  Note: It is not the role of technology transfer to identify quality attributes and process parameters and their relationships; this is the basis of the control strategy which is typically defined during process development. The control strategy is required to be available to the receiving unit to provide the scientific rationale which supports the manufacturing process and supporting analytical methods.

- Process flow diagrams, unit operations, parameters and material attributes, design space (where used), process trends, and detailed historical data (including the sequence and justification of any process changes), and risk assessments and identified CPPs (e.g., set points and ranges) and CQAs to ensure the control strategy is executable

- Facilities capabilities assessment relative to the ability to implement and consistently execute requirements of the control strategy

In addition to the basic requirements described above for the technology transfer package, additional items should be incorporated as the technology transfer progresses.

The technology transfer team should review the completed technology transfer package and request SME input, as appropriate, to ensure that all relevant knowledge is captured. The team should also summarize the available knowledge and conduct a high level technical risk assessment.

The technology transfer team should ensure that the receiving unit understands the information that they have available and can obtain clarifications as necessary. The knowledge transfer should take into account the level of expertise at both the sending unit and the receiving unit.

For transfers to an external partner or CMO, the transferring organization should determine what kinds of intellectual property it would consider to be confidential and determine how to redact documents appropriately or extract the key information needed by the receiving unit for transfer. See also Section 1.2.3.

The technology transfer package information will be used by the team members to work on a detailed process description and list of requirements for the receiving unit to execute the process. For example, the receiving unit process description may include scale-up information not provided by the sending unit. The team should then work through this list as part of a gap analysis, to identify, assign scores (if needed), and propose mitigation activities for gaps. The information gathered from the process description and gap analysis should be incorporated into a detailed technology transfer plan.

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3.3 Identify Risks, Conduct Risk Assessments, and Develop Technology Transfer Plan

Following approval of a proposal for the technology transfer and exchange of the technology transfer package, the transfer team should perform risk assessments on identified gaps between the process or procedure requirements and the receiving unit capabilities and practices.

The technology transfer package can be used by both units to initiate risk assessments and begin developing the technology transfer plan for process or method implementation, along with the information, data, or work that will be required to ensure the technology can be successfully executed at the receiving unit. These assessments should compare the process or procedure history with the receiving unit capabilities and operations to identify gaps or misalignments that will require modifications.

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In addition, the receiving unit should perform assessments of the following:

- Process/method performance and validation packages
- Raw materials/reagents for supply risk
- Impact of incoming technology
- Impact of changes needed as part of the implementation

This planning phase concludes once the sending and receiving units have completed the required assessments and have initiated plans to either remediate identified risks or receive risk acceptance from an appropriate governance.

Planning and execution of a technology transfer should include a comprehensive risk assessment process (i.e., risk identification, risk analysis, and risk evaluation as per ICH Q9 [4] or similar), followed by risk control (i.e., risk reduction and risk acceptance) and development of a detailed action plan to mitigate risks.

As part of a technology transfer, QRM is the documented procedure by which differences between sending unit and receiving unit can be identified and managed. Risk assessment is the process by which the criticalities of these differences are qualitatively or quantitatively estimated and prioritized. Once key gaps that require mitigation of risk are determined, a specific action plan for each gap should be defined and performed as part of the technology transfer execution.[3]

The aim of QRM is to lower risks to acceptable levels. Where this is not possible or readily achievable, the risks are presented to the stakeholders for their approval to proceed "at risk" to the process qualification stage. Any differences between the sending unit and the receiving unit have the potential to increase risk to the receiving unit's ability to demonstrate product realization, cause delays, increase technology transfer costs, and have negative business impacts. Sufficient time should be provided before technology transfer execution to identify and address differences. The sending unit and the receiving unit should perform these activities to help to improve knowledge transfer. These activities can promote teambuilding and improve future communication during later stages of a technology transfer.

It is usually preferable to address issues prior to the process qualification exercise, as opposed to during. For instance, if a process change is required due to conditions/constraints at the receiving unit, risk could be better defined and potentially mitigated by conducting studies in the small scale process model to examine the effects of the change.

It is advised that risk identification, risk analysis, and mitigation be conducted as early as possible since some risk mitigation activities may have long lead times and thus could significantly delay technology transfer. Although site selection is not discussed in this Guide, facility fit and site expertise are important factors to consider as they can have significant impact on the timeline and costs.

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This risk assessment should be used to define the overall transfer approach.

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3.3.1 Identifying Risks

As noted in Section 1.2, effective knowledge management (including tacit knowledge) is critical for effective risk management. Teams are advised to critically review information and knowledge utilized in risk identification, e.g., gaps in development reports.

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Identification of risks is typically performed by appropriate SMEs and ad hoc members from the sending unit and the receiving unit (i.e., process/technical experts, manufacturing, quality, facility and process engineers, automation, materials/procurement, HSE, analytical, Chemistry, Manufacturing, and Controls (CMC) regulatory, etc.). This

[3] In the case of site to site transfers, an assessment and mitigation of differences in the QMS between sending unit and receiving unit should be performed. Typically, this is done as part of the quality audit. Details of these activities are outside the scope of this document.

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is usually achieved through a series of joint and breakout meetings with the goal of capturing all differences or similarities between the sending unit and the receiving unit and may include a trained facilitator. Where available, data sources and level of uncertainties should be captured.

Effective risk identification/gap analysis requires availability or completion of the following:

- A sufficiently complete technology transfer package reviewed prior to meetings
- Dedicated input from team SMEs, e.g., for execution of risk assessment practices
- Detailed simulated runs of the process covering topics such as those listed below. Gaps noted during the simulated run exercise should be captured and further evaluated through the risk management program. It is recommended that the team work step-by-step through actual batch records or other descriptive documents such as those for:
    - HSE standards and regional requirements
    - Equipment, utilities, and facility detailed descriptions and drawings
    - Instrumentation, control, and automation
    - Operating procedures (sample handling, timing of key process actions)
        > For example, how long it takes to collect a bioreactor sample and take a measurement, the instrumentation used for off-line tasks, avoiding product shear through high speed stirring/vortexing
    - Resources projections and availability (people/FTE)
    - Materials and consumables, including review of suppliers and ability to supply (quality and quantity), extractables, and leachables
    - Analytical support requirements
    - Cleaning requirements for equipment (including ancillary processing equipment) and facility
    - Process modeling to help identify facility engineering modifications
    - Documented control strategy, including understanding of product CQAs, the unit operations, their CPPs and material attributes, and (if used) the design space to confirm the control strategy is executable
    - Opportunities for continual improvements
- Process and method reviews or walkthroughs by sending units and receiving units to jointly identify all differences or gaps that should be assessed
    - This exercise can also be beneficial to knowledge transfer and provides opportunities for the sending unit to observe and ask questions regarding critical manufacturing operations
    - The walk throughs can utilize current documentation, such as batch records, and systems, such as automation sequences, to guide the team stepwise through the process or method
- Assessment of the capability and expertise of the receiving unit (in addition to resource gaps)
    - Appropriate training programs should be developed to address potential technical or operational gaps

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The level of risk should be reviewed and endorsed by the respective risk owners on the technology transfer team. Residual risk (i.e., risk remaining after remediation) should be proactively communicated to leadership. The role of leadership is to either accept the residual risk or require migration actions prior to proceeding to subsequent phases of the transfer.

The risk identification/gap analysis is a living document which should be updated on an ongoing basis, as further information becomes available.

3.3.2 Risk Analysis and Evaluation

Risk analysis and evaluation is a process by which the criticalities of identified risk are qualitatively or quantitatively estimated and prioritized (see ICH Q9 [4] and the *ISPE Guide Series: Product Quality Lifecycle Implementation (PQLI®) from Concept to Continual Improvement* [11]). Risk analysis and evaluation should be applied to identified risks that potentially have a direct or indirect impact on HSE, CQAs, CPPs, and process performance. Where there is insufficient knowledge or data to judge comparability of the potential risk between the sending unit and receiving unit, a systematic, proactive method for evaluating how that risk could impact the process (e.g., FMEA) should be used to assess the risk and whether mitigation is needed.

Steps to assess risks typically involve identifying potential failures, likelihood and severity of each potential failure, ability to detect failure, followed by determining the overall risk using some form of qualitative (as exemplified in Figure 3.2) or quantitative estimation methodology. The rationale for ratings assigned for likelihood, severity, and detection should be documented. Once the overall risk level is estimated for each gap, the team should determine the need for risk mitigation based on acceptability of risk and then prioritize activities.

Based on the example risk estimation matrix provided in Figure 3.3:

- An overall risk level is **low** could translate to an acceptable risk in which case mitigation is not required; however, the technology transfer team may propose mitigation to further reduce the risk.
- At risk level **medium**, the risk may be acceptable and either justification is documented, or mitigation is required.
- If the risk level is **high**, risk is not acceptable, and mitigation is required to reduce risk to an acceptable level. The team should determine appropriate actions, e.g., facility/equipment modification or acquisition, laboratory studies, or gathering of additional information.

Figure 3.3: Risk Estimation Matrix

Likelihood of Occurence	Severity: Minor	Severity: Moderate	Severity: Critical
Certain	Medium	High	High
Average	Low	Medium	High
Remote	Low	Low	Medium

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Identified risk mitigation activities should be captured in the technology transfer plan with specific deliverables (what, by whom, and when). Risk should be reassessed after mitigation actions are established; risk that remains after mitigation is the residual risk. Residual risk should be captured in the technology transfer plan and communicated to management.

The results of the completed residual risk assessment should be documented in the technology transfer package.

Based on the assessment of risk, the required actions may vary widely, for example:

- Simple transfer of the written procedure with no formal qualification required
- Exercise that involves confirming the system suitability criteria are achieved in the receiving unit
- Detailed analysis of the difference in ways of working between the units (operational risk assessment) and formal qualification that the method is delivering fit for purpose data in the receiving unit

For drug product and drug substance, the goals of the risk assessment/risk identification would normally include:

- Confirming that the technology transfer risk assessments considers how quality attributes, process parameters, and material attributes will be achieved for both drug substance and drug product and that any implications on the control strategy are assessed
- Ensuring the link between the drug product CQAs and the drug substance material attributes and process parameters are understood to ensure the technology transfer will be appropriately achieved, and then identifying gaps and the potential impact
- Identifying and prioritizing the technology transfer strategy and required resources

Examples of quality attributes and process parameters to considering during the risk assessment are shown in Tables 3.1 and 3.2, respectively. Table 3.3 provides examples of HSE variables to review.

Table 3.1: Examples of Quality Attributes to Consider During the Risk Assessment

Drug Substance	Biologics Drug Substance	Solid Oral	Parenteral
• Assay • Bulk density • Color • Impurities • Limits of Detection (LOD) • Particle morphology • Particle size • Residual solvent • Solution clarity	• Protein content • Aggregates • Host cell protein • Biopotency • Endotoxins • Sterility • Peptide map • Residual DNA • Acidic species	• Blend potency • Blend uniformity • Bulk density • Content uniformity • Disintegration • Dissolution • Final potency • Flowability • Foreign matter • Particle size	• Appearance • Assay • Endotoxins • Identity • Impurity • O_2 in headspace • Particulate matter • pH • Sterility • Volume

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Table 3.2: Examples of Process Parameters to Consider During the Risk Assessment

Drug Substance	Biologics Drug Substance	Solid Oral	Parenteral
• Agitation speed • Distillation time • Drying time • Equipment configuration • Order of addition • Pressure • Reaction temperature • Reaction time • Regulatory starting material flowability • Stoichiometry	• Air sparging rate • Culture transfer rate • Dissolved oxygen • Gas humidity • Hold time • Incubator temperature • Nutrient feed rate • pH • Seeding density • Temperature	• Agitation speed • Airflow • Ejection force • Holding time • Impeller speed • Lubricant particle size distribution • Mill speed • Mixing time • Sieve size • Specific volume	• Airflow • Drying time • Filtration time • Gas humidity • Holding time • Mixing speed • N_2 flow rate • Order of addition • Shelf temperature • Sterilization temperature

Table 3.3: Examples of HSE Variables to Review

Category	Example	Notes to Consider
Environmental	Waste water permitting	Often a long lead time item
	Waste stream management	Local interim storage
Operator exposure	Personal protective equipment requirements	Requires training and inventory management
	HSE driven material flows and isolation requirements	Co-storage of new materials required for transfer
Safety	Inerting chemical plants	Required for volatile solvent handling
	Hazard analysis of new non-routine operations	Building this into every phase of the transfer

There are various methods and approaches to capture the risk identification/gap analysis output. Examples are provided in the appendices:

- FMEA – refer to Chapter 12 (Appendix 5)
- Fishbone diagram (Ishikawa) – refer to Chapter 13 (Appendix 6)
- Mind mapping diagram

On completion of the risk assessments and management plans, a review of the overall technology transfer plan should be performed to ensure that all required mitigation plans are included. Specifically, evidence of adequate risk reduction should be included in the plan.

3.3.3 Technology Transfer Plan

During development of the technology transfer plan, additional team members may be added; their roles and responsibilities should be clarified.

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The items listed below are recommended for inclusion in the technology transfer plan; each team and project will likely need additional information specific to that project:

- Scope and objectives
  - Define the point where technology transfer ends (e.g., successful operation of a given number of clinical production runs, completion of process validation)
  - Define any necessary modifications to the process
- Resources and budget
  - FTEs aligned to required SMEs and competencies, e.g. project management (see Section 3.1), materials, capital
- Timeline and milestone dates
  - Sequence of activities, stage gates or review points, other key milestones
- Roles and responsibilities
  - Clarify/update as needed
  - Identify additional FTE resources needed from both sending unit and receiving unit (e.g., supply management, technical services, SMEs)
  - Sending team on-site support at the receiving unit
- Deliverables and acceptance criteria
- Documentation requirements, such as protocols and reports
- Assumptions, constraints, risks
  - Elaborate on what was presented in the technology transfer charter/proposal
  - Include information gathered during the risk identification/gap analysis
- Detailed risk mitigation plan to address items identified in the gap analysis
- Work plan and goals for the technology transfer process
  - For example, facility modification if needed, equipment installation and testing, buffer/materials preparation, shakedown and engineering runs
- Work plan and goals for development studies to be implemented at either the receiving unit or sending unit, for example:
  - Small scale and/or pilot scale demonstration runs to gain experience with the process
  - Development/characterization of process changes that will be implemented at the receiving unit
  - Studies to model acceptable equipment performance (mixing, temperature set point differences)

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- Satellite studies to be conducted in parallel with the initial full scale batches

- Equipment details, e.g., need for ancillary equipment, critical parts or supplies, any demonstration studies required

- Supply chain and cold chain logistics – lead times, shipment, and licenses in preparation for process qualification

- Training requirements

- Updated governance structure and reporting obligations

The technology transfer plan should be updated to accommodate new information or significant changes to the timeline or resources. Any new risks encountered should be assessed and mitigated.

3.4 Operational Readiness

The goal of the operational readiness phase is to achieve a state of readiness required to move into the qualification phase, as described in Section 3.5. Both the process and methods are required to be in a state of control prior to entering qualification phase. A critical review of the operational readiness should be performed and formally documented.

Analytical methods are required to be qualified prior to full scale process qualification in order to be able to measure process performance. In some cases, method qualification may be concurrent with process qualification; in this case, the risk to the overall qualification risk must be documented and accepted by management.

Closure of actions defined by gap analysis and corresponding risk assessments drive the requirements to complete this phase.

Operational readiness activities required may include:

- Additional process or method development

- Equipment installation, modification, or re-validation

- General commissioning, qualification and validations as needed (refer to *ISPE Baseline® Guide: Commissioning and Qualification* [21])

- Engineering runs

- Procurement

- Training runs or training of staff

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Having identified gaps and risks, then subsequently prioritizing activities, technology transfer operational readiness may include the following:

- Facility modifications identified in risk identification/gap analysis, including appropriate equipment validations

- Testing of full scale manufacturing equipment to ensure proper function, e.g., mixing studies, buffer or water runs, cleaning studies, etc.

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In the case of cleaning, it is recommended that a comprehensive review of the requirements (e.g., soil types, required cleanability limits, and methods of detection) are compiled into a planning document. It is highly likely that changes in equipment type, scale-up, and process optimization implemented during the transfer will require new cleaning and monitoring methods, sample types, and corresponding analytical method qualifications.

As noted in Sections 2.2 and 3.3.1, effective knowledge management is crucial throughout all the previous phases. Operation readiness requires generation of documentation at the receiving unit required to implement and control changes necessary for implementation of the transferred process and/or methods. Diligence in document review and reconciliation in this phase is of paramount importance prior to proceeding to the qualification phase.

Examples of required documentation are as follows:

- Flow chart of process steps
  - Quantities of all materials and stoichiometry of the starting materials, intermediates, and reagents
  - Specific processing conditions (e.g., times, temperatures, and pressures)
  - Sequence of activities (e.g., order of addition of reagents)
  - Expected theoretical yield and output weight/volume of product
  - CPPs, i.e., factors impacting product quality
  - Permitted tolerance ranges for key parameters (e.g., yield impacting parameters)
  - Detail of other materials and by-products generated and their fate
  - Mass balance information
- Development and approval of receiving unit manufacturing instruction documents (batch records, SOPs, automation recipes, etc.)
  - A clearly defined structure for review and approval of the manufacturing documents should be agreed upon by the sending unit and receiving unit, including expectations for duration and number of reviews. A well-organized system for tracking and managing edits throughout the process should be used by the teams.
- Implementation of small scale laboratory models at receiving unit
  - As part of the transfer, small scale models of some or all the unit operations may be transferred to the receiving unit. These can be used for process training before large scale runs begin, for additional process development work identified in the risk identification/gap analysis, and can also be used for efficient troubleshooting during the initial large scale runs. It is helpful to perform small scale runs before the large scale campaign begins, to demonstrate that the small scale model performs as it is expected to at the receiving unit.
  - A suggested practice is to use the small scale model in a “satellite” mode alongside the initial large scale runs with the same buffers or media. This can help to troubleshoot problems that may be observed at scale by providing a control for some of the parameters.
  - An overall assessment comparing the small scale runs to the production runs should be documented or referenced in a campaign summary report.

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- Process training
  - Training should be conducted by SMEs prior to the start of the GMP campaign. It may be helpful to have some elements of this training led by personnel from the sending unit where they may have more extensive process experience. It also may be useful to have personnel from the sending unit as support staff during operation of first full scale batch.
- Models
  - Various models can be used by companies to develop pharmaceutical products. The rigor and documentation associated with the technical transfer of such models should be commensurate with the complexity and risk associated with their implementation and intended use.
  - Where models are established only for the purpose of understanding a process and its unit operations, it is usually sufficient to transfer the model as part of the knowledge base for the product, as these models are not part of ongoing control decisions during production. Where models are established as elements of a control strategy for ongoing production, consideration should be given to scale, sampling, and instrument dependencies as applicable, in addition to documenting and highlighting these potential challenges as risks of site implementation. In addition, procedures to calibrate, verify, maintain, if applicable validate, and update models should be established, as appropriate to the nature of the model being used.
- Creation of real time processing models and data trending template.
  - Data trending models are a valuable tool for monitoring an ongoing process in real time. The models are created from development data and updated with scale processing data for the lifetime of the process.
- Identification of in-process controls for monitoring
- Cleaning requirements per process step, including required analysis and forward processing criteria
- Generation of sample plan

In a GMP environment, some of these activities may require additional risk assessments and change controls as a part of the activity.

Additionally, the operational readiness phase of the technology transfer project will include the generation of the documentation and/or electronic systems that will be used to execute the technology at the receiving unit. This activity should be supported through the information and deliverables that were included as a part of the technology transfer package generated in the pre-transfer planning phase.

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The operational readiness phase of the technology transfer is complete when the process and analytical procedures are qualified. The sending and receiving units should be confident in the receiving unit's ability to successfully execute analytical procedures or manufacturing processes.

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Following successful completion of the technology transfer operational readiness phase, the project will move into the process/procedure qualification phase of the project.

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3.5 Process (Procedure) Qualification

As noted in Section 3.3, a documented review of all readiness criteria and transfer requirements in the plan should be completed in a gated review with management oversight prior to proceeding to the process (procedure) qualification phase.

This phase may be executed through a demonstration of the technology at the receiving unit and a comparison of the process and product and data generated to ensure it aligns with expectations. This activity may be performed as a part of a formal qualification or validation exercise, in which case the data generated may be useful for supporting the regulatory approval processes.

The risk assessment from the technology transfer planning phases should be signed off and documented in accordance with the QMSs for both the sending and receiving units prior to proceeding to process (procedure) qualification. For further information, see ICH Q9 [4] and the *ISPE Guide Series: Product Quality Lifecycle Implementation (PQLI®) from Concept to Continual Improvement* [11]. As previously noted in Section 3.3, risk assessments are most effective when utilized in phase gate reviews.

This phase should involve completion of the assessment of the data and product produced as a part of the transfer execution and identification of required mitigation plans to remediate any issues observed during qualification. The sending unit and receiving unit should define a plan for routine support of the technology at the receiving unit.

Execution of a qualification protocol should not begin until the protocol has been reviewed and approved by all appropriate departments, including the appropriate quality units. Any departures from the protocol should be made according to established quality procedures.

A report documenting and assessing adherence to the written qualification protocol should be prepared in a timely manner after the completion of the protocol. This report should:

- Discuss and cross-reference all aspects of the protocol.
- Summarize data collected and analyze the data, as specified by the protocol.
- Evaluate any unexpected observations and additional data not specified in the protocol.
- Summarize and discuss all non-conformances, such as deviations, aberrant test results, or other information that has bearing on the validity of the analytical procedure/process.
- Describe in sufficient detail any corrective actions or changes that should be made to existing procedures and controls.
- State a clear conclusion as to whether the data indicates the analytical procedure met the conditions established in the protocol and whether the analytical procedure is considered to be in a state of control. If not, the report should state what should be accomplished before such a conclusion can be reached.
- Include all appropriate department and quality units review and approvals.
- Provide the basis for continued process monitoring and control strategies.

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It is recommended that a statistician or person with adequate training in statistical techniques help develop the data collection plan and any statistical methods and procedures used in measuring and evaluating the process or analytical method. If the process or analytical method qualification does not meet the acceptance criteria, the cause of the failure should be investigated and addressed before repeating the exercise.

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3.6 Finalize Transfer and Perform Review

The final actions of the project should be focused on performing a review (sometimes called an after action review) to collect specific knowledge from the technology transfer. This should consider how each stage in the transfer went, how effective the communication and team work were, and whether any issues experienced during the qualification exercises were related to the effectiveness of the knowledge transfer.

The review may identify the need for a remediation plan and any ongoing support required from the sending unit. Where appropriate, key knowledge from the review should be shared with the teams involved in the technology transfer.

The complete evaluation of the success of the technology transfer through analysis of data or product (e.g., unexpected variability) generated as a part of the transfer execution will involve the sending and receiving units evaluating the performance of the technology, as well as the laboratory or manufacturing facility, and determining if any corrective action is required prior to declaring the transfer successful.

3.6.1 Planning for Transition of Support and Disengagement of Sending Unit

This can potentially include accountabilities and responsibilities for process or method improvements identified as a part of the transfer or through lifecycle management. There should be a clear transfer of responsibility to the receiving unit.

Agreement should be defined regarding the intended level of continuing support and involvement from the sending unit members of the technology transfer team.

Examples of aspects that should be considered for ongoing operations include:

- The receiving unit will operate the specified process.
- A change control procedure should be in place at the receiving unit.
- Process deviations should be noted and evaluated for communication with the sending unit.
- Process changes (continuous process improvement/lifecycle management) may be needed and initiated by either the sending or receiving unit. The initiated change must have prior agreement and be performed through a change control procedure. In some cases, process changes may require re-validation of the process. Global change control (e.g., when multiple sites are making the same product) is outside the scope of this document.
- Process monitoring should be performed by the receiving unit. Any anomalous process trends should be evaluated for communication with the sending unit. If the product is manufactured at multiple sites, regular manufacturing review meetings should be conducted to troubleshoot processing problems between sites.

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3.6.2 Support of Regulatory Activities

Post-transfer activities can include support of regulatory inspections or questions as a result of submissions associated with the transfer. Members from either the sending or receiving unit may be needed to provide assistance with regulatory activities by authoring documentation for filings or participating in site inspections.

3.6.3 Lessons Learned

An ongoing review during the transfer and/or a review at the conclusion of transfer phases that captures best practice and shared lessons learned should be performed. This should be documented and accessible to other teams from a continuous improvement perspective.

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Where lessons learned are not captured during the execution of the transfer, a lessons learned meeting should be held as soon as possible after the completion of the process qualification. The meeting should provide an opportunity to review the technology transfer project and overall GMP performance. At a minimum, the meeting should involve members of the technology transfer team and focus on the successes and challenges faced throughout the project. It may be valuable to break out into specific areas that need to be addressed, for example:

- Process
- Documents
- Analytics
- Quality
- Project management

The output from the meeting should be documented and shared with the technology transfer team.

The sending and receiving units should formally document lessons learned and determine future actions that would improve the success and efficiency of transfer projects or product commercialization.

Teams should strive to distinguish knowledge that is known (e.g., CQAs, CPPs, formal reports, Annual Product Review summaries, SOPs, etc.) from that which can be described as tacit (i.e., the knowledge that may be crucial for a successful transfer but is not actually written down.) Such tacit knowledge might include:

- Historical knowledge
- Informal knowledge
- The way an operator or a laboratory analyst performs certain steps
- Details of the order of operations, e.g. powder addition rates in the preparation of intermediates
- The unusual operation of items of equipment
- The actual way an operation is carried out (rather than just what the SOP says)
- Ranges within which operations proven to be effective
- Interventions
- Other (non-critical) quality attributes or process parameters that might have an impact on, for example, ease of operation rather than directly impact product quality
- Converting data into knowledge (e.g., data points can be plotted on a control chart and the knowledge is gained by assessing trends, limits, robustness, etc.)
- Use of work arounds (which, in practice, should be built into formal procedures)
- Taking learning from new events that arise (e.g., ensuring CAPAs do deliver the prevention as well as the correction)

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It should be noted that ICH Q10 [1] recommends knowledge to be managed over the whole product lifecycle – see enablers in the diagram from ICH Q10 [1] (Figure 1.1) and also how technology transfer fits in the lifecycle.

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ICH Q10 [1], Section 1.6.1, states the following regarding knowledge management:

> *"Product and process knowledge should be managed from development through the commercial life of the product up to and including product discontinuation. For example, development activities using scientific approaches provide knowledge for product and process understanding. Knowledge management is a systematic approach to acquiring, analysing, storing and disseminating information related to products, manufacturing processes and components. Sources of knowledge include, but are not limited to prior knowledge (public domain or internally documented); pharmaceutical development studies; technology transfer activities; process validation studies over the product lifecycle; manufacturing experience; innovation; continual improvement; and change management activities."*

It is noteworthy that technology transfer is included in the above paragraph.

The following ISPE concept papers provide useful supporting information and refer to technology transfer and knowledge management:

- *ISPE Concept Paper: Implementing Knowledge Management in Bioprocess* [12]
- *ISPE Concept Paper: The Role of Process Capability in Monitoring Product Quality* [22]

The *ISPE Cultural Excellence Report* [13] also has some indicators regarding the importance of knowledge.

It is important for the team to distinguish between observations and true lessons learned. The key distinction is the effectiveness and sustainability of the actions taken related to the observation. The following points are offered as a tool for teams to distinguish lessons learned from observations:

- Lessons learned could include observation, prioritization, and an action and monitoring plan.
- Observations without actions can result in reoccurrence of the respective event. Depending on the impact of reoccurrence, CAPAs may be developed and tracked in a company's QMS.
- Collect "work arounds" can be short term solutions but are often an unsustainable means to maintain performance.

Communication of the observations to the appropriate stakeholders, particularly to the receiving unit, should be confirmed prior to conclusion of the technology transfer project. In addition, it is recommended that the receiving unit manage a decision process for which observations warrant action plans and tracking. Similarly, if there is a functional area that manages technology transfer throughout the company, the same process should be followed to facilitate continuous improvement of the technology transfer business process.

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Lessons learned documentation may be in addition to corrective actions captured and tracked through a QMS.

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3.6.3.1 Transfer of Ownership and Closure

Upon completion of technology transfer, a final assessment of the remaining risks should be conducted and documented as part of technology transfer close-out. A review should be conducted by the technical transfer team in order to document and agree that all critical issues have been resolved, knowledge transfer is documented, and any outstanding items have been assigned for owners to resolve. Successful technology transfer should ensure that all high risks have been addressed and justification has been documented as to why some residual risks are acceptable and agreed by the receiving unit. At the time, any additional support that is required should be agreed to by the receiving unit. An important step to improve future technology transfers is for the receiving unit to provide feedback to the sending unit. Depending on the cycle times of technology transfer, production schedules, and regulatory agency filings, this feedback may occur during the transfer, throughout PPQ and during regulatory agency driven events such as filings and inspections.

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During the closure phase, teams should review the success criteria developed for the respective transfer (see Section 2.3). It is useful for teams to prepare an executive summary in which the transfer is declared successful; a technology transfer can be considered successful if the knowledge transferred allows the receiving unit to:

- Routinely reproduce the transferred product, process, or procedure against a predefined set of acceptance criteria which support the control strategy agreed with the sending unit
- Demonstrate through clear documentation that the main elements described above have been satisfied
- Demonstrate that the requisite business needs have been met

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4 Technology Transfer of Analytical Methods

4.1 Introduction

This chapter highlights those aspects of a technology transfer process specific to analytical methods.

The goal of an analytical method technology transfer is to ensure that knowledge from the sending unit is translated into an effective analytical method control strategy (that supports the overall product control strategy) in the receiving unit. When a receiving unit is required to perform an analytical method transfer, the company should execute an SOP to ensure:

- Consistent practices are followed
- Exchange of knowledge between the sending unit and the receiving unit is clear and robust
- Data produced by the receiving unit is fit for purpose, equivalent to data from the sending unit, and defensible from both quality and regulatory standpoints

Method transfer may be required for several reasons, including:

- Initial introduction of an analytical method into a unit that has not previously executed an equivalent method
- Introduction into a manufacturing facility by a development group
- Transfer from a manufacturing facility or development facility to another
- Transfer into a third party testing unit

The level of effort, formality and documentation of the method transfer, and any testing (performed to demonstrate the success of the transfer) should be commensurate with the criticality of the respective analytical data. For example, companies may not want to spend the same amount of time on transfer specific activities for products in the early versus late stage of its lifecycle. Late stage method transfers should be rigorously performed in order to ensure success of the method transfer, in support of product release and/or inclusion of the data into regulatory registration filings.

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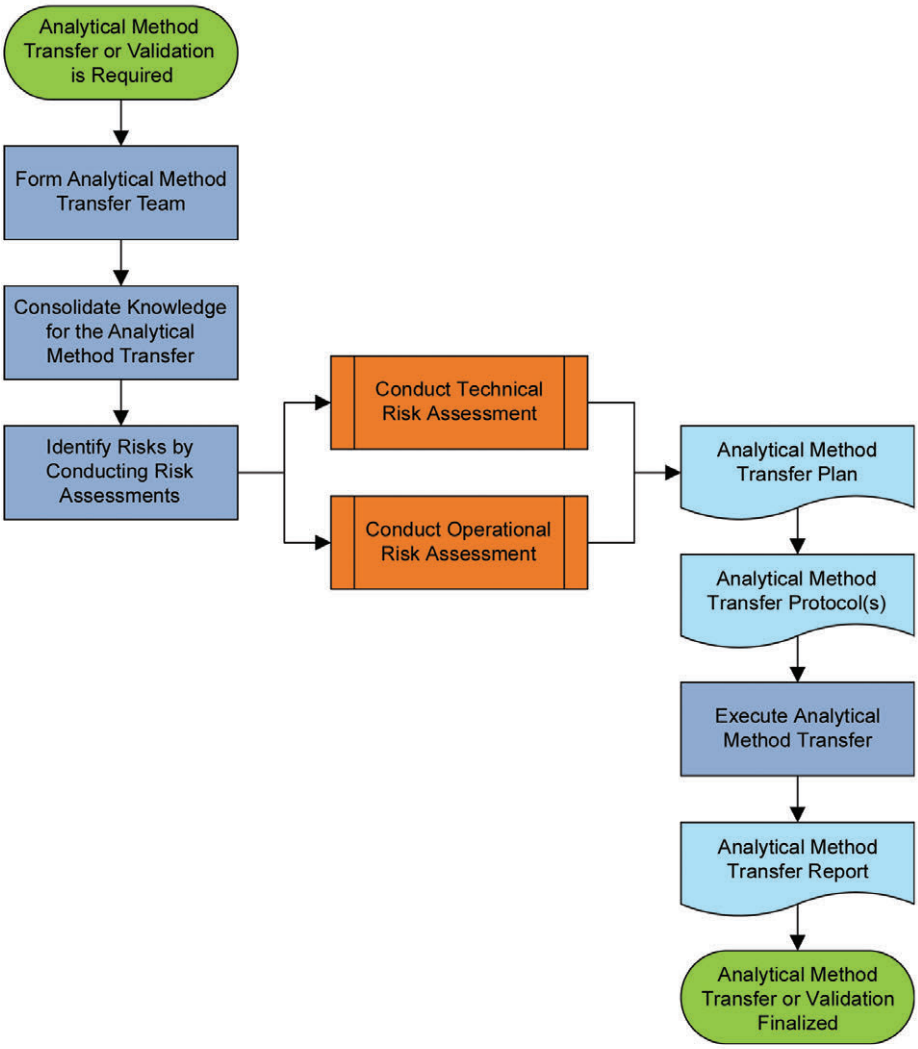
4.2 Technology Transfer Process

This section highlights those aspects of a technology transfer process that are specific to analytical methods. Figure 4.1 provides a pictorial of the process described in this chapter.

Figure 4.1: Analytical Method Transfer Process



4.2.1 Form an Analytical Method Transfer Team

A technology transfer team should be set up with membership from both the sending and receiving units. When the scope of the transfer includes more than one receiving unit, all impacted parties should be part of this team. This team should include the following members:

- Sending unit's analytical method transfer coordinator and/or management with:
  - Familiarity with the development history of the analytical method
  - Experience in its operation

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- Receiving unit's analytical method transfer coordinator and/or management with:
  - Understanding of the environment in which the analytical method will be operated
  - Understanding of the constraints and capabilities that exist in that environment
- SME from the organization that originally developed the method (e.g., the method developer) or is otherwise familiar with the development history, if available
- Analytical and QC staff SMEs and their counterparts at the receiving unit

Additionally, consider including the QA approvers of the documents in Quality as part of the team.

Where the transfer of the analytical methods is part of a broader manufacturing process transfer, the analytical team should be a sub-team of the wider technology transfer team. An effective communication process should be established between the analytical leader(s) and the members of the wider technology transfer team.

4.2.2 Consolidate Knowledge for the Analytical Method Transfer

For analytical methods, key information that needs to be collated to ensure effective knowledge transfer usually includes:

- Stage of the product lifecycle that the product is in
- Documented history of experience with the analytical method(s) including any development and validation reports
- Summary reports of any issues identified from data trends, analytical method deviations, and out of specification results associated with the methods being transferred
- Available information on the known and/or required performance criteria of the analytical method, e.g., invalid rates, acceptable uncertainty in the measurement data (Analytical Target Profile (ATP))[4], reporting, detection, and quantification limits
- Documented analytical method(s) and typical chromatograms or spectra (where not included in the analytical methods)
- Critical method controls or variables, where they have been identified
- Analytical equipment requirements, including data handling, software versions, etc.
- Details of analytical reference standards/reagents/columns and media
  - Where they are critical to the analytical method(s), performance details of the suppliers or grade should be shared
  - Information on availability and stability considerations, where known, should also be shared
- Set of samples that have been selected for confirmation or comparability testing; these samples should suitably challenge the analytical method(s) and include the applicable sample matrices as appropriate

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[4] The concept of ATP parallels the concept of QTPP as described and defined in ICH Q8(R2) [3]. The ATP defines the objective of the test including the maximum measurement uncertainty allowable for the reportable result. An ATP is likely to exist for methods developed using QbD principles. For additional information, refer to Harrington et al., 2018 [23].

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- Summary of change controls/revisions that have been applied to the analytical method(s) since its initial validation
- Overview of the regulatory status of the analytical method
- HSE requirements associated with operation of the analytical method
- Description of the importance and implications of (or contributing to) the control strategy for both drug product and drug substance

It is also important to evaluate, per method, the appropriate analytical method transfer strategy and document it appropriately. Refer to Table 4.1 for examples of approaches for analytical method transfer.

Table 4.1: Examples of Approaches for Analytical Method Transfer

Approaches	Description	Points to Consider
Waiver of Method Transfer Study	Can be considered for all method transfers but requires scientific justification since testing is not performed	• The method should already be in use for a similar sample type at the receiving unit • The personnel at the receiving unit should have experience with the testing technique using the equivalent equipment, systems, and software
Method Co-Validation	When the receiving unit participates in co-validation with the development laboratory	• Assessment of method reproducibility between sites should be performed • Validation protocol should be specific to the co-validation strategy; responsibilities should be defined
Method Verification	Consider for compendial methods (e.g., pH, osmolality, etc.)	• At least one assay should be performed in the receiving unit using all appropriate sample types with appropriate success criteria (i.e., specification or historical trends) to verify performance
Qualitative Method Transfer Study	For qualitative product specific methods (e.g., appearance, identity by near-infrared spectroscopy, etc.)	• Comparative test data is usually not required; transfer is typically performed by training • Performed for all late/commercial stage products; could also be deemed required for clinical stage products depending on risk
Quantitative Method Transfer Study	For quantitative product specific method (e.g., assay, bioassay, impurities, etc.)	• Comparative test data is usually required • Performed for all late/commercial stage products; could also be deemed required for clinical stage products depending on risk

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4.2.3 Identify Risks by Conducting Risk Assessments

For analytical method transfers, risk assessments may be divided into two types:

- **Technical risk assessments:** Assessment of all high level technical risks associated with introducing the analytical method into the receiving unit. Technical risks include aspects such as analytical issues with the method, frequent failures to meet system suitability and assay acceptance, difficulties in training staff, etc.

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- **Operational risk assessments:** Detailed mapping exercise aimed at understanding all potential sources of variability in the operation of the analytical method and developing appropriate controls. This may be documented per the appropriate laboratory quality system.

It may be possible to abridge risk assessments if methods, procedures, equipment, and/or equipment parameters are identical between the sending and receiving units.

4.2.3.1 Conduct a Technical Risk Assessment

The technical risk assessment should identify risks associated with utilization of the method in the new environment of the receiving unit. Table 4.2 lists questions to facilitate the risk assessment and provides suggested remediation opportunities.

Table 4.2: Technical Risk Assessment – Questions to Ask and Suggested Remediation Opportunities

Questions to Ask	Suggested Remediation
Are there any known issues with the operation of the analytical method? (e.g., assay failure rates, dependencies on single source reagents, complex extractions requiring advanced training, etc.)	Seek remediation to these issues from the sending unit or method developer prior to proceeding.
How well has the method performed over time?	If there is a poor performing assay, ensure remediation plans are in place.
Does the receiving unit perform any methods or technologies that are similar? (e.g., residual solvents, glycan profiles, etc.)	If the test or technology is new, seek a training plan that ensures an understanding of how the assay is performed at the sending unit.
Is the specification limit or process performance likely to be different in the receiving unit? If so, what are the implications for the capability of the analytical method?	Seek to better understand why there may be differences; put a remediation plan in place or accept these differences.
Does the receiving unit have the technology and equipment required to perform the analytical method? If not, will the analytical method need to be transferred to a third-party unit?	If the technology or equipment is new, obtain the equipment and seek training from the sending unit. Vendor training may also be beneficial.
Does the receiving unit personnel have the skills, training, and education required to perform the analytical method?	Seek appropriate training from the sending unit, hire staff with prior experience, and ensure training of multiple staff to ensure redundancy.
Are there any gaps in the analytical method validation/ performance information?	Gaps should be addressed with the individual(s) who performed method validation.
Are there any unusual preparations or storage requirements for samples, standards, or reagents/ special reagents?	Seek to ensure appropriate training, appropriate storage, and alignment with HSE practices.
Are controlled substances used? If so, how will they be stored?	Work with the receiving unit's HSE department to put the appropriate controls in place before receipt.
Does the receiving unit understand how to obtain the sample for testing? (e.g., how to expel the contents from the device, how to reconstitute, etc.)	Work with the sending unit to obtain the appropriate training and establish the appropriate procedures to ensure consistent practices.
Would it benefit the receiving unit to perform a test run on a known standard to ensure readiness for transfer?	Work with the sending unit to select a standard and/or sample for testing to ensure system readiness prior to starting the transfer.

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Table 4.2: Technical Risk Assessment – Questions to Ask and Suggested Remediation Opportunities (continued)

Questions to Ask	Suggested Remediation
Are there any critical consumables needed for the performance of the test methods to be transferred? (e.g., sample filters for dissolution testing to avoid API adsorption during sample preparation, coated HPLC vials and/or pipette tips to avoid protein binding etc.)	Work with the sending unit to identify any critical consumables needed to perform the assay at the receiving unit. Obtain these consumables prior to performing the assay. If the consumable cannot be obtained (i.e., due to import issues), justify why an equivalent is acceptable.
Are there any special environmental requirements for the test laboratory related to performing any of the methods? (e.g., controlled humidity for the testing of inhalation products or moisture sensitive products, etc.)	Work with the sending unit to ensure all information related to environmental requirements are understood. Address any risks.
Is the receiving unit familiar with real time analysis technologies and model prediction and how to react to the analytical result?	Work with the sending unit to obtain the appropriate training and establish the appropriate procedures to ensure consistent practices.

The analytical method transfer may be considered low risk when:

- For a new product with a composition (e.g., molecule/chemical and formulation) that is comparable to that of an existing product and/or the concentration of active ingredient is similar to that of an existing product and is analyzed by analytical methods with which the receiving unit already has experience
- The same as or very similar to an analytical method already in use
- Personnel who performed the method development, method validation, or routine method analysis of a product at the sending unit are moved to the receiving unit

If the risk is considered low, a technology transfer using comparative test data may not be required (see Table 4.1).

In some cases, the risk assessment may identify that an analytical method is not fit for transfer and method improvement actions, such as method redevelopment, may be required before progressing with the technology transfer. This situation may also occur upon examination of the data in a post-transfer review.

4.2.3.2 Conduct an Operational Risk Assessment

Where the possibility exists that a receiving unit generates data that is not "right the first time" or is of a more complex nature than the receiving unit has experience, an operational risk assessment can be performed. This risk assessment is used to identify, in detail, the differences between how the analytical method is operated at the sending unit versus how it will be operated in the receiving unit.

It is good practice to use tools, such as process mapping and Ishikawa (fishbone) diagrams, to systematically identify different variables that may exist in the sending and receiving units. Refer to Chapter 13 (Appendix 6) for an example Ishikawa diagram provided in Figure 13.1.

On completion of the technical and operational risk assessments, a plan should be developed to address any gaps in understanding and to complete the development of a local operating instructions for the analytical method that will ensure that it produces fit for purpose data in routine use.

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4.2.4 Readiness to Ensure Receiving Unit "Right the First Time" Execution

The analytical method transfer team needs to determine the number of assays to be performed and sample types to test during training and/or readiness. Prior to performing any live transfer activities, it is recommended to perform one or more assays (number based on past experience) at the receiving unit. To evaluate unit readiness and to enable the design of readiness training and experiments, a test-a-batch exercise may be performed. This exercise is used to obtain alignment around the following requirements for the receiving unit:

- Access to the current method
- Access to and performed review of any supporting documentation (e.g., method validation report, method development report, control charts, etc.)
- Appropriate equipment that are qualified as applicable
- Appropriate software, templates, scripts, etc. that are verified and qualified
- Appropriate materials, standards, and samples at the unit ready for use
- Qualified staff
- Appropriate quality systems required to support analysis in place
- Access to any additional sample handling training for testing
- Proficiency in similar analytical methods (e.g., receiving the technology for the first time versus prior experience with it)
- Performed assessment of the complexity of the analytical method to be transferred (e.g., pH versus bioassay)
- Performed assessment of the lifecycle state of the method (validated versus non-validated)
- Understanding the testing that the method is for (e.g., testing for stability indicating methods depend on the degradation pathway)
- Predetermination of the number of readiness assays that may be completed at the receiving unit if training occurs at the sending unit

If readiness results are determined to be unacceptable, a plan should be developed to assess the root cause including, but not limited to, training, method performance, equipment, materials/samples, and calculations.

Depending on the complexity of each analytical method, it may be useful for representatives from the receiving unit to directly observe the operation of analytical methods at the sending unit in order to ensure effective transfer of all tactical knowledge. This training enables the receiving unit team to put any tactical knowledge and risks into context and helps to build an understanding of the analytical method on an operational level. Additionally, it may be useful for the sending unit team to observe the receiving unit in order to build a good understanding of the equipment and establish capability at a practical level at the receiving unit once available. These interactions can help to establish good communication between members of the transfer team. The observations should be used to build an understanding of any differences in how the analytical method will be operated in each of the two units and can be of significant benefit in ensuring a successful transfer.

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4.2.5 Analytical Method Transfer Plan

An analytical method transfer plan, with a project plan to evaluate timing and resources, is recommended for the transfer of two or more methods. This document can serve as the holistic overview of what is required for method verifications and/or validations, method qualification, and method transfers. In some cases (e.g., when the analytical method transfer is part of a process/manufacturing transfer), the analytical transfer plan may be included as part of the overall technology transfer plan.

The implementation strategy should account for the following factors:

- Experience and knowledge of the receiving unit
- Degree of familiarity of the receiving unit with the methodology or technology used
- Specification of the product
- Complexity of the analytical method

The method transfer plan should include details of how many repetitions will be performed and how the performance of the analytical method will be assessed, e.g., by review of system suitability data or method precision. The number of repetitions should be adequate to provide sufficient statistical confidence that the method is producing data that meet the agreed measurement uncertainty target and data comparison.

At this stage, it may be useful to define the maximum acceptable uncertainty in the data (as per the ATP, if available). This can be established by setting equivalency criteria using a specific statistical test for demonstrating comparability between units (e.g., transfers). It requires that the two-sided 90% confidence interval (or two, one-sided 95% confidence bounds) for the difference in units fits entirely inside pre-established bounds. A potential dataset may include historical assay control data if the sending unit is using a similar sample type.

The degree of familiarity of the receiving unit with the methodology or technology may also influence the level of effort required in the transfer process.

Table 4.3 provides suggested sections for an analytical method transfer plan.

Table 4.3: Suggested Contents for Analytical Method Transfer Plan

Section Number	Section Title
1	Purpose
2	Roles and Responsibilities
3	References
4	Analytical Method Training
5	Analytical Method Transfer Strategy
6	Analytical Method Validation Strategy (as required)
7	Qualification of Reference Standards
8	Assessment of Critical Reagents
9	Analytical Method Validation History
10	Sample Testing Matrix

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4.2.6 Analytical Method Transfer Protocol

An analytical method transfer protocol should be generated to address each method being transferred either individually or collectively. The cumulative knowledge available should be used to establish a transfer protocol that ensures appropriate experiments are performed to demonstrate that the analytical method produces data in the receiving unit, which are fit for purpose.

Criteria and analytical method performance indicators that allow for a science and risk-based decision about the ability of the analytical method to consistently meet the target measurement uncertainty should be established. The criteria should include a description of any statistical methods to be used in analyzing collected data and any provision for addressing deviations from expected conditions and handling of nonconforming data. Data should not be excluded from further consideration without a documented, science-based justification.

A comprehensive list of samples requiring analysis should be compiled and reviewed by both the sending unit and the receiving unit. The review should include an assessment of the performance of the respective analytical method with each sample type. Attention should be given to in-process samples which may have matrix effects on the performance of the assay; additional method development work may be required either prior to, or concurrent with, or part of the qualification runs. If the method is supposed to be stability indicating, appropriate stability samples need to be included in the transfer.

For each test involved in the transfer, the type of transfer should be clearly articulated. Transfer types are varied and may comprise:

- Paper exercise/scientific justification, i.e. waiver (e.g., one based on familiarity of the receiving laboratory with the methodology as applied to a similar sample)
- Method verification
- Local validation of particular aspects of the method
- Full side by side comparison of data generated on the appropriate sample by each of the sending and receiving units

The transfer protocol needs to contain predetermined success criteria and be approved prior to execution. Support to write and approve the protocol should be provided by team members from both the sending and receiving unit, involving SMEs, Quality, statisticians where appropriate, etc.

A transfer protocol may not be required for method verification of a compendial method or a qualitative method if the transfer plan clearly documents how these will be executed (e.g., sample types, success criteria, etc.).

Table 4.4 lists suggested sections for an analytical method transfer protocol.

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Table 4.4: Suggested Contents for Analytical Method Transfer Protocol

Section Number	Section Title and Key Components
1	Purpose • Method(s) name and number • Purpose of the method(s) • Summary of the method(s)
2	Scope • Description of the assays to be run • Sample types assessed • Reference to the analytical method transfer plan
3	Materials and Equipment • For materials: item or part number, supplier, grade, storage requirements, and expiration dates • For equipment: manufacturer, model, serial number, and calibration/verification due dates
4	Experimental Design • Sample test plan, overall testing scheme, and success criteria
5	References
6	Appendices

Further recommendations for the analytical method transfer protocol include:

- Engaging support from the method developer and statisticians to set the study design and success criteria
- Using historical data to fulfill the success criteria if justified in the protocol versus performing comparative testing (e.g., sending and receiving units testing the same samples and then comparing the data)
- Selecting sample types that are representative of the materials being tested at the receiving unit and documenting the rationale to exclude any samples types – recommendations for selection of sample types include:
  - Consider lots of the same formulation, different formulations/strengths, and different sample types
  - Use the reference standard or assay control as one of the samples for bias assessment
  - For stability indicating methods, include a stability sample, preferably one with a relevant level of degradation based on method qualification or validation
  - For products with multiple strengths, bracketing may be appropriate
  - For qualitative assays, include a negative control sample where possible
  - If authentic samples are not available, surrogate samples may be used
  - Avoid the use of commercially distributed materials, unless other samples are not available
  - Set aside ample retain samples in all instances to ensure an ample volume of samples for the transfer
  - Consider the impact of any shipments of samples between units and mitigated accordingly
- Justifying the use of any alternate equipment or reagents at the receiving unit

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- If a method revision is required prior to the approval of the protocol, providing a draft of the method along with the protocol for approval

- Including both the receiving and sending units in protocol review and/or approval; protocol review should include the receiving unit staff, sending unit managers, and statisticians

- Utilizing mathematical model transfer algorithm between instruments where applicable

The samples selected for confirmation or comparability testing should suitably challenge the analytical method. For example, it is recommended to utilize samples with impurity levels close to acceptance and/or threshold criteria in order to evaluate the receiving unit's ability to reach the same quality decisions as the sending unit. This may be achieved by using expired samples. If no such samples are available (or it is not possible to import expired samples), it may be appropriate to spike the samples with known amounts of impurities (if a homogeneous sample can be assured) or use system suitability test solutions as samples or another suitable alternative challenge.

Note: If testing will be using a GLP/GMP batch that had been previously released, the risks associated with the potential to generate data that are different from those used to release the batch should be considered and discussed.

As the focus of the exercise is to demonstrate that the analytical method is working effectively at the receiving unit, consideration should be given for those situations in which comparative testing on multiple samples may add value to the knowledge obtained during the method transfer process. For example, there may be no need to use more than one sample or batch for impurity testing unless the batch composition (i.e., distribution of the impurity within the batch) is expected to have an impact on the analytical result. Conversely, experimental design can consider bracketing of sample types of similar composition; for example, adequate method transfer may be able to be demonstrated using the high and low of multiple strengths of a product of a common blend.

4.2.7 Execute the Analytical Method Transfer

The receiving unit staff performs the transfer or qualification experiments as outlined in the analytical method transfer protocol. Data review and assessment of the results should be performed with the sending unit staff, preferably as the method transfer is being executed to prevent any delays in understanding issues with the data being generated throughout protocol execution. The sending unit should be notified of any invalid assays and unexpected results.

Care should be taken to ensure that the integrity of the samples is unaffected by the shipping process between units. The transfer protocol should define which reference standards are to be used in any testing and from where they are to be obtained.

The defined local operating instructions should be followed during execution of the transfer protocol. The transfer exercise should be performed under normal conditions by the personnel routinely expected to perform the analytical method. Normal operating conditions should include the utility systems (e.g., air handling and water purification), material, personnel, environment, and instrument operating procedures. Any testing performed during the execution of the transfer should be performed by operators that have been fully trained in all relevant aspects of the required techniques.

A review of the transfer should be performed by the sending unit and the receiving unit. Questions to consider during the review include:

- Is the data being produced in the receiving unit considered fit for purpose?

- Is the analytical method operating reliably in the receiving unit?

- Are there any improvements to the analytical method to consider?

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4.2.8 Analytical Method Transfer Report

The receiving unit should generate a report outlining the pre-approved protocol success criteria and an assessment against the criteria for each of the analytical methods involved in the transfer.

If the success criteria are not met, it is important to follow the receiving unit's quality system to investigate. Consideration should be given to such aspects as training, equipment, reagents, environmental conditions, success criteria (i.e., not set correctly), etc. to determine the root cause. All exceptional conditions outside of the pre-approved protocol should be documented and justified, including all invalid assays (e.g., assay acceptance failures, technical errors, equipment failures, etc.).

If an analytical method transfer is unsuccessful, an investigation should be performed per the appropriate QMS and next steps should be determined.

An example method transfer report with protocol success criteria and results is provided in Table 4.5.

Table 4.5: Example Analytical Method Transfer Report with Protocol Success Criteria and Results

Performance Characteristic	Sample Type	Success Criteria	Results	Met/ Not Met
Equivalency	Reference Standard	The bounds of the 90% CI for the difference in the receiving unit mean from the historical mean must fall entirely within: % Attribute 1: ± 0.95% % Attribute 2: ± 0.92%	90% CI for the difference between the receiving unit and the sending unit is: % Attribute 1: **[-0.019, +0.58]** % Attribute 2: **[-0.60, -0.00055]**	Met
Intermediate Precision	Assay Control	The intermediate precision (SD) for the n determinations obtained in the receiving unit must be: SD ≤ 2 for both Attribute 1 and Attribute 2	**Attribute 1 %CV = 2.2% (2.2%) SD = 0.7** **Attribute 2 %CV = 1.2% (1.6%) SD = 0.7**	Met
Reproducibility	Assay Control	The assay control for each determination must not be more than 2.0% RSD	**2.4% RSD**	Not Met
Note: CI, Confidence Interval; CV, Coefficient of Variation; RSD, Relative Standard Deviation; SD, Standard Deviation.				

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4.3 Points to Consider

4.3.1 Site Selection

The receiving unit should be included in any site selection process when it comes to technology transfers. Typically, focus is placed on the manufacturing area; however, capital equipment and instrument costs can be high in the QC space as well as expense costs for training, travel, consumables, etc. These costs should be considered up front to ensure that the overall technology transfer budget will be met.

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4.3.2 Governance

If feasible, the overall method transfer process can be assisted by having a team that includes the product quality leader, analytical sciences, quality, and the sending and receiving unit coordinators. Relevant line management should also be included to review the agreed upon project plan and milestones and to ensure planned timelines are met.

If issues arise, this governance team should be informed in a timely manner in order to provide the appropriate escalation and support to resolve the issue.

4.3.3 Areas to be Aware of for Analytical Method Transfers

Timing with Respect to PPQ Runs

Optimally, analytical method transfers should be finalized prior to PPQ runs to ensure that the data generated at the receiving unit can be used real time and are reliable. If it is not completed in time, the sending unit may need to execute this testing to ensure PPQ success.

Comparison of Data in Co-Validation

When performing co-validation as a method transfer approach, it is important to ensure that a comprehensive review of all the data, as compared to the sending unit, is completed. In some instances, validation criteria can be met despite true data differences (e.g., bias or significant differences in variability) between the sites. These differences should be investigated.

Bias in Results Between Units

In some cases, one unit may tend to have a bias in results. For example, the results from a unit may be higher or lower than the other unit. This data should be closely assessed as the method transfer progresses to ensure that the method, equipment, training, etc. did not contribute to these differences. A potential action step in assessing the root cause is for the sending unit to visit the receiving unit and observe the assay setup and assay performance.

4.4 Additional Sources of Information

For additional information regarding analytical method transfer, refer to the following:

- Article titled “Transfer of analytical procedures: a panel of strategies selected for risk management, with emphasis on an integrated equivalence-based comparative testing approach”, Agut et al., 2011 [24]
- EMA Guideline on bioanalytical method validation [25]
- EMA Guideline on the use of near infrared spectroscopy by the pharmaceutical industry and the data requirements for new submissions and variations [26]
- EMA Volume 4, Part 1, Chapter 6: Quality Control [27]
- FDA Guidance for Industry: Analytical Procedures and Methods Validation for Drugs and Biologics (July 2015) [28]
- FDA Guidance for Industry: Dissolution Testing of Immediate Release Solid Oral Dosage Forms (August 1997) [29]
- USP <1092> The Dissolution Procedure: Development and Validation [30]

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- Draft USP <1220> The Analytical Procedure Lifecycle, published for public comment in Pharmacopeial Forum [31]
- USP <1224> Transfer of Analytical Procedures [32]
- USP <1225> Validation of Compendial Procedures [33]
- USP <1226> Verification of Compendial Procedures [34]
- WHO Technical Report No. 961, Annex 7, WHO guidelines on transfer of technology in pharmaceutical manufacturing [14]

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5 Technology Transfer of Drug Substance

5.1 Introduction

This chapter provides specific guidance for technology transfer of drug substance processes for large molecules (e.g., recombinant proteins, monoclonal antibodies) or small synthetic molecules. This guidance is intended to be relevant to all technology transfers between sending and receiving units. For this reason, some aspects included in this chapter may not be applicable to specific transfer programs that are usually dependent on the stage in the product lifecycle. For example, early stage transfers (from one R&D group to another), transfers of supporting laboratory scale processes, and transfers of limited process scope may not require all the elements presented.

Technology transfer teams should review the guidance provided and determine which are appropriate for their situation. This chapter can be read as a stand-alone guide for drug substance technology transfer. Further relevant information is included in the complete Guide; references to other chapters in the Guide are therefore included.

5.2 Technology Transfer Process

The focus of this chapter is on the product and process specific requirements of the technology transfer (from the initial high level technology transfer proposal to operational readiness and process/procedure qualification) as analytical method requirements are covered in Chapter 4. Where relevant, aspects that are specific to only large molecule or small molecule drug substance will be indicated.

Information on forming a technology transfer team and developing a charter is presented within Chapters 2 and 3.

5.2.1 Initial Risk/Feasibility Assessment

An initial risk assessment will be required to determine whether the technology transfer of the process and product to the receiving unit is feasible (see also Section 2.2). Prior to obtaining a detailed technology transfer package from the sending unit, sufficient information to perform an initial, high level assessment of the process at the receiving unit should be obtained. At this stage, especially if the process is being transferred to an external company, it may not be possible to share very much information due to confidentiality restrictions and documents may be redacted.

During this feasibility assessment, a skeleton cross-functional team at the potential receiving unit (representing process, analytical, operations, HSE, materials management, regulatory, and quality) should perform an initial risk assessment based on the targeted markets and desired volumes against the product specification, mass balance, raw materials, and the block/process flow diagram. For example, potential safety hazards, toxicity of the materials, corrosive nature, unique analytical requirements, raw material supply, and plant capacity are factors that should be assessed as part of this early stage risk assessment.

The rationale for evaluating product specification, mass balance, raw materials, and block/process flow diagram in the initial risk assessment are as follows:

- Depending on the stage in the product lifecycle, the regulatory strategy required to meet the business strategy should be part of the initial focus of the technology transfer. For example, if a product specification has been filed at the sending unit, this will form the basis of the technology transfer to ensure the product at the receiving unit meets the targeted market requirements, with justification of differences from existing pharmacopoeia requirements. The CQAs in the product specification are not the only items needing to be successfully produced as other quality attributes may be difficult to achieve in a new manufacturing facility.

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- A mass balance defines the plant capacity highlighting where equipment changes may be required. For example, for large molecules, equipment changes are likely to be required if the product titer is low at harvest and the drug substance final product concentration is high. A mass balance is also required to determine the most suitable receiving unit based on size of bioreactor/number of batches to meet clinical demands. For small molecule API manufacture, the mass balance serves to provide process stream composition based on both analytical data and computational analysis. As such, balanced chemical equations identifying all side- and by-products of the process chemistry need to be part of the initial risk assessment.

  - Assessment of the raw materials establishes critical supply chain, analytical complexity, and any safety constraints.

  - The block/process flow diagram provides an overview of the process defined at the sending unit to gain better understanding of the plant requirements.

Information on identifying risks, conducting risk assessments and establishing the technology transfer plan (including initial knowledge transfer) is presented within Chapter 2 (Section 2.2) and Chapter 3 (Section 3.3).

5.2.2 Consolidate Knowledge for Technology Transfer Package

Once the initial risk assessment has been performed and the receiving unit has been identified, the formal technology transfer can proceed. Confidentiality agreements may be required before the technology transfer package can be sent to an external company. A structured approach to sharing process and product information should be agreed upon upfront. Use of a controlled/defined file sharing system helps to limit large email traffic and allows for secure transfer of information and organization of information type (e.g., development reports, batch records, analytical information) between the sending and receiving units. Technology transfer activities need to follow the company requirements for data sharing and integrity.

A checklist of information/documents for large and small molecule processes is provided in Chapter 8 (Appendix 1). Some of the listed items may not be necessary or available, depending on the type of transfer being executed (e.g., internal versus external, early clinical versus commercial) and the type of molecule. This checklist should only be used as a tool to guide accumulation of the required knowledge as some APIs may have unique aspects requiring information specific to that technology transfer.

The remainder of this section outlines elements containing information about the process being transferred. Process diagrams and overviews provide a starting and reference point. These are then supplemented with the control strategy and other more highly detailed instruction for operating the process. Historical process data summaries, reports, and other process information provide necessary context and reference points for comparison between the sending unit process and the newly transferred process. Once sufficient information is available, the receiving unit can then use modeling tools to understand process fit and potential plant or process modifications required (see Section 3.4).

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5.2.2.1 Process Information

- **Block Flow Diagrams (BFDs)** are a Lean tool used to improve processes. For technology transfers, they ensure that the process steps are understood, in the correct order, and identify all process streams including waste and by-products. They can also serve the same function as the Process Flow Diagram and since they are typically easier to develop, can be used to quickly compare processing options. An example BFD is shown in Figure 5.1.

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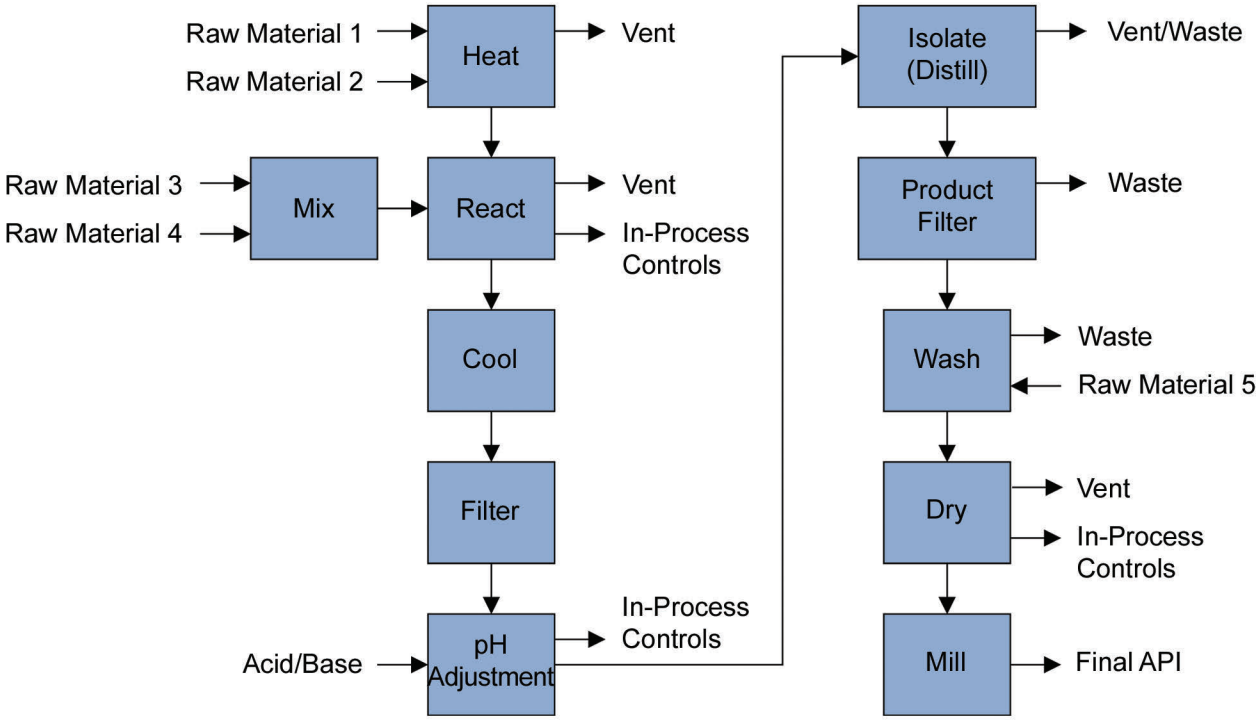
Figure 5.1: Example Block Flow Diagram – Small Molecule Upstream Process



- **Process Flow Diagrams (PFDs)** aid visualization of key data, e.g., processing times, in-process testing, etc. PFDs from all previous installations of the process (different sites or scales) can be helpful, especially for supporting updates to regulatory filings. Examples of PFDs are shown in Figures 5.2 and 5.3.

Figure 5.2: Example Process Flow Diagram – Large Molecule Upstream Process
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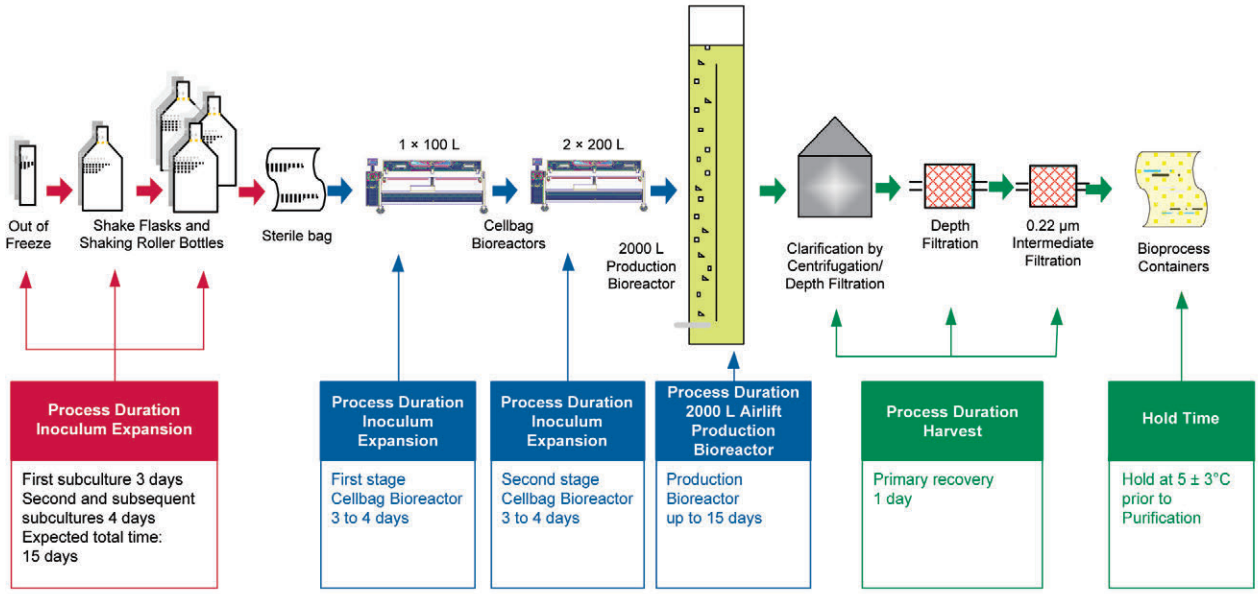

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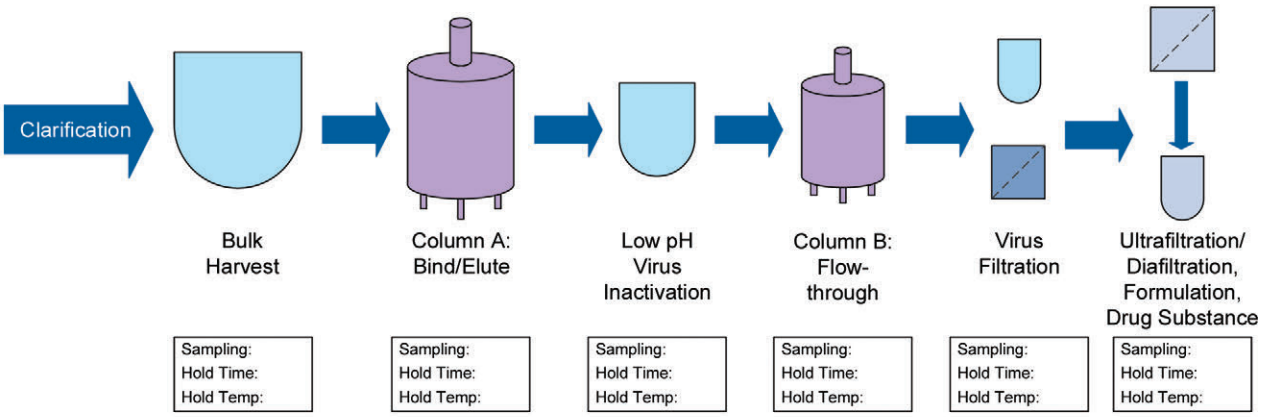
Figure 5.3: Example Process Flow Diagram – Large Molecule Downstream Process



Depending on which stage of the product lifecycle is being transferred, there may be a substantial amount of current and historical information available from the sending unit. Sources of useful information from the sending unit about the current process being transferred include:

- **Process overview descriptions** can be helpful when written in a site and scale-independent way, so that the particulars of the existing implementation(s) are not confused with the generic requirements or validated ranges. Examples of information to be provided include:
  - Tabular data for process setpoints
  - Acceptable ranges for unit operations as well as media and buffer preparations
  - Descriptions of operating procedures and strategies
  - Critical and other important process parameters, CQAs, and KPIs for the product – these should be clearly identified to build scientific process understanding at the receiving unit
- **Manufacturing batch records** (blank and/or examples of executed documents where relevant) can be the starting point for the receiving unit to generate their internal batch records. For sites that operate with paperless fully automated systems, it can sometimes be difficult to extract this information into a human-readable format to be passed along to a receiving unit. Alternative ways of information sharing may be required such as generating screenshots.
- **SOPs** from the sending unit provide information about detailed operation procedures (e.g., particle sizing for small molecules), although these may be related to site specific equipment and procedures. Any relevant or applicable SOPs already in existence at the receiving unit should be identified, as these maybe used with no or minimal modification.
- **Drug substance/process control strategy** and the link between drug product CQAs, CPPs, potential CPPs, and drug substance manufacturing requirements (see Chapter 10 (Appendix 3)). An example in-process sampling form in support of the control strategy is provided in Table 5.1. An example control strategy for CPPs is provided in Table 5.2 (see also Chapter 7). An example control strategy for quality attributes/CQAs is shown CQAs is provided in Table 5.3.

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Table 5.1: Example Template for Sampling Summary

Sample Point	Type (In-Process Controls/ Product)	Number of Samples Pulled	Sample Amount	Sample Container	Sample Hold Times	Sample Storage Requirements

Table 5.2: Example of CPP Justification for Small Molecule Process

Critical Process Step	Process Parameter	Target Range	Criticality	API CQA Impacted	Control Scheme
Raw Material Charging	Order of Addition	Not applicable	Critical	Assay/Impurities	Batch Record
	Raw Material 1	XX-XX kg	Critical	Assay	Precise measurement required; All other charges determined from Raw Material 1
	Raw Material 2	XX–XX kg	Key		
	Mix Speed	XX rpm	Non-Critical		
	Temperature	XX–XX°C	Critical	Impurities	Online control via jacket temperature with high pressure steam
	Time at Temperature	< XX hrs	Key		Hold times demonstrated are longer than the target range
	Raw Material 3	XX–XX kg	Critical	Assay	Precise measurement required
	Raw Material 4	XX–XX kg	Key		
	Mix Speed	XX rpm	Non-Critical		
Reaction	Rate of Addition Raw Materials 3 and 4	> XX lpm	Critical	Assay/Impurities	In-line orifice Flow measurement device
	Mix Speed	XX rpm	Critical	Assay/Impurities	Control system with calibration prior to campaign
	Temperature	XX–XX°C	Critical	Assay/Impurities	Online control via jacket temperature with high pressure steam
	Time at Temperature	< XX hrs	Key		
Note: This table is based on Figure 5.1.					

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Table 5.3: Example of In-Process Testing Control Strategy

Process Step	Attribute/Unit	Acceptance Criteria	Recording of Results	Quality Aspect/ CQA	Filed In-Process Control?
Column 1	DNA	≤ 1,000 pg/mg	QC batch record Laboratory management system	Quality aspect in-process (CQA for drug substance release)	No
Column 2	Host cell protein	≤ 100 ng/mg	QC batch record Laboratory management system	CQA (no further clearance steps)	Yes
Column 3	% aggregates	≤ 2%	QC batch record Laboratory management system	CQA (no further clearance steps)	Yes

- **CPV data** generated to support process parameter trending and any statistical process control analysis that has been performed as part of quarterly/annual product reviews.
- **Process observation** for transfer of the often tacit process knowledge that arises from hands-on operating experience. It can be a worthwhile exercise to observe the process operations with the existing manufacturing instruction documents (batch records, SOPs, etc.) and look for any additional elements that may not be well documented (e.g., addition of component #4 always causes a color change, pictures of typical cell morphology). Alternatively, other ways to enable sharing of this knowledge include facilitating on-site support at the receiving unit by SMEs from the sending unit and generating videos of small scale batches at the sending unit.
- **PAT**, as relevant.
- **Storage and shipping conditions** for final drug substance, which includes container type, packaging requirements, and any specific transportation requirements or validation.
- **Process intermediate and final drug substance stability** information to identify stability constraints that affect hold times between steps or timing of product release for shipping to other processing sites (e.g., fill finish).
- **Process specific deviations, investigations, and CAPAs** from GMP batches.
- **Extractables/leachables/elemental impurities knowledge** (risk assessments or testing data relevant to the process) and to clarify which disposable equipment is suitable for use in the process.
- **Spreadsheet-based process models** are important tools, if available, to facilitate process fit at the receiving unit. These can be used to identify suitably scaled equipment, e.g., for predicting column sizes depending on a range of harvest titers or for identifying volume handling issues.
- **Product and reagent safety documentation** detailing any potency concerns (in terms of operator handling) and cleanability issues. Safety information should be cross checked between the sites as part of the technology transfer since this can be company specific.

The current process at the sending unit will have been established based on various studies and production of GMP and non-GMP batches at different scales. Historical data is valuable to ensure product comparability based on established assay acceptance criteria (see Chapter 4) as well as process comparability, support for potential

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deviations, understanding of risk, suitability of small scale data, process sensitivities, establishment of the control strategy, etc. Such data is likely to include:

- **Historical process manufacturing reports** that describe previous or existing site specific instances of the process so that the receiving unit has tangible examples of how the generic process requirements have been implemented previously (e.g., media may have been successfully prepared using water at different temperatures, buffers may have been prepared in a fixed tank or via in-line dilution, bioreactors may have been run with different sparger configurations, etc.). These should include, as applicable, process and equipment settings, cycle and turnaround times, cleaning/regeneration settings, yield, capacity, and intermediate specifications.

- **Historical process and analytical data –** this may be formatted in a number of ways including statistical process control trend charts, tabulated means and ranges of KPIs, as well as in-process trends for continuous data (such as cell growth, pH control, freezing temperature profiles, etc.). Examples of visual data (such as column chromatograms, pictures of cell suspensions, etc.) should be included where relevant.

- **Process development reports –** especially for transfers from initial process development at the sending unit to GMP manufacturing at the receiving unit. These reports can be helpful for technical support staff at the receiving unit to understand how the process evolved and where its sensitivities may lie.

- **Virus clearance data** for molecules of mammalian origin.

- **Small scale model qualification** may be useful to establish a tool for monitoring the GMP process at the receiving unit and determining which, if any, historical process characterization data is applicable at the receiving unit.

- **Operating instructions and data** from small/laboratory scale models of the process.

- **Process characterization reports –** these present results from Design of Experiment studies or other studies evaluating the design space around process parameters and resulting CQAs. Such assessments provide the basis for defining CPPs and Proven Acceptable Ranges. This informs the process control strategy, guides the establishment of Normal Operating Ranges for the process, and provides further understanding of where process sensitivities exist.

- **Process validation reports –** demonstrating the acceptable ranges and robustness of the process at the sending unit.

- **Reports from prior transfers** of this process, especially where they may contain significant lessons learned.

- **Annual product reviews** to capture process/analytical trending and capabilities and out of specification events.

- **Regulatory license documents** (relevant sections) and existing commitments for the process that may have resulted from previous regulatory inspections.

- **Change control summary** for the product/process which would cover all specification changes, process changes, equipment changes.

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5.2.2.2 Materials Information

Early alignment of raw materials or consumables used in the process will facilitate transfer. There is likely to be an extensive list of approved chemicals and consumables already in use in GMP manufacturing at the receiving unit. Materials information include:

- Specifications of raw materials

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- Critical material attributes
- CQAs for raw materials that can impact the final API (for small molecules)
- Bill of materials for the process
- Media and buffer formulations, including titrants used for pH adjustments
- Other materials/consumables (e.g., filters, bags, gaskets/O-rings, tubing, sieves for sifters, screens for milling) – including lead times
- Identification of new materials such as disposables/consumables and understanding of lead times (see Table 5.4 for an example)
- Current suppliers and catalog numbers – primary and secondary sources, including audit findings
- Identification of acceptable alternative materials (see Table 5.5 for an example)
- Identification of receiving unit restrictions on suppliers or materials
- Cell bank, origin of cell bank, cell bank testing (large molecule only)
- Appropriate general safety documentation
- Excipient risk assessments

Table 5.4: Example Comparison of Materials between Sending and Receiving Units

Component Specified by Sending Unit	Closest Equivalent at Receiving Unit	Comments
Millipak® 20 (0.2 µm) filter	Millipak® 60 (0.2 µm) filter	Equivalent acceptable
Supplier A depth filter for harvest	Supplier B depth filter for harvest	Equivalent acceptable
Sartopore® 2 (0.2 µm + 0.1 µm) filter	Pall Fluorodyne Kleenpak® (0.1µm) filter	Equivalent acceptable, same membrane type
1 L and 500 ml Teflon® bottles for drug substance bulk fill	High-density polyethylene Nalgenes® for drug substance bulk fill	Teflon® bottles to be sourced
Sodium phosphate dihydrate	Sodium phosphate monohydrate and heptahydrate available	Buffers to be reformulated and evaluated at small scale
L-Histidine base	Multi-compendial grade routinely sourced from supplier 1	Process specific requirement from new supplier required
L-Histidine monohydrochloride		

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Table 5.5: Example Template for Raw Materials Summary

Compound	Category	Type (New/ Existing)	Site of Manufacture	Grade	Packaging	Supplier	Distributor Used	Qualification Status	Quality Agreement Status
Raw Material 1	Starting Material or API								
Raw Material 2	Critical								
Raw Material 3	Critical								
Raw Material 4 – Salt	Critical								
Raw Material 5 – Acid/Base	Non-Critical								
Raw Material 6 – Solvent	Non-Critical								
Raw Material 7 – Solvent	Non-Critical								

5.2.2.3 Analytical Method Information

To support in-process and drug substance testing at the receiving unit, early establishment of analytical methods is a critical activity (see Chapter 4). At a minimum, the following information is required:

- Compendial (e.g., USP [35]) or other qualified methods used for all raw materials and final drug substance
- In-house methods used for all raw materials and final drug substance
- Method qualification and validation protocols/reports at sending unit
- In-process sample control specifications and methods
- Setup and qualification of raw material and reagent testing at the receiving unit – this can be a long lead item

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5.2.2.4 Facility/Equipment Specifications and Capabilities Information

Information on specific process details that affect the facility/equipment for the manufacturing process should be highlighted early on in the technology transfer to allow any facility modifications at the receiving unit to be priced out, planned, and implemented (see Chapter 11 (Appendix 4)). This information includes:

- Piping and Instrumentation Diagrams (P&IDs) for equipment that may be specific to the process
- Equipment User Requirements Specification (URS) and/or Functional Requirements Specification (FRS)
- Automation/process control requirements for specific process operations that are to be replicated at the receiving unit (e.g., pH control strategy, nutrient feed control strategy, controlled rate freezing, ultrafiltration/diafiltration, chromatography)

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- System size and capacity – e.g., suitability of available vessels, piping diameters, pumps, etc. for the incoming process scale
- Capability for temperature control if needed
- Requirements for particular sensors or analytical equipment where differences in types (sensitivity, methodology, etc.) or suppliers may result in skewed measurements, equipment materials of construction, compatibility studies for process materials and solutions
- Cleaning procedures (with emphasis on any difficult to clean residues in the process), development studies, validation reports, and/or recommendations
- Safety assessments and/or information on hazardous waste or other waste streams specific to the process that may require special disposal or neutralization consideration
- Product/process safety assessments including information on requirements for Personal Protective Equipment, exposure limits, containment requirements, etc.
- Instrumentation requirements (pH meters, control systems, PAT technology used, etc.)

5.2.3 Technology Transfer Operational Readiness

Gap Analysis/Risk Mitigation

Once the technology transfer package has been obtained, the receiving unit can then initiate a process gap analysis in order to compare how the process was operated at the sending unit to how the process will be operated at the receiving unit. This will identify:

- Where there are gaps in process knowledge
- Where process changes may be required to fit the new facility
- Whether any engineering modifications are required for the new facility

Section 3.3.2 describes approaches to risk analysis and evaluation and the tools that can be used. By considering the severity (impact), occurrence (probability) and detection, a risk priority number can be derived. This can be used as a basis for determining the level of risk and whether mitigations are required. Examples of process gap analyses are provided in Section 5.4. The gap analysis is often the first opportunity for the transfer team to intensively review the process together, and thus can serve as a good learning exercise for the receiving unit.

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Transfer and Implementation of Laboratory Scale Process

If possible, the laboratory scale version of the process should be transferred prior to (or concurrently with) the full scale process transfer. This provides the receiving technical unit the opportunity to work with the process hands-on and catch any details that may have been overlooked in the technology transfer package. Prior to implementation of an engineering batch at the receiving unit, a full laboratory scale run of the process is advisable as soon as possible to help in scaling up the process, to minimize risks and establish acceptance criteria (see case studies presented in Chapter 10 (Appendix 3)).

The knowledge gained from this exercise can be used for drafting preliminary process descriptions and as a basis for scale-up. Technical SMEs from the sending and receiving units can align on process details. The laboratory scale process may also be used to qualify raw material suppliers or to evaluate revised buffer formulations. For small molecule APIs, laboratory scale/pilot runs can be conducted outside the GMP facility for evaluation/consistency in order to demonstrate reproducibility under GMP conditions.

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Low Risk Transfers

If there is sufficient knowledge of the operating procedure of the receiving unit, based on previous transfers, a transfer straight into the manufacturing facility is possible. However, it remains advisable to implement a laboratory scale process that can be operated in satellite mode (using the manufacturing process intermediates, materials, and solutions at each step) during initial full scale operations. This will assist in troubleshooting any excursions or failures that might occur at scale (see Chapter 3).

In this situation, there may be useful experimental work to support transfers, as described in the following example of risk mitigation.

Example of Risk Mitigation

In a large molecule purification process, buffer pH and conductivity are important measurements for ensuring the required product quality and impurity clearance on chromatography steps. However, the use of different meters between the sending and receiving units may result in small differences in measurements. One way to mitigate this is to send buffer samples between sites and compare measurements. In this way, differences in measurements can be considered and if applicable, offsets applied in documentation at the receiving unit. Any such experimental work should be documented to provide traceability and justification of what may appear to be a specification change on transfer.

5.2.4 At Scale Process (Procedure) Evaluation and Qualification

This is the principle of demonstrating the process at full scale at the receiving unit. Proceeding to full scale runs may require approval via the stage gate reviews to demonstrate that the required preparation has been completed. When these runs are intended to be fully GMP and released for use (clinical or commercial), it is good practice to document the decision making leading to the conclusion of readiness.

The initial run(s) at full scale may be designated as engineering runs or commissioning runs. These may be fully GMP runs intended for release, depending on the perceived risk and mitigation strategies. Engineering runs are defined here as trial full scale runs performed in the same equipment to be used for GMP manufacturing using the intended procedures and without the intent to be released for use. Either an engineering run or a GMP run can be considered relevant as part of demonstrating the appropriate performance of the process (qualification). Such test runs are recommended prior to running PPQ batches.

- Engineering runs/testing
  - If an engineering run is processed, the data should be reviewed prior to the first GMP batch to identify any potential issues. An engineering run provides opportunity to ensure that the data generated would meet acceptance criteria (if performing process validation) or is comparable to previous campaign data. Depending on the nature of the transfer, it may be appropriate to have specific acceptance criteria agreed upon prior to the engineering run. Schedules to proceed to GMP runs may be constrained and if the data look borderline acceptable, a clear decision-making process will be necessary.
  - Developed manufacturing instructions should be used during the engineering run and updated with necessary changes if applicable.
  - In-process sampling can be used for testing, including evaluating the status of assay procedures to be used. Samples may also be used for any further studies identified which may be run concurrent with the first GMP batch (e.g., resin reuse).

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- Qualification/validation
  - The main objectives of validation include ensuring that all facilities, processes, systems, procedures, and equipment that may affect the quality of the product operate reliably and reproducibly and that the process is run under a state of control.
  - A validation master plan should be generated by the receiving unit prior to validation activities to help draw conclusions as to whether the overall process is validated.
  - Equipment qualification should be completed in advance of the process validation. Typically, cleaning validation may be performed concurrently with process validation.
  - Process validation protocols should be approved prior to initiation of validation runs. The protocols will summarize specific parameters and tests that need to be monitored and compared to the protocol acceptance criteria.

5.2.5 Finalize Transfer and Perform Review

Evaluation of the technology transfer should be performed to determine if the transfer stage can be formally completed/closed out; some guidance is given in this section. Information on technology transfer success criteria is presented in Section 2.3.

5.2.5.1 Evaluate Transfer Performance Against Goals

In order for the technology transfer to be considered successful, the project manager needs to establish clear boundaries and goals for the outcome of the project. Typically, a commercial-commercial technology transfer is considered to be complete upon successful validation at the commercial scale of the receiving unit with the required regulatory filing. For clinical products, the technology transfer may be considered complete after the manufacture of the clinical batch supply and filing of the investigational new drug. The goals and boundaries should be defined and agreed upon with management using a formal process such as a charter (see Section 3.1.2).

Upon completion of the planned full scale runs, the data needs to be carefully reviewed against the performance goals identified at the start of the technology transfer process, to evaluate the success of the process transfer.

Key elements may include:

- Process data which demonstrates successful execution of the control strategy
- Satisfactory in-process and drug substance product quality
- Successful operation of the specified process over an agreed number of batches
- Material usage within the predicted ranges
- Successful operation of the plant and equipment with regard to operability, labor, and plant occupancy times
- Successful manufacture of drug product from the drug substance manufactured at the receiving unit
- A remediation plan, if necessary, should be agreed to prior to wrapping up the technology transfer activities

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5.2.5.2 Project Close-Out

There are likely to be unique goals for each technology transfer. However, a process and product comparability report should be written to demonstrate that the transfer was successful, with justification of any differences which may have occurred due to use of different equipment, different analytical methods etc. Any major changes or deviations should be documented and justified. This report may form the basis of the next stage in the product lifecycle (e.g., from clinical to commercial batch) or may be required for a filing update.

A lessons learned activity between the sending and receiving units, preferably via face-to-face meetings, is useful to identify what went well and what could be improved (covering e.g., program management, technical aspects, analytical/QC, GMP documentation, QA, raw materials). The output should be documented as a summary report to provide a basis for subsequent technology transfer between units and to identify future improvements for such a program which can then be adequately addressed and implemented.

5.3 Points to Consider

For large molecules, development of a platform process (for use with multiple products where possible) facilitates process transfer, whether between internal sites or to a CMO. After transfer of the first platform process, the receiving unit will then have some knowledge of process operating conditions. In addition, as a process transfers into a CMO from a process development sending unit, the sending unit will gain more knowledge of the CMO capabilities, which facilitates future process development.

A technology transfer policy and SOP are recommended at the sending unit and receiving unit to define the framework of the transfer, to include, e.g., where decision points should be made and which functional groups should be involved at each stage of the transfer.

During process development for large or small molecules at a sending unit, with the intent of transferring to a CMO selected for partnering, it is recommended to work with the CMO during process development to ensure the process will fit the manufacturing facility in order to reduce risks. The CMO will have standard operating conditions for equipment (such as flow rates on chromatography steps) or there may be equipment constraints in volume handling and there may be a defined timeframe for the manufacturing slot.

To ensure more rapid large molecule process transfers, the use of animal component free, chemically defined media is preferred. The use of media such as hydrolysates or that contain animal components can introduce more variability into a process, which can affect process comparability between sites. If animal components are present, these must be of suitable quality for use in a GMP facility, with good traceability of supply. Any raw materials of consumables new to the receiving unit may cause delays due to establishment of the supply chain (e.g., new supplier agreements may be required) and availability of appropriate paperwork (demonstrating appropriate quality).

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5.4 Example Gap Analyses for Small Molecule and Large Molecule Processes

The tables below show simple examples from process/facility gap assessments. Example gap assessments are provided for a small molecule process in Table 5.6 and for a large molecule process in Table 5.7. Each transfer team will need to work together to establish a risk assessment matrix suitable for the program and the site quality requirements. Teams may find it helpful to start the assessment by developing a detailed PFD of the existing process by stepping through a process batch record, identifying each piece of equipment, vessel, engineering/facility requirement, product manipulation, product contact, process solution, etc. that is encountered throughout that process step. This list may then be used to generate the first columns in the gap assessment table. Once the mitigation list has been prioritized, actions should be assigned with realistic due dates to ensure the process is ready for transfer.

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Table 5.6: Example Gap Assessment – Small Molecule Process

Category	Parameter	Sending Unit State	Receiving Unit State	Gap/Risk Description	Risk*	Mitigation Plan
Process	pH	Performed on in-process sample to ensure pH range	Needs to be performed since this is a critical parameter	• Instrument type used in the EU not readily available in the US • Sample location critical	High	• Perform method qualification with new pH analyzer • Design equipment for sample location
Facility	Utilities	High pressure steam required	Not available at pressure needed	Capital investment	Medium	Upgrade steam
	Milling	• Segregated room • Enclosed milling/ packaging system • Dust explosion prevention	• Segregated room available • Milling/ packaging systems are separate • Dust explosion category higher than facility design	• Capital investment • Possible regulatory filing impact • Safety impact	High	• Upgrade of dust explosion prevention needed • QA evaluation of separate systems impact • File separate milling/packaging systems • Permitting impact
Equipment/ Operations	Raw material difficult to charge	• Special charging system • Raw material sourced with special packaging	Charging system not adequate	Capital investment	High	• Procurement controls • Equipment design/ qualification
	Filtration	Uses centrifuge	Need to use filter/ dryer	Ability to filter and wash product adequately	High	Perform laboratory testing and qualification
Raw Materials	Metered charge of one raw material to reaction	Critical parameter to process	• Type of metering required • Materials of construction	• Capital investment • New procedures required	Low	• Ensure equipment design suitable for the process requirement • Equipment qualification • Write SOPs for proper implementation of the metered rate
Analytical	Instrument types	Uses Vendor 1 HPLC	Only has Vendor 2 HPLCs	• Successful method transfers • Project timing	Medium	• Perform method equivalence study • Order/qualify new instrument
HSE	Containment	Fully enclosed processing	• Filter is an open top centrifuge • Dryer is manual discharge	API toxicity classification requires full containment	High	Ensure plant design adequately addresses all HSE exposure concerns

*Risk (Low/Medium/High) may also be evaluated using other risk management tools such as assessment of severity/occurrence/ detection to generate a Risk Priority Number (RPN).

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Table 5.7: Example Gap Assessment – Large Molecule Process

Category	Element/ Process Parameter	Sending Unit State	Receiving Unit State	Gap/Risk Description	Risk*	Mitigation Plan
Mixing	Buffers with high concentration of reagent X require adequate mixing	Experience with this type of buffer preparation at smaller scale	No experience with similar buffer types	No SOP for defining mixing parameters: time/ RPM/pump rate	Medium	Establish SOP and test during shakedown
Facility	Room configuration for post-nanofiltration	Segregated room; nanofiltrate collected in class C room with a higher air pressure	No segregated room available	Ability to maintain segregation between pre and post-nanofiltration	Medium	• Operate as closed system, make connections in biosafety hood • Consult with QA
Equipment/ Operations	Time critical addition and mixing of chemical reagent	Small volume, visual assessment of mixing	>10x scale-up to vessel	Potential for incomplete reaction or reaction "hot spots"	High	Computational fluid dynamics modeling of reaction vessel to determine appropriate mixing conditions
Equipment/ Operations	Tangential flow filtration system holdup volume	Small system, minimal holdup relative to product volume	Available system is substantially larger, 1/3 of product will be in holdup volume	Temperature increase (extended time in system piping) and possible effect on target protein	Medium	• Test in engineering run • Backup plan to switch to smaller portable vessel
Raw Materials	Disposable bags	Vendor A	New Vendor B, limited experience with this item at sending unit	• Delay in raw material, leaks or other performance issues • Potential risk from leachables	Medium	Evaluate during engineering runs
Operations	Available column sizes	Column sized to perform two operations per day at 20°C	Three cycles will be required based on available column	• Limited stability of column eluates will require storing eluates at 8°C then pooling and returning to ambient for next operation • Potential effects on product, and timing/logistics	Medium	• Evaluate at small scale • Schedule realistic processing times and evaluate performance

*Risk (Low/Medium/High) may also be evaluated using other risk management tools such as assessment of severity/occurrence/ detection to generate a Risk Priority Number (RPN).

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6 Technology Transfer of Drug Product

6.1 Introduction

As a drug product transitions through each stage of the clinical program and a higher degree of confidence in the clinical profile of the product materializes, the strategy should shift to focus upon commercialization of the dosage form. This should include defining process scale-up and transfer plans to the intended commercial manufacturing site. This activity traditionally occurs between the phase 2 and phase 3 clinical stages, in order to ensure phase 3 clinical and pivotal stability supplies are produced within a commercial facility. This also can be advantageous in supporting lifecycle validation activities for the product and helping to build manufacturing experience in a commercial facility. In addition, there may be a need to transfer a drug product to a new commercial manufacturing site post-registration and launch. Although more information may be available while transferring a commercial product (filed release specifications, CPPs, etc.), the general knowledge needed and steps to successfully transfer a product remains the same.

6.2 Technology Transfer Process

This chapter highlights those aspects of a technology transfer process specific to drug product. Analytical requirements are covered in Chapter 4. Note that many of the principles of drug substance transfer, covered in Chapter 5, are applicable to drug products.

Information on forming a technology transfer team, developing a charter, consolidating knowledge for transfer, and agreeing on a high level technology transfer proposal is presented within Chapters 2 and 3.

In general, a technology transfer can be described by answering the following questions:

- What is the current process/product?
- What is the desired final product/process, based on any constraints/criteria related to the new site?
- What needs to be done to prove that the final product is correct and the process is robust?

The technology transfer is defined by how these questions are answered.

6.2.1 Identify Risks, Conduct Risk Assessments, and Develop Technology Transfer Plan

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An initial risk assessment should be conducted to determine whether the technology transfer of the process and product to the receiving unit is feasible. This high level risk assessment may be challenging to complete if the product is being transferred to or from an external company and information is difficult to share due to confidentiality agreements.

A cross-functional team consisting of, at minimum, process, analytical, supply chain, operations, HSE, and quality should evaluate the data available to determine feasibility. Items for consideration include:

- Projected volume and plant capacity
- Target markets
- Safety hazards
- Special equipment, PAT, and analytical needs

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For aseptic processes, additional consideration should be given to facility capabilities to maintain an aseptic area. The assumption is that the receiving facility will have the appropriate controls in place to produce aseptic products (e.g., HEPA filtration, air classification, gowning procedures, etc.).

Based on a positive outcome from this analysis, conducting an in depth risk assessment with technical experts from both the sending and receiving units is highly recommended. This is an opportunity to dive into the technical details of the manufacturing process and understand the product and the relationship between manufacturing steps and critical attributes (if known). It is highly advisable, if possible, to conduct this assessment in a face-to-face environment, to better facilitate the transfer of knowledge. The risk assessment should also capture gaps in product knowledge/ process understanding, or highlight the additional information needed from the external site to guide the technology transfer plan.

Information on identifying risks, conducting risk assessments, and establishing the technology transfer plan is presented in Chapter 2.

6.2.2 Consolidate Knowledge for Technology Transfer Package

- **Block Flow Diagrams (BFDs)** are a Lean tool used to improve processes. For technology transfers, they ensure that the process steps are understood, in the correct order, and identify all process streams including waste and by-products. They can also serve the same function as the PFDs and since they are typically easier to develop, can be used to quickly compare processing options. Simple and detailed examples of BFDs are provided in Figures 6.1 and 6.2.

Figure 6.1: Example of Simple Block Flow Diagram – Direct Compression Coated Tablet

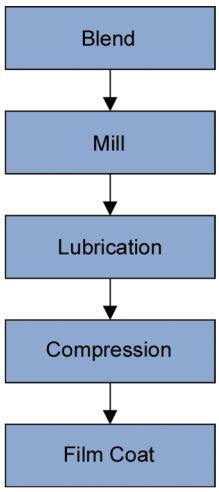

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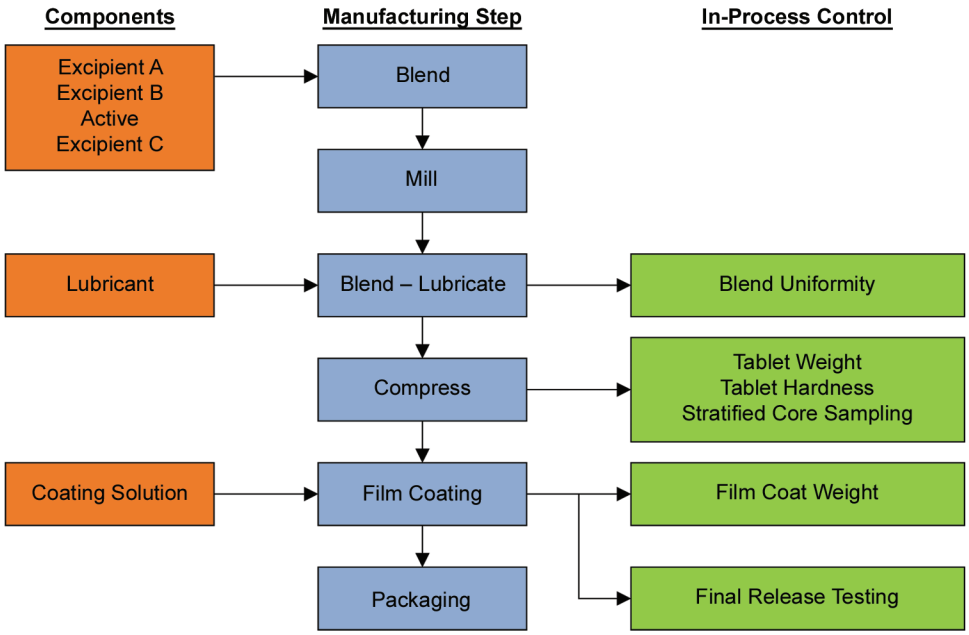
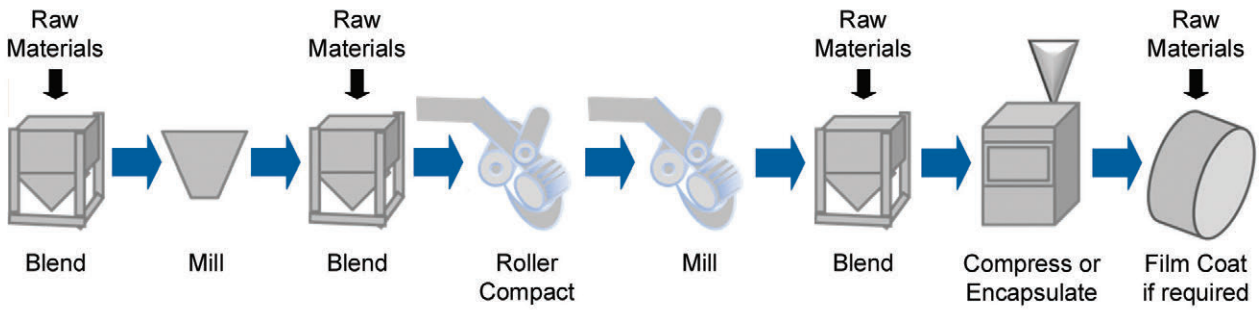
Figure 6.2: Example of Detailed Block Flow Diagram – Direct Compression Coated Tablet



- **Process Flow Diagrams (PFDs)** aid visualization of key data, e.g., processing times, in-process testing, etc. PFDs from all previous installations of the process (different sites or scales) can be helpful, especially for supporting updates to regulatory filings. An example PFD is shown in Figure 6.3.

Figure 6.3: Example Process Flow Diagram – Dry Granulation Product



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Depending on which stage of the product lifecycle is being transferred, there may be a substantial amount of current and historical information available from the sending unit. The following list of points indicate useful information about the current process being transferred and can be used when requesting documentation:

- **Manufacturing batch records** (blank and/or examples of executed documents where relevant) can be the starting point for the receiving unit to generate their internal batch records. For sites that operate with paperless fully automated systems, it can sometimes be difficult to extract this information into a human-readable format to be passed along to a receiving unit. Alternative ways of information sharing may be required such as generating screenshots.

- **Drug product/process control strategy** and the link between drug product CQAs, CPPs, potential CPPs, critical aspects, and drug product manufacturing requirements.

- **Risk Assessments** that may have been conducted, including any excipient analysis work or packaging analysis.

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- **Process observation** for transfer of the often tacit process knowledge that arises from hands-on operating experience. It can be a worthwhile exercise to observe the process operations with the existing manufacturing instruction documents (batch records, SOPs, etc.) and look for any additional elements that may not be well documented (e.g., addition of excipient #3 always causes the solution temperature to drop). Alternatively, on-site support at the receiving unit by SMEs from the sending unit can facilitate sharing of this knowledge.

- **PAT** tools, as relevant.

- **Storage and shipping conditions** for drug product to include container type, packaging requirements, freeze/thaw cycle data, and any specific transportation requirements or validation.

- **Process intermediate stability** information to identify stability constraints that affect hold times between steps (e.g., blend hold time prior to tableting, suspension hold time prior to filling).

- **Extractables/leachables/elemental impurities/materials compatibility knowledge** (risk assessments or testing data relevant to the product/process) and to clarify which equipment is suitable for use in the process (e.g., product compatibility with silicon tubing used to transfer solution from bulk tank to filling nozzle).

- **Spreadsheet-based process models** are important tools, if available, to facilitate process fit at the receiving unit. These can be used to identify suitable equipment parameters, e.g., for predicting cycle time for secondary drying in a lyophilized product.

- **Product safety documentation** detailing any hazard concerns (in terms of operator handling) and cleanability issues.

- **Equipment cleaning** procedures or cleaning validation information.

- **Historical process and analytical data –** this may be formatted in a number of ways including statistical process control trend charts, tabulated means and ranges of KPIs, as well as in-process trends for continuous data (such as compression force data, pH control, temperature profiles, etc.). Examples of visual data (such as chromatograms, pictures of phase separation, etc.) should be included where relevant. This should also capture a process baseline including process settings and maintenance routines such as scheduled elastomer replacements.

- **QA documentation –** if available, this may include deviation reports, laboratory investigations, and incident reports. For a commercial product, this could include annual reports to regulatory agencies.

- **Process development reports –** especially for transfers from process development to manufacturing. These reports can be helpful for technical support staff at the receiving unit to understand how the process evolved and where its sensitivities may lie.

- **Setup/operating instructions and data** from all scales of the process (small/laboratory scale models of the process through to any intermediate/large scales available).

- **Process characterization reports –** these present results from Design of Experiment studies or other studies evaluating the design space around process parameters and resulting CQAs. Such assessments provide the basis for defining CPPs and Proven Acceptable Ranges. This informs the process control strategy, guides the establishment of Normal Operating Ranges for the process, and provides further understanding of where process sensitivities exist.

- **Product characterization reports –** these present results from Design of Experiment studies or other studies evaluating the product characteristics and how they are impacted by processing parameters (e.g., % of amorphous/crystalline material in a lyophilized product, droplet size in an ointment, membrane thickness for a functionally coated tablet).

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- **Process validation reports –** demonstrating the acceptable ranges and robustness of the process at the sending unit. This is usually not available early in product lifecycle. There may also be information around areas where heightened monitoring would be recommended. While the receiving unit needs to do their own validation, this can be useful baseline information.

- **Reports from prior transfers** of this process, especially where they may contain significant lessons learned.

- **Regulatory license documents** (relevant sections) and existing commitments for the process that may have resulted from previous regulatory inspections.

6.2.3 Technology Transfer Execution/Pilot Scale Batches

Scale-up or pilot batches are often used to gather initial information using the new equipment train while minimizing use of raw materials and API. It is important to design these batches with the final scale/equipment in mind. Depending on the lifecycle phase of the product being transferred, these batches are an opportunity for building and improving upon batch records, control strategies, etc. (if not already established). These can also be an opportunity to characterize the product attributes with a new process, so it is important to have a thorough sampling plan in place before starting manufacture.

For example, for a direct compression tablet process, it would be advisable to collect stratified tablet core samples across the run to determine if there is a tendency toward segregation or a lack of uniformity. For a liquid product, it may be desirable to collect multiple samples during API dissolution to gain an understanding of the dissolution profile. Alternately, PAT can be used if applicable. A statistician can help determine how many samples need to be tested to gain understanding of each unit operation. It is helpful to work with the analytical team/sample testing laboratory so that there is awareness of the scope of work required and can trigger a resources/prioritization discussion if needed.

The team should work together to produce a document containing the sampling plan, what tests are desired from each sample, and where those samples should be sent.

Pilot batches are also useful in addressing gaps that were identified in the transfer risk assessments. For example, process understanding Design of Experiments that may not be practical on a large scale may yield product understanding across a range of conditions, e.g. impact of coating parameters (spray rate, gun-to-bed distance, pan speed) on the permeability of a functional membrane coating.

If the commercial equipment is not available, a similar equipment train should be used. The results from this type of experimental work can be confirmed with a smaller number of runs on the commercial equipment/at the commercial scale, saving time and resources while generating appropriate data to support regulatory filings. Key criteria (release specifications and in-process controls) should be matched across scales (e.g, target water content in a lyophilized product or hardness of a tablet core).

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Modeling should be evaluated for help with scale transfer, and appropriate characterization work should be done, including in-process tests (e.g., keeping the dwell time constant in a tablet press across scales/equipment, testing tablet thickness). For aseptic processes, consider using these batches to aid in filter validation, container/closure integrity testing, and developing protocols for inspection processes.

Depending on the phase of the product being transferred, it may be worthwhile to conduct another risk assessment or gap evaluation once the pilot lots are complete, in preparation for moving to the full scale work.

6.2.4 Engineering Runs/Pre-Process Qualification Runs

Where an engineering run is processed, the data should be reviewed prior to proceeding to batches to be released to identify any potential issues. An engineering run provides opportunity to ensure the data generated will meet acceptance criteria if performing process validation (or is comparable to previous campaign data.) Depending on the nature of the transfer, it may be appropriate to have specific acceptance criteria agreed upon prior to the engineering

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run. Schedules to proceed to qualification runs may be constrained and if the data looks borderline acceptable, a clear decision-making process will be necessary.

If there is further need to scale-up, proceeding to full scale runs may require approval from key stakeholders via stage gate reviews to demonstrate the required preparation has been completed. Where these runs are intended to be fully GMP and released for use (clinical or commercial), it is considered good practice to document the decision making leading to the conclusion of readiness. The initial run(s) at full scale may be designated as engineering runs or may be full GMP runs intended for release, depending on the perceived risk and mitigation strategies. Engineering runs are defined here as full scale trial runs performed in the same equipment to be used for GMP manufacturing using the intended procedures and without the intent to be released for use. Either an engineering run or a GMP run can be considered relevant as part of demonstrating the appropriate performance of the process. At this point, the basic control strategy should be known and implemented (e.g., tablet weight control, ointment tube seal integrity testing).

Developed manufacturing instructions should be used during the engineering run and updated with necessary changes if applicable.

In-process sampling can be used for testing, including evaluating the status of assay procedures to be used. Samples may also be used for any further studies identified, i.e., prototype stability studies.

A development report should be compiled to document the process, parameters, findings from development work (edge of failure, sensitive processes), control strategy, etc. prior to moving into qualification runs.

6.2.5 Process (Procedure) Qualification

Pre-requisites for qualification activities in terms of facility, equipment, skill capability, and documentation should be established. The timing of qualification activities may be dependent on the completion of these pre-requisites. Pilot batches are frequently a pre-requisite that is used to increase process and product understanding, as described above.

Qualification of Technology Transfer/Process Qualification Runs

Prior to validation activities, a plan should be generated by the receiving unit to determine the validation strategy. Process validation protocols should be approved prior to initiation of validation/qualification runs. The protocols should summarize specific parameters and tests that need to be monitored and compared to the acceptance criteria.

The following points should be addressed by the completion of validation:

- Control strategy
- Master and executed batch records
- Cleaning validation
- Training and SOPs
- Sampling requirements to build product and process understanding
- Stability protocol
- Analytical methods
- Acceptance criteria for in-process controls and finished product testing
- Storage and packaging conditions

Information on finalizing the transfer and performing a review is presented within Chapter 2.

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7 Quality Aspects of Technology Transfer

7.1 Introduction

This chapter provides guidance on quality aspects to consider for technology transfer, including:

- Quality representation on the technology transfer team
- QbD and control strategy
- QRM
- Analytical method comparability and stability strategies
- Process validation strategies
- Change management
- Execution

This chapter is intended to be relevant to all technology transfers; therefore, some aspects may not be applicable to specific transfer programs. For example, early stage transfers (clinical versus commercial), may not require all the elements presented. Technology transfer teams should review the information provided and determine which are appropriate for their situation.

7.2 Quality Representation on the Technology Transfer Team

Quality needs to be considered throughout the technology transfer process; therefore, it is essential to ensure quality is represented in the core technology transfer team. The core team should include representation from QA and analytical/QC as well as SMEs to address technical aspects of product quality, analytical comparability, and stability requirements with sufficient product knowledge to participate in quality risk assessments. It is important to ensure close communication with the quality unit throughout planning and execution phases to ensure alignment on strategies, timelines, and resource allocations.

7.3 Quality by Design and Control Strategy

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QbD principles should be considered when the control strategy is developed; these include QTPP, CQAs, CPPs and design space (as applicable). For more information on these principles, refer to the following ICH [36] quality guidelines:

- ICH Q8(R2) Pharmaceutical Development [3] states:

  "demonstration of greater understanding of pharmaceutical and manufacturing sciences can create a basis for flexible regulatory approaches"

- ICH Q9 Quality Risk Management [4] – QRM can be used to justify a control strategy for technology transfer and to identify and control potential quality issues during development and manufacturing. Effective QRM provides regulators with greater assurance of a company's ability to deal with potential risks.

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- ICH Q10 Pharmaceutical Quality System [1] – Technology transfer is specifically called out as a lifecycle stage goal. It states:

  "The goal of technology transfer activities is to to transfer product and process knowledge… This knowledge forms the basis for the manufacturing process, control strategy, process validation approach and ongoing continual improvement."

- ICH Q11 Development and Manufacture of Drug Substance (Chemical Entities and Biotechnological/Biological Entities) [5] provides further clarification on Q8(R2), Q9 and Q10 [3, 4, 1] as they pertain to drug substance.

Note: It is recognized that there are ongoing developments with industry guidelines and this Guide reflects an understanding of them as of the publication date. It is also recognized that draft ICH Q12 Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management [37] was endorsed by members of the ICH [36] Assembly and released as a draft on 16 November 17. The draft addresses the commercial phase of the product lifecycle with respect to ICH Q8(R2), Q9, Q10 and Q11 [3, 4, 1, 5], and describes how increased product and process knowledge can be used to justify less extensive regulatory oversight.

CQAs are the basis of the control strategy and should be available to the receiving unit to provide the scientific rationale for manufacturing process controls and supporting analytical methods. In combination with product quality risk assessments, CQAs ensure adequate controls are in place to produce consistent product quality.

To increase the likelihood of successful technology transfer, changes should be minimized as follows:

- Materials (including raw materials, starting materials, intermediates, reagents, primary packaging materials for the drug substance, resins, disposables, etc.) should be well understood and, ideally, as similar to the materials used at the sending unit as possible to minimize any differences in drug substance/drug product quality.

- Differences in the design of the manufacturing process (e.g., sequence of purification steps for large molecules, order of addition of reagents, etc.) should be limited to differences in scale to minimize any potential product quality impact between manufacturing sites.

- Any changes to in-process controls (including in-process tests and process parameters) and release testing should be limited to enable comparison of data to the sending unit.

- Analytical methods should be consistent between sites. If analytical methods need to change, then ensure equivalency studies are performed so results can be compared between sending and receiving units.

- Consider the various levels of controls within the manufacturing process, specifications, in-process controls, batch records, and automation controls. Any differences need to be clearly documented and justified.

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7.4 Quality Risk Management

As described in Chapter 1, QRM per ICH Q9 [4] is an extremely important part of a technology transfer project. Successful technology transfer should ensure all high risk areas have been considered, appropriate mitigation has been put in place, and that any residual risks are assessed and agreed as acceptable between sites. The QRM process should be documented.

- A detailed risk assessment (e.g., using tools such as cause effect matrices, Ishikawa (fishbone) diagrams, FMEA, gap analysis, etc.) should be put in place so that risks are reduced to an acceptable level and controlled through a defined risk mitigation plan. This could involve experimentation and/or facility/equipment modifications and, when successfully implemented, will help to minimize risk and negative factors in the transfer process.

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- Most technology transfer projects require various risk assessments. Existing risk assessments may need to be adapted to acknowledge changes in risk during the technology transfer.

- Introduction of new technology to replace obsolete systems or equipment is a common challenge during technology transfers. A risk assessment evaluating new risks caused by new technologies is recommended.

- Transfers frequently need to be made into existing equipment and facilities. The assessment of fit-for-use when the receiving unit equipment is not identical to the sending unit is required to ensure all significant risks are mitigated. The fit-for-use assessment would include any process scale-up or scale-down required to complete the transfer.

- Segregation risks assessments are needed to confirm that proper segregation measures have been incorporated into the facility design (e.g., layouts, process and product flows, etc.) or via procedural controls. The requirements are mostly driven by the type of product or system; for example, sterile, low bioburden, closed processing, etc. In the case of a multi-product facility, a cross-contamination risk assessment is recommended. Refer to the *ISPE Baseline® Guide: Risk-Based Manufacture of Pharmaceutical Products (Second Edition)* [20] for more information.

- Contamination risk assessment (in alignment with EU GMP Annex 1 [38]) should be conducted. Controls are implemented to ensure risk of contamination is low for the transfer.

- Consider use of a consolidated risk register to track mitigation of each high risk item.

7.5 Analytical Comparability/Similarity and Stability Strategy

Analytical comparability is a consideration at any point after the first toxicology lot, when there is a change in the manufacturing process which may potentially impact product quality. During technology transfer, it is advantageous to minimize process changes as much as possible to ensure analytical comparability is maintained. Analytical comparability is reported in the clinical trial amendment for clinical programs and is reported in the marketing application. Comparability should be demonstrated according to ICH Q5E [39] for biotechnological/biological products.

Analytical comparability strategies are tailored to each product and each change. The comparability strategy incorporates the existing knowledge of the product and process as well as the risks associated with the change itself (such as a major process change or a change to a single unit operation). In general, products in the early lifecycle stage have limited non-clinical and clinical data associated with the pre-change product and the safety or efficacy profile of the product have not yet been defined. In these situations, comparability testing is generally not as extensive as for a late stage or approved products. In many cases, the analytical comparability data is sufficient justification to not require clinical trials on the post-change for changes made at later lifecycle stages (post-pivotal and commercial). Therefore, the expectations and requirements for comparability increase as the product matures throughout the lifecycle stages.

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The analytical comparability strategy depends on the significance of the process changes and their potential to impact product quality, safety, and efficacy. Considerations for potential impact at the following manufacturing stages include:

- Cell culture changes – consider attributes that may be influenced by cell bank, cell growth, media composition, etc., (e.g., primary sequence, glycans, size, and charge heterogeneity)

- Purification changes – consider attributes that may be influenced by the performance of the unit operation and processing conditions (e.g., impurity clearance, shear forces, light exposure, temperature changes, etc.)

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- Drug product changes – consider attributes that may be influenced by process conditions and product handling, including primary and secondary containers, shear forces, light exposure, or temperature changes (e.g., aggregates, oxidation, deamidation, leachables, extractables, or other process or product impurities due to formulation degradation, processing conditions, or primary containers)

- Combination products (devices) – consider changes that may impact device functionality and the interface with the drug product

- Synthetics – consider process changes that may impact lot release testing

In cases where drug substance is being transferred, the impact to drug product quality should also be assessed. Minimally, it should be determined whether the resulting drug product should be assessed at lot release and placed on stability. Regulatory requirements should also be considered when determining the comparability and stability strategy.

Stability studies at the recommended storage condition confirm the integrity of the product through expiry. Studies at accelerated stability conditions and/or stress stability conditions are often conducted to satisfy regional filing requirements as well as to support product impact outside of the recommended storage condition. While stability at accelerated and/or stress stability conditions are not useful indicators of product expiry due to the uncertainty associated with analytical models, the evaluation of accelerated and/or stress stability condition degradation rates and profiles is useful for comparability purposes; the conditions may elicit differences in the post-change product which are not evident at lot release or on stability at the recommended storage condition. The comparability strategy should be endorsed by appropriate levels of governance for the technology transfer.

The comparability assessment criteria should be established prospectively. The product quality risk assessment and the QTPP should be utilized to provide additional information for establishing a comparability strategy based on the product CQAs. Analytical testing can include lot release tests as well as characterization tests depending on the change and potentially impacted CQA. The number of lots required to establish comparability is largely determined by the lifecycle stage. Earlier stages should be able to justify fewer lots and later stages (i.e., commercial) may require more to demonstrate consistency.

It is important to determine the sample type and account for any differences between analytical methods and testing sites when setting the comparability assessment criteria. Sampling handling can also significantly impact results, such as shipment of samples for sub-visible particle testing. Analytical methods should be qualified and/or validated for all sample types, keeping in mind that new sample types may be a requirement of the technology transfer control plan

In the event a comparability test result yields an abnormal result, teams will need to determine if there is an impact of the aberrant product quality attribute on patient safety and/or product efficacy. There are a variety of factors that may contribute to an abnormal comparability result and the approach for the subsequent evaluation needs to be tailored to the situation. At a minimum, the abnormal result should be evaluated for practical significance using scientific judgment and considerations for the product safety and efficacy profile. The standard guidance for investigations involving comparability lots are the same guidance provided for any manufactured lot. A lot selected for comparability may be a lot rejected for clinical or human use. Based on the reason for lot rejection, the technology transfer team can assess the suitability of the results generated and determine if they are still adequate for the intended purpose of process comparability with appropriate justification and quality function approval.

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7.6 Process Validation/Process Performance Qualification Strategy

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Process validation for technology transfer should align with the process validation concepts and lifecycle approach as described in, e.g., ICH Q7 [2], FDA Guidance on process validation [6], EU GMP Annex 15 [8], EMA Guideline on process validation for finished products [7], and EMA Guideline on process validation for biologics [10].

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The lifecycle concept links process design, qualification of the commercial manufacturing process, and CPV. PPQ and process validation are used interchangeably in this document.

The process validation strategy is based on the process development history, process characterization approach, prior experience at clinical and commercial scale, and product quality risk assessment. The validation strategy includes the validation studies that will be executed and justification for family, bracketing, and/or matrix approaches. Homogeneity/uniformity studies should be included to demonstrate consistency within a batch with appropriate sampling/testing plan and statistical analysis.

Process knowledge and risk assessments are used to justify the number of lots required to successfully demonstrate that the process and product can be reliably and consistently manufactured. In many regulatory regions, a minimum of three successful, consecutive lots are used. However, other regulatory bodies may accept more (or less) lots depending on the knowledge available for the product. When determining the number of lots to use, considerations should also include the number of lots required to demonstrate analytical comparability and stability. It is advisable to directly consult regulatory bodies on the strategy for the number of lots required to demonstrate successful transfers.

Prior to executing PPQ runs, the control strategy (including operating ranges, process parameters, and in-process controls) should be established. The acceptance ranges and acceptance criteria should be documented in a pre-approved protocol. Analytical methods should be validated or qualified. Manufacturing equipment, utilities, and facilities should be qualified. Raw materials need to be sourced from qualified vendors and released prior to use. Consider using different lots of critical raw materials to demonstrate process robustness against material variability sources. Similarly, if performing PPQ on drug product, consider using different lots of drug substance to demonstrate that expected variability within a controlled range of drug substance attributes does not impact the drug product quality and process.

7.7 Change Management

During a technology transfer, there is a clear need to document changes being incorporated into the process or system.

Change management ensures that changes are tracked, assessed, implemented, and documented in a structured manner. The adoption of a system that effectively tracks, evaluates, and manages changes between the sending and receiving units ensures that changes are designed and executed to meet CQA and QTPP requirements. In addition, robust change management will support future investigations as part of the technology transfer process. The structured documentation of changes (planned or not) will help the team to interpret discrepancies between expected and actual values during transfers (Caraballo, 2015 [40]).

Considerations for change management include:

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- Change management usually starts when GMP aspects are designed or executed. The scope of change management extends to materials, components, product specifications, and process descriptions. All specifications, analytical methods, and process changes should be strictly controlled.

- Changes are usually classified by the relative impact to product quality, safety, or regulatory submissions (Deeks, 2018 [41]):

    - A low risk or low impact change tends to be simple and easy to implement, with no impact to regulatory filings.

    - Medium impact changes are more complex and may require additional validation activities and may impact regulatory filings.

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- A high impact change may affect regulatory filings or multiple manufacturing sites. High impact changes may take years to be fully implemented. Significant changes either occurring during or as a component of the technology transfer may pose risk to achieving quality compliance and put timely completion of transfer activities at risk. The necessity of changes of this nature needs to be critically reviewed and, if implemented, potential impacts on quality needs to carefully controlled.

- Once the process is validated, a stricter change control management system is in place where qualified representatives of appropriate disciplines review proposed or actual changes that may affect the validated status of the process. The intent is to determine actions needed to ensure the process is maintained in a validated state.

- Quality and supply agreements play an important role when executing transfers between CMOs (Gibson and Schmitt, 2014 [42]). Refer to FDA Guidance for Industry – Contract Manufacturing Arrangements for Drugs: Quality Agreement [16] for guidance on quality agreements. The requirements for change management should be clearly established in the agreement.

7.8 Execution

The execution of technology transfer is closely aligned with the validation process for regulated products. A Validation Master Plan contains the validation approach, strategy, sampling plans, number of runs, testing methods, procedures, process descriptions, and overall data requirements to demonstrate that the product can be manufactured in a controlled manner and that it meets the expected quality attributes. It usually involves as many production batches as necessary to generate the data required for validation, comparability assessments, stability requirements, and regulatory submissions. Because there is a need to execute multiple batches, challenges may arise when deviations or batch failures happen. The quality system elements of deviations and CAPAs should be in place during technology transfer execution.

When process deviations occur, an investigation is initiated to assess root cause. If the root cause for the deviation is related (intrinsic) to the process, the process validation may be impacted. Usually, in order to demonstrate process control, consecutive batches are manufactured as evidence. An intrinsic process deviation resulting in product failure would likely require repeating the run because the consecutiveness of batches would be interrupted. If the deviation and root cause are extrinsic to the manufacturing process, then the consecutiveness of validation runs is not normally interrupted. These decisions are made on a case by case basis given the large variety of manufacturing processing steps, however, they follow certain principles illustrated by these questions:

- Is the root cause for the deviations intrinsic or extrinsic to the process?

- Were product quality attributes negatively impacted by the deviation?

- Can corrections be implemented to avoid recurrence?

- Is the required corrective action(s) within the process design as described, or referenced in the PPQ protocol?

For each deviation, a formal investigation needs to be completed and corrective actions identified.

The practice of executing engineering or test runs before validation runs has value, especially for high value or complex processes. An engineering run would incorporate phase appropriate GMP elements during its execution. (Caraballo, 2015 [40]) For example, if GMP relevant data is collected during execution of engineering runs (e.g., process hold times, cleaning validation parameters), then GMP principles for data integrity, training, and documentation control should be in place.

The execution of technology transfer is usually documented in batch records, validation protocols, and technology transfer summary reports. These documents are managed under GMP using the documentation management element of the quality system.

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8 Appendix 1 – Checklist of Information/ Documents for Large and Small Molecule Technology Transfer

Checklist		Large Molecule	Small Molecule
General			
	Technology Transfer Summary Report	x	x
	Risk Assessment	x	x
	Lessons Learned	x	x
Materials			
	Bill of Materials	x	x
	Starting Material		x
	Starting Material Route of Synthesis		x
	Critical Materials	x	x
	Cell Bank/Origin of Cell Bank	x	
	Current Suppliers	x	x
	Current Specifications and CQAs	x	x
	Material Certificates of Analysis/Release Data	x	x
	Storage Requirements	x	x
	Raw Materials Qualification Reports		x
Product			
	Development Reports	x	x
	Development Run Data	x	x
	Process Description	x	x
	Specification(s)	x	x
	Historical Product Data	x	x
	CQAs and Justification	x	x
	Particle Size		X
	Polymorphs		x
	Solubility Data		x
	Toxicology Data		x
	Extractable/Leachable/Adsorption Data	x	x
	Bioavailability/Bioequivalence Data		x
	Stability Report/Data	x	x
	Ongoing Stability	x	x
	Cleaning Limits	x	x
	Cleaning Batch Record	x	x
Process			
	Block Flow Diagram	x	x
	Process Flow Diagram	x	x
	Physical Property Data	x	x
	Mass Balance	x	x
	Heat Balance		x
	Yields	x	x

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Checklist		Large Molecule	Small Molecule
Process (continued)			
	Cycle Time	x	x
	CPPs, Key Parameters, and Ranges	x	x
	Control Strategy	x	
	Master Batch Record(s)	x	x
	Historical Batch Data	x	x
	Trend Analysis/Statistical Process Control	x	x
	In-Process Samples	x	x
	Hold Points/Times	x	x
	Packaging	x	x
	Waste/Vent Characterization		x
	Equivalency Studies	x	x
	Qualified Rework	x	x
	Qualified Recycled Materials	x	x
Health, Safety, and Environment (HSE)			
	Lower Explosive Limits	x	x
	Safety Data Sheets for all Materials	x	x
	Material Classification	x	x
	Process Hazard Risk Analysis	x	x
	Personal Protective Equipment Requirements	x	x
	Permits Issued	x	x
Facility and Equipment			
	P&IDs	x	x
	URS/Design Specifications	x	x
	Utilities Requirements	x	x
	Control System(s)	x	x
	Automation	x	x
	Instrumentation	x	x
	Material Compatibility	x	x
	Sample Locations	x	x
	Qualification Data/Reports	x	x
Quality/Regulatory			
	Quality Agreements	x	x
	Supplier Audits	x	x
	Product Out of Specification	x	x
	Change Controls	x	x
	Process Deviations/CAPAs	x	x
	Process Validation Reports	x	x
	Regulatory Filing	x	x
	Inspection Reports	x	x
	Annual Product Reviews	x	x

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9 Appendix 2 – Case Studies: Biologics

9.1 Summary

This chapter presents case studies that illustrate key technology transfer concepts presented in this Guide, with an emphasis on utilizing CMOs.

The key components of any technology transfer are the technical challenges and effective program management. With any technology transfer program, technical challenges can arise that could subsequently impact the planned timelines, process performance, and product quality. These challenges should be addressed in a collaborative way between the sending and receiving units to minimize delays in the program and frustrations arising between sites.

In this chapter, five case studies are presented, the first two from a client perspective and the subsequent three from a CMO perspective. These examples demonstrate a range of challenges which occurred during technology transfer programs and how they were subsequently resolved.

The molecules in these case studies are primarily monoclonal antibodies (mAbs) but the lessons learned are generally applicable across all molecular classes. Manufacturing processes for mAbs have become relatively well-defined and are often based on platform technologies that can be effectively leveraged; there is a generally broad knowledge base. In addition, the advent of single-use technologies is facilitating transfers by minimizing changes in equipment and operating parameters. The amount of drug substance manufacturing experience that smaller companies typically have is modest, often at different scales and with limited development history and process knowledge. Transfers represent the opportunity to right-size a process and implement process improvements with an eye to minimizing changes that might impact product comparability. Technology transfers also potentially offer an opportunity to assess new technologies, such as continuous manufacturing and PAT, if the CMO has these capabilities and is skilled in implementing them in a timely and risk-controlled manner.

9.2 Main Challenges

One of the main challenges for successful technology transfer utilizing a contract manufacturer is selecting the right CMO and developing a good partnership. From an operational perspective, this involves ensuring capacity (i.e., having manufacturing slots available within an acceptable time frame) and ensuring appropriate process and analytical capabilities necessary for project success. Demand for manufacturing slots will vary over time, but demand for capacity continues to grow and is a potential constraint that is a priority planning consideration. Recent experience with these case studies has resulted in successful transfers (i.e., completion of first GMP batch in approximately 12 months). Other selection criteria, such as compliance history, commercial experience, scale, and cost, are also important.

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Another main challenge for a small to mid-size company executing technology transfer is how to deal with a lack of development history or process knowledge, either due to limited manufacturing experience or insufficient process development. Performing a gap assessment prior to initiating the transfer and then sharing this assessment with the CMO at initiation (to gain alignment and ensure transparency) can be critical for success. Early involvement of manufacturing teams during technology transfer and development phases is also critical for successful implementation of the process at scale.

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9.3 Overview of Technology Transfer Phases

As described in Chapter 3, there are six phases of technology transfer:

1. **Project Initiation:** Form technology transfer team and develop charter
2. **Knowledge Transfer:** Consolidate knowledge for transfer and agree on high level technology transfer proposal
3. **Project Delivery Planning**: Identify risks, conduct risk assessments, and develop technology transfer plan
4. **Operational Readiness**
5. **Project Implementation:** Process (procedure) qualification
6. **Project Close-Out:** Finalize technology transfer and perform review

The approach presented in this chapter takes a modified approach that has been successfully utilized by small to mid-sized companies. The three general stages, as defined in this chapter from the client perspective, are (1) CMO selection, (2) project initiation, and (3) project execution. In relation to the six phases presented in Chapter 3:

CMO selection precedes establishment of a technology transfer team but is a necessary precursor for outsourcing.

- Project initiation incorporates aspects of phases 1 to 3.
- Project execution includes aspects of phase 3 as well as phases 4 and 5. Phase 6 can range in scope from completion of a pilot or engineering batch to submission of a regulatory filing as well as a lessons learned exercise between the client and the CMO.

From the CMO's perspective, stage gates can be defined within the detailed project plan per their procedures to ensure that activities (such as document transfer, preparation of GMP documentation, etc.) occur in a timely manner to support manufacture, as shown in Figure 9.1. This is also a good example of the need for a small to mid-sized company to have flexible and adaptable operational and quality systems. Typically, they will not have formal technology transfer procedures and it will be advantageous and efficient to utilize and leverage the CMO's procedures, recognizing that these will likely differ between different CMOs.

Figure 9.1: Example Technology Transfer Process Flow
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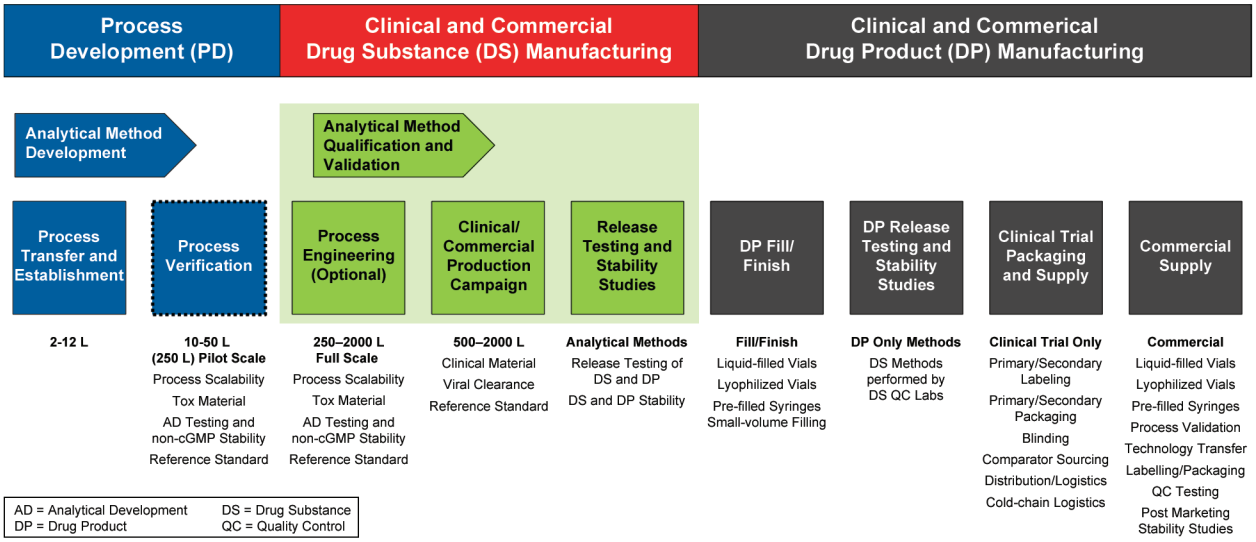

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9.3.1 CMO Selection

The CMO selection process is initiated by developing a list of candidates based on industry knowledge and experience that meet baseline criteria, including scale, capability, and compliance history. A detailed Request for Proposal (RFP) should be developed that includes an appropriate amount of process detail, analytical method descriptions, the product specification, a bill of materials, and requests for any development work. Target dates for major deliverables should also be included, such as any requirements to have toxicology or clinical studies initiated by a certain date or if there are any commitments to achieve a target date for an investigational new drug filing. Adequate process detail is necessary to ensure that the proposals that are received in response to the RFP are comparable and permit an appropriate comparison. It is also important to initiate the CMO selection process with adequate time for refining responses to the RFP. In general, there might be at least one to two cycles of proposal refinement that will allow an accurate comparison between CMOs and facilitate contract negotiation. In addition, an appropriately detailed RFP permits the CMO to properly assess their ability to run the process and be successful within project timeline expectations.

For example, an RFP can be structured as follows:

1 Introduction
- 1.1 Background
- 1.2 Objectives
- 1.3 Response to RFP

2 Scope of Services Requested
- 2.1 Technology Transfer
- 2.2 Analytical Services
- 2.3 Engineering and Clinical Batches
- 2.4 PPQ Batches
- 2.5 Process Validation
- 2.6 Stability
- 2.7 Additional Services

3 Supporting Information (development reports, batch records (if available), PFDs, specifications, test methods, bill of materials, etc.)

For the first case study presented in Section 9.5.1, this approach started with engagement of six candidate CMOs (although this number can vary dependent on experience and project needs). The proposals received from these candidates were reviewed and a head to head analysis was performed on work scope, cost, and timing. From this assessment, two lead candidates were identified and due diligence audits were performed, resulting in a lead and a backup CMO. For the second case study that followed approximately one year after initiation of the first case study, a modified selection process was performed and is described in more detail in Section 9.5.2.

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9.3.2 Project Initiation

Once the CMO is selected, contract and quality agreement negotiations are immediately initiated, utilizing CMO templates for efficiency. Negotiation of these documents takes time and getting them in place as soon as possible is important for establishing the partnership and avoiding unnecessary conflict during the early stages of the relationship. Utilizing CMO templates can also accelerate the process.

A project kickoff meeting should be scheduled as soon as possible at the CMO and is critical for establishing the personal connections and lines of communication that will be necessary for project success. The composition of the client project team may vary, but will typically include a project leader, a project manager, process and analytical SMEs, a CMC regulatory representative, and a quality representative. The CMO project team will be comparable and complementary. Establishment of personal connections between these team members and their counterparts at the CMO are invaluable in anticipation that there will be challenges during the technology transfer and manufacturing that will require good working relationships to ensure appropriate and timely resolution. Face-to-face meetings are critical for establishing relationships and lines of communication.

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The third aspect of project initiation is document transfer and sharing of the QTPP, CQAs, and CPPs that define the control strategy. These documents are important because they:

- Define the intended use and product quality profile
- Ensure that technology transfer process development/optimization begins with the end in mind and that client and CMO goals and expectations are aligned
- Define those product characteristics or CQAs that must be monitored and controlled during technology transfer

9.3.3 Project Execution

Project execution follows a standard course of events. Many CMOs are experienced in executing technology transfers because it is a key component of their business model. Therefore, for small and mid-sized biotech companies, working closely with the CMO's process and procedure is generally necessary and ultimately more efficient. The following is an overview of activities undertaken in both case studies (presented in Sections 9.5.1 and 9.5.2):

- Process assessment and identification of the need for process development/optimization for facility fit
- Identification and immediate ordering of long lead time materials/consumables
- Small scale/development studies, as appropriate (see Chapter 5)
- Method transfer and qualification, as appropriate (see Chapter 4)
- Development of GMP documentation, including batch records, test methods, specification, etc.
- Pilot scale run, if needed
- Engineering run, if needed
- GMP run(s)
- Analytical comparability
- Regulatory submission

9.4 Risk Management

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Although there is a defined process for CMO selection that plays a key role in reducing risk, no formal risk assessment (technical, operational, or business) was performed by the client for initiating technology transfer in these case studies. Client SMEs experience working with CMOs and due diligence audits were used to informally assess risk. However, formal risk assessments are performed by the client as part of change control and are subsequently formally introduced through collaboratively performed FMEAs as the project advances toward pivotal study or PPQ.

The requirements of the selection process that reduce risk are as follows:

- Commercial experience
- Appropriate scale (current and future)
- Capacity (availability of manufacturing slots)

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- Demonstrated ability to expand capacity/capabilities
- Compliance history
- Technical skill/experience

These requirements were met through the selection process that included a combination of the following:

- Proposal submission based on the detailed RFP
- Site visit
- Due diligence audit
- Professional experience

This could generally be considered a standard practice for small companies.

A key component of the proposal is an adequately detailed project plan that is utilized by both client and CMO project managers to manage timelines and inherently manage risk associated with those activities by identifying critical path activities. Capacity and technical experience play important roles in meeting deadlines and ensuring that the ultimate goal of the technology transfer, production of the first GMP batch, is successful and on time.

Once a contract is signed, the project is initiated with a face-to-face team meeting at the CMO. The purpose of this meeting is to formally present the project goals and objectives and initiate relationships between SMEs and other project team members who will work closely together throughout the technology transfer and possibly beyond. The establishment of this personal connection has been found to be extremely valuable when issues and problems arise. Document transfer will also occur around this time, preferably before the kickoff meeting, but as soon as possible. The scope of documentation may include executed batch record(s), test methods and qualification reports, specifications, bill of materials, development reports, etc. (see Chapter 5).

Risk is further managed throughout the technology transfer process with regular project meetings to facilitate communication. This will include weekly teleconferences and face-to-face meetings at critical junctures. Information can also be shared through new and evolving technologies that can provide real time operational data as the process moves into manufacturing. Effective communication is critical throughout the process; regular interaction between SMEs is encouraged with the important caveat that project leaders and/or project managers are involved to ensure coordination and appropriate follow up.

Formal risk management tools will subsequently be used to support process characterization and process validation, but a quantitative risk assessment for a small company may have limited utility and the process relies more on involvement by more senior and experienced personnel.

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9.5 Technology Transfer Case Studies

The first two case studies present specific technical, operational, and project management challenges that were faced from a client perspective during the technology transfer of two products, how these challenges were resolved, and what the outcome was for each. The following three case studies are presented from the CMO perspective to illustrate technical challenges arising during technology transfer from small and mid-sized companies.

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9.5.1 Biologics Case Study #1: Transfer of a Phase 2 mAb Drug Substance with Significant Clinical Potential

The first two case studies are interrelated in that they involve two products, one CMO company, and three CMO sites. The first case study describes a product in early phase 2 development that held significant clinical potential to move to a pivotal study. With the advent of breakthrough therapy designation and accelerated product development, the timelines for process optimization and the timeframe for gaining manufacturing experience to achieve a robust and reproducible manufacturing process is significantly reduced. Therefore, the process was transferred from the client's internal early phase manufacturing site to a US-based commercial CMO (site A) fairly early in clinical development.

This CMO was selected for the following reasons:

- Commercial experience
- Appropriate scale – 2000 L single-use bioreactors (SUB)
- Available capacity – manufacturing slots available with ~12 months lead time
- Good compliance history – FDA inspected as well as multiple other regulatory agency inspections
- Technical skill/capability
- Cost competitive

The client's internal manufacturing train was 250 L and 1,000 L, so transfer and scale-up to a 2,000 L scale was deemed low risk. However, manufacturing at the client site was performed with a low productivity cell line. Development of a new higher productivity cell line and establishment of a new Master Cell Bank (MCB) was nearing completion at the start of the technology transfer. Limited development work was performed at the client site and technology transfer was initiated prior to having completed full process development. Utilization of a Research Cell Bank (RCB) for initiation of technology transfer was also required since the MCB was not available yet.

Changes were made to the downstream process to enable facility fit and to improve process efficiency. A focused development and optimization approach leveraging CMO expertise and experience was taken for the downstream process to enable timely transfer of the process to GMP manufacturing. Changes in the downstream process included addition of a pH treatment step, change in depth filters, change in sequence of operations (Sartoclear® P filtration step moved from post to pre-SP-Sepharose® column chromatography), change in viral filter, and removal of a dilution step prior to loading the SP-Sepharose® column. An engineering run was performed to shake down the process and ensure process changes were effective and successfully scaled up to the manufacturing scale (2,000 L). The technical challenges and the corresponding resolutions for this technology transfer are summarized in Table 9.1.

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Table 9.1: Biologics Case Study #1 – Technical Challenges and Resolutions

Challenge	Resolution
New, higher producing cell line that was not immediately available; no prior in-house manufacturing experience with new cell line	Started development work with RCB, then switched to MCB for confirmation runs; agreed on enhanced RCB testing to allow shipment to CMO
Limited process optimization performed prior to technology transfer	Performed focused process optimization work at CMO in parallel, utilizing the Ambr® system for upstream process; Downstream process optimization included filter sizing studies, column load/cycling studies, etc.
Analytical changes, including: • Addition of Polysorbate 80 test method • Transition from sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) to capillary gel electrophoresis (CGE) • Incorporation of in-process buffer matrices into method qualification	Leveraged platform assays when possible and utilized bridging studies to incorporate method changes
Viral clearance study issues – gaps in EU compliance at CMO and testing laboratory	Limited certification of EU compliance (testing but not manufacturing EU certification) was available for regulatory submission. Follow up Qualified Person certification required to ensure EU compliance

Result: Successful transfer and scale-up of manufacturing process to 2,000 L, resulting in comparable product quality and process performance.

Since this was the first time the client had worked with this CMO, there was the learning curve associated with gaining knowledge of the facility, the equipment available, any equipment limitations, and the CMO's quality and risk management procedures. Release testing for both drug substance and drug product was performed at the CMO, thereby reducing time and cost associated with transferring qualified methods to the drug product CMO at this stage of development. The operational challenges and resolutions for this technology transfer are summarized in Table 9.2.

Table 9.2: Biologics Case Study #1 – Operational Challenges and Resolutions

Challenge	Resolution
Manufacturing process changes requested by client based on manufacturing experience; client needed a specific filter with low filter load to reduce host cell proteins	Incorporated facility fit and manufacturing process changes into process optimization studies
Change in culture bags/filter trains; CMO had different culture bags and the timeline for introducing the client culture bag was lengthy and required a change to meet timelines	Technical teams performed studies to support the change to the CMO's preferred culture bag/filter train
Managing deviations during manufacture and testing	Fully engaged client and CMO QA organizations worked to resolve deviations/CAPAs in a timely manner
Client's lack of understanding of CMO's batch disposition/review process	CMO flexibility permitted on time disposition of batch by real time reviews/resolution and frequent meetings; accelerated release documentation transfer to drug product CMO permitted a smooth transition

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Result: Operational challenges were readily and effectively managed, permitting successful manufacture of drug substance and on time release testing of drug product and submission of regulatory filing.

Effective project management is critical for timely project success; communication and flexibility are key aspects of a successful partnership. The project management challenges and resolutions for this technology transfer are summarized in Table 9.3.

Table 9.3: Biologics Case Study #1 – Project Management Challenges and Resolutions

Challenge	Resolution
Managing unexpected events and subsequent timeline changes at CMO during technology transfer: • CMO informed client that a new suite was being built to improve flexibility late into technology transfer—the required suite construction, certification, and equipment qualification resulted in a timeline delay • Addition of process development/optimization work and expanded in-process testing was required to enable rapid transfer	• Regular meetings of core team and SME sub-teams to ensure timelines were adjusted and technical issues were addressed • Readjustment of scope to prioritize and manage costs/timeline
Lack of a joint steering committee to address issues early in technology transfer	• Engagement with senior site management to address issues • Establishment of a formal joint steering committee structure to facilitate issue resolution

Result: Streamlined communication channels and conflict resolution pathways; adjustment of expectations and improved planning/scheduling

What Worked:

- Technical competence and experience of CMO's process development/analytical development groups with technology transfer and scale-up to speed development
- Defining changes required up front and implementing changes during initial development to monitor product quality
- Coordination of analytical development/QC activities to complement product development activities
- Dedicated project management and regular face-to-face and web-based meetings/updates – core team and sub-team (upstream process, downstream process, and analytical) meetings
- Effective project management processes; document tracker with deliverables and utilization of online collaboration website for document sharing
- Spirit of partnership and flexibility; commitment to improve and resolve issues
- Studies co-completed during transfer to generate required data to sufficiently reduce manufacturing risk
- Timeline achieved – ~15 months from project initiation to release of bulk drug substance; slight delay, but acceleration of drug product release testing permitted meeting regulatory filing goals

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9.5.2 Biologics Case Study #2: Transfer of a Late Clinical Phase mAb from One CMO to Another CMO in Preparation for PPQ and Future Commercial Manufacture

The second case study involves the mAb intermediate component of an antibody-drug conjugate in late stage clinical development. Technology transfer was driven by the need to move the mAb manufacture from the originally intended commercial CMO because of that company's decision to decommission the bioreactors that were planned for commercial manufacture of the product. In addition, there was no viable alternative available at the original CMO, so identification and selection of a new CMO was required. Based on the prior experience with the CMO (site A) described in Biologics Case Study #1, another site in that CMO's network (site B) was selected for the following reasons:

- Although site A was originally approached for this project based on the prior positive experience, capacity constraints indicated that they could not meet the required ~12 month timeline. The client was then directed to site B, a new facility in the CMO's network that had capacity and could accommodate the timeline.
- Although the distance to site B, which was outside the US, was considered a potential challenge, the proximity of a CMO development site in the US (site C) was deemed a compensating attribute. This CMO development site was where supporting optimization and small scale studies would be performed.
- CMO site B's host country offered financial incentive to perform pharmaceutical R&D activities in country.
- Having multiple products with a single CMO was anticipated to provide efficiencies from a contractual, quality, and overall management perspective. However, there is also an associated risk with depending on a single CMO; this risk was mitigated by the fact that the CMO had a global footprint with multiple manufacturing sites in different countries that could potentially serve as alternate sources.

At the time of initiating the technology transfer activities, the client was still in the process of establishing a new cell line suitable to meet regulatory expectations for commercialization. Utilization of a RCB for initiation of technology transfer was required because the MCB was unavailable due to delays in testing and challenges from differences in requirements for the receipt of new cell lines at the various CMO sites. Implementation of a modified filtration step following low pH viral inactivation was also required. However, due to time constraints, optimization prior to manufacturing was not possible. Based on the CMO's experience with the filter, it was considered low risk, and it was agreed that optimization could occur at a later date. Finally, a new bulk drug substance container was introduced to accommodate a change in storage temperature from 2–8°C to ≤ -65°C. The technical challenges and the corresponding resolutions for this technology transfer are summarized in Table 9.4.

Table 9.4: Biologics Case Study #2 – Technical Challenges and Resolutions

Challenge	Resolution
Cell line issues: • New cell line still in development at the time of transfer • Manufacturing and testing delays at third party cell bank manufacturer impacted schedule • Inconsistent requirements between CMO sites for the receipt of new cell banks	Established criteria for receipt of RCB for development purposes versus MCB for GMP purposes; allowed flexibility to begin small scale work with RCB and transition to MCB once available
Implementation of alternate filtration step identified prior to transfer	Agreed that optimization would be performed at a later date due to CMO's technical experience with the new filter
Change in bulk drug substance container and storage temperature	Risk assessment determined risk to be low based on vendor white paper demonstrating compatibility with aqueous based solutions and performance of a short-term stability study to demonstrate compatibility with the new container closure system

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Result: Successful transfer and scaling of manufacturing process at 2,000 L, resulting in comparable product quality and process performance

While the client had experience working within this CMO's network, this was the first time working with this specific site location. There were site specific requirements as well as regional government regulations that posed additional operational challenges. The CMO site change required both a process scale change and various process changes to accommodate facility fit. The operational challenges and the corresponding resolutions for this technology transfer are summarized in Table 9.5.

Table 9.5: Biologics Case Study #2 – Operational Challenges and Resolutions

Challenge	Resolution
Facility fit: • Scale and process changes required for facility fit • Equipment constraints required process changes (harvest vessels, single-use chromatography skid, no centrifuge, etc.)	Performed small scale modeling exercise followed by focused small scale upstream and downstream confirmation run at CMO site C
Optimization of the cell expansion process, reducing the number of media required from four to one	Evaluated process change in process development at CMO site C and transferred optimized process for manufacturing to CMO site B
Media feed component transportation hazard classifications – a simple media feed component with no restrictions for transport within the US was now considered hazardous goods with strict requirements and restrictions on international shipping	Evaluated raw material supply chain for potential regional sourcing options and adjusted raw material lead times based upon transportation, safety, and importation requirements

Result: Implementation of process modifications resulting in successful transfer of manufacturing process and production of analytically comparable mAb intermediate

Result: Identification of alternative raw material sources within the region, mitigating supply chain risk and improving purchasing lead times

In recent years, CMOs have been promoting the idea of a full-service provider (or one stop shop) whereby clients can seamlessly progress a molecule from development through full commercialization within one established network. While attractive in principle, it is important to understand that many of these networks have been established through merger and acquisition. While the business side of the merger and acquisition activities may progress quickly, it often takes time to fully integrate sites across different geographies, cultures, and corporate experience. The more knowledgeable a client is about each of the sites they plan to utilize within the network, the higher the probability of success. Communication and effective project management are the keys to success. The project management challenges and the corresponding resolutions for this technology transfer are summarized in Table 9.6.

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Table 9.6: Biologics Case Study #2 – Project Management Challenges and Resolutions

Challenge	Resolution
Ensuring effective project team organization and structure between CMO sites A, B, and C	Established sub-teams in addition to the core project team
Ensuring effective communication between three CMO sites and multiple client sites	Regular meetings (online and face-to-face) for both core and sub-teams with documented minutes and action items; additional regular project management meetings to prepare as well as to respond to open items
Integration of legacy procedures from prior CMO mergers into unified CMO procedures across three sites is a work in progress	Integration and harmonization are corporate priority for CMO

Result: Individual site requirements necessitated additional time to complete site specific paperwork; however, all CMO sites agreed to expedited document reviews which allowed all processing to remain on schedule.

Result: Streamlined communication, accountability, and responsibility established over time as an effective partnership developed.

What Worked:

- Technical competence and experience in technology transfer with well-defined process documentation
- Small scale modeling and feasibility studies confirmed process operations; full scale engineering run confirmed small scale process and provided manufacturing with operational experience
- Productive and appropriately scheduled meetings, including face-to-face kickoff, regular sub-team, and regular project management preparation and follow up meetings (proper communication and alignment is key)
- Person-in-place on-site at CMO site B helped develop relationship and allowed sponsor to gain first-hand experience in how the facility operates; person-in-place also clearly aided in the transfer of information, clarification, and understanding of all unit operations with the CMO
- Client QA involvement early in the process provided opportunity to gain process knowledge prior to moving into GMP manufacturing
- Achieved timeline goal – ~12 months from project initiation to bulk drug substance fill

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9.5.3 Biologics Case Study #3: Transfer to a CMO for Evaluation at Laboratory Scale, Prior to Transfer to Pilot and GMP Scale

An mAb purification process was developed at a small biotechnology company and transferred to a CMO for evaluation at laboratory scale, prior to scale-up to a non-GMP pilot and first GMP clinical batch. The process consisted of a Protein A chromatography step, low pH virus inactivation, and pH adjustment followed by cation exchange and anion exchange chromatography steps.

At the sending unit, the product was eluted from Protein A with acetate buffer pH 3.2 and the product eluted from the column within the required range for the viral inactivation step. The eluate was pH adjusted to pH 5.0 for the cation exchange step using 1 part product to 4 parts titrant. Prior to manufacture of the first GMP batch, a laboratory scale and pilot scale evaluation was performed at the receiving unit. During the laboratory scale evaluation at the receiving unit, two issues arose with the Protein A step in the purification process.

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A root cause analysis (Ishikawa diagram) was performed on the Protein A step. A number of items were highlighted for further investigation between the two units, relating to elution buffer and titrant preparation. During the transfer, only the names of the buffers were shared and not the preparation details. Two key differences were identified:

- At the sending unit, the elution buffer and the titrant were prepared using the trihydrate salt, compared to use of the anhydrous salt at the receiving unit.
- At the sending unit, the elution buffer was titrated with hydrochloric acid, compared to titration with acetate acid at the receiving unit which resulted in much higher buffering capacity.

The challenges and the corresponding solutions for this technology transfer are summarized in Table 9.6. Table 9.7 summarizes the challenges and resolutions prior to the pilot scale batch.

Table 9.7: Biologics Case Study #3 – Challenges and Resolutions

Challenge	Resolution
The pH of the Protein A eluate, although in the viral inactivation range, was much lower than expected	The two issues were linked. Elution buffer was formulated at the receiving unit using trihydrate salt and acetic acid to achieve the required acetate concentration. The titrant buffer was prepared at the receiving unit using trihydrate salt. The new formulations and the original formulations were compared side by side on two Protein A cycles. With the new formulations, the pH of the eluted product was now above the required viral inactivation range. Further evaluation of the step was performed at the receiving unit resulting in adjustments to the acetate concentration in the elution buffer and titrant.
Approximately double the volume of titrant was required to adjust the pH of the Protein A eluate to pH 5.0	

Result: Formulation of the elution buffer with the trihydrate salt and adjustment of the acetate concentration resulted in the eluted product pH achieving the required pH range for viral inactivation. Preparation of the titrant with the trihydrate salt and adjustment of the acetate concentration resulted in the expected volume of titrant being added for pH adjustment to pH 5.0.

What Worked:

- Good communication between technical SMEs at sending and receiving units: the unexpected performance of the Protein A step and potential causes were flagged for the receiving unit, so details of buffer and titrant preparation could be compared quickly and adjusted as the laboratory scale evaluation was ongoing
- Formulation of the elution buffer facilitated preparation at scale in the GMP facility

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9.5.4 Biologics Case Study #4: Transfer to CMO for Evaluation at Laboratory Scale, Prior to Transfer to non-GMP Pilot Scale

This case study relates to transferring a process from a small biotechnology company with process development capabilities to a CMO with process development, scale-up, and GMP capabilities. A good relationship had been established between the two parties from collaborations on previous process transfers, process development, and GMP batch manufacture.

In this case study, the product was a very large, challenging antibody molecule (immunoglobulin M) which could not be purified using resins already available at the CMO. The plan was to evaluate the process at laboratory scale and scale-up to non-GMP pilot scale before manufacturing a GMP batch.

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As there were no batch records or development reports available from the sending unit, the process was transferred from the sending unit using a transfer document developed at the CMO. In addition to process details, this transfer document included a request for information from the sending unit on product stability at high or low pH, any tendency to aggregate, evidence of instability in particular buffers, solubility issues at high or low conductivity, and highest concentration that the product has been demonstrated to be stable. In addition, SOPs at the CMO were outlined in this transfer document, including flow rate ranges (based on equipment available), resin cleaning conditions, and generic hold times. This helped to align procedures at both sites.

The process transferred to the CMO included a Protein A step, anion exchange (AEX) chromatography, and ultraviolet (UV) treatment. Final formulation development was ongoing during the transfer. Initially, the process was evaluated at 10 L bioreactor scale at the CMO, then scaled up and evaluated in two non-GMP pilot batches. SMEs from the sending unit were present during key processing steps at the receiving unit to be able to discuss, advise, and help resolve technical issues real time. As more data was generated at the CMO site, the sending unit was making changes in parallel to address arising issues. Table 9.8 summarizes the challenges and resolutions for this technology transfer as both units worked to establish a purification process suitable for scale-up.

Table 9.8: Biologics Case Study #4 – Challenges and Resolutions

Challenge	Resolution
New Protein A resin for CMO which required a new (vibrational) packing method	CMO evaluated the vibrational method with on-site input from the vendor.
Much lower than expected recovery on Protein A column during laboratory scale evaluation at CMO	Reduced loading capacity and flow rates at CMO. Sending unit performed further evaluation to align with CMO data. Further evaluation on scale-up at CMO to define final loading capacity.
Higher than expected conductivity on elution from the Protein A column during laboratory scale evaluation at CMO, which would impact performance of AEX step	Introduced ultrafiltration step at CMO to reduce conductivity. Subsequently, an alternative Protein A elution buffer was evaluated at sending unit, which eliminated the requirement for an ultrafiltration step.
New AEX column type for CMO, no flow rates transferred from sending unit	CMO used manufacturer's recommendations, with on-site support from vendor on scale-up.
AEX column gave inconsistent process performance results on scale-up and poor impurity removal at CMO	Sending unit developed alternate AEX step (traditional resin), which increased throughput and improved impurity removal.
Molecule was too large for 20 nm nanofiltration; sending unit recommended UV treatment, but no experimental work had been performed prior to transfer to CMO	CMO hired UV equipment; this required evaluation at non-GMP pilot scale due to material quantities required. Developed step, but resulted in increased % aggregates.
On scale-up at CMO, product concentration post-AEX step was too low for UV treatment	Introduced a concentration step prior to UV treatment, using CMO platform conditions.
Unknown impact of UV treatment on virus reduction	GMP virus reduction study was performed by the CMO to define operating conditions.
No third step was initially developed for aggregate removal at the sending unit	CMO and sending unit evaluated two options in parallel, third chromatography step further established at sending unit and evaluated on scale-up at CMO.
Limited analytical data at sending unit as impurity methods had not been established	Use of CMO platform methods for impurities.
Limited analytical data at CMO as product quality methods had not been transferred from sending unit	Samples sent from CMO to sending unit for product quality analysis.

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Result: The purification process was evaluated over a 12-month timeframe, at laboratory and non-GMP pilot scale. The final batch was manufactured with the revised, three chromatography step process. However, although product quality and impurity levels in drug substance were acceptable, the cumulative recovery was <10%. Consequently, cost of goods was excessive. At this point, the program was placed on hold by the client, due to other more promising molecules in their pipeline and the process was never transferred to GMP manufacturing.

What Worked:

- Throughout the technology transfer process, both the sending and receiving units gained a huge amount of experience and maintained a good working relationship
- Technical competence at both units – data compiled real time for sharing and discussing, SMEs worked together to resolve issues real time
- New technologies were implemented which required support from vendors

9.5.5 Biologics Case Study #5: Transfer Involving a Small Biotechnology Company who Owned the Product (Fusion Protein), from One CMO (Early Clinical Phase) to Another CMO for Scale-up and Commercialization

For the final case study, the process was transferred from one CMO where the process had been developed and manufactured to a second CMO with the intent of scaling up the process, due to increased material requirements beyond the equipment capabilities of the first CMO.

A SharePoint site was set up between the biotech company and the second CMO as a way of exchanging confidential information, including process development reports, executed batch records, in-process testing, drug substance testing, and trend analysis of the GMP batches for upstream and downstream processes. During the laboratory scale evaluation at the receiving unit, an issue arose on the second chromatography step in the purification process. Table 9.9 summarizes the challenges and resolutions.

Table 9.9: Biologics Case Study #5 – Challenges and Resolutions

Challenge	Resolution
Tight timeframe – 10 L laboratory scale evaluation being initiated only one month after receipt of technology transfer information package, a non-GMP 400 L pilot scale batch was initiated 10 weeks later, with the first GMP engineering batch in the larger CMO manufacturing facility seven months later.	Experimental work was initiated under a "Letter of Intent" and the formal Amendment was signed approximately two months later detailing the stages of work Increased number of teleconferences
Process data was shared in multiple documents, including batch records from the CMO and a summary development report generated by the biotechnology company	Development laboratory at the second CMO prepared process descriptions based on the data available, which were reviewed by SMEs from the biotechnology company before proceeding with laboratory work
Product did not elute as expected from the second chromatography column, breakthrough was observed during a post-load wash, which significantly impacted the % recovery	Investigation resulted in an error in chromatography resin type used

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An initial investigation was performed. Process step details were compared between the batch records from the sending unit, the process description generated at the receiving unit, and data recorded in the laboratory notebook. Initial root causes considered were related to buffer pH and conductivity specifications as the ranges transferred were broad. However, all buffers prepared at the second CMO were within the required ranges.

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Results of the initial investigation were discussed with technical SMEs at the sending unit. The SMEs questioned the type of resin which had been used. While the right functional group was present, the fast flow version of the resin had been used instead of the high performance version of the resin. The loading capacity was higher on the high performance resin, hence the breakthrough observed on the fast flow resin. This detail had been missed during the review of batch records at the second CMO and also during review of the process description by the biotechnology company.

Result: The process was further evaluated in a second laboratory scale run and the second chromatography step performed as expected with the high performance resin. The pilot batch proceeded as planned and the process has since been successfully commercialized.

What Worked:

- As part of the technology transfer program, a second laboratory scale evaluation had been planned prior to the first pilot batch. The correct resin was used successfully for the second chromatography step, performance was comparable to that from the previous CMO GMP batches. Although the timeline was tight, there was benefit in running a second laboratory scale evaluation

- Access to technical SMEs at the receiving unit helped to resolve the technical issue quickly, with no impact on the timeline

9.6 Conclusions/Lessons Learned

Following completion of these technology transfer projects, a lessons learned exercise was performed in collaboration with the CMOs. This was both a positive exercise for recognizing things that were done well as well as an opportunity for continuous improvement for both the CMO and the client. Based on this exercise, the following were deemed necessary for ensuring a successful technology transfer and ultimately establishing a successful relationship with a CMO:

- Relationship building: partnership, taking a collaborative approach to resolve issues, understanding any cultural differences

- Communication: key to help resolving issues, always better face-to-face

  - In the early days of the technology transfer relationship, when gaps and potential challenges are being identified at the receiving unit, the relationship can become fraught as both sites seek to understand ways of working together, different procedures and equipment used at different sites, etc.

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- Flexibility: willingness to adapt and compromise and being prepared to expect the unexpected

- Aligning expectations: being clear upfront, timeframes for work, definition of a successful transfer, when transfer is considered complete, alignment terminology (glossary of terms is useful)

- Accountability: being clear upfront on responsibilities, using a RACI matrix early on in the program

- Investing time in the details:

  - Providing complete details of raw materials such as chemicals, media and resins used in a process (grade, supplier, catalog number)

  - Providing buffer formulations including titrants used; formulate buffers to avoid pH adjustments in GMP manufacturing

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- Before transferring a process, consideration of how robust the process being transferred is and whether it needs to be optimized especially with regards to cost of goods
- Establishing analytical methods to support process transfer or discussing what in-house assays could be implemented at the receiving unit
- SMEs present on-site during key steps to help resolve issues
  - Unexpected results can occur at any point, which can require additional supporting work
  - This is especially helpful if client and CMO are located on different continents in different time zones
- Consideration of how the technology transfer information is transferred
  - Information may be present in a number of different documents which will be unfamiliar to the receiving unit
  - Face-to-face meetings with process experts or on-site attendance at the sending unit will facilitate process understanding

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10 Appendix 3 – Case Studies: Small Molecule

10.1 Summary

This chapter focuses on key challenges associated with global API technology transfers and provides examples of potential technical, cultural, and organizational gaps between the sending and receiving units. Addressing these challenges and gaps can enhance program management of technology transfers across markets.

The technology transfer of APIs follows similar steps as those presented in the case studies for biologics in Chapter 9 and as explained in detail in this Guide. In this chapter, three case studies for small molecule API technology transfers are presented. The first two case studies show challenges related to non-technical differences between the sending and receiving units. Both the sending and receiving units in these case studies were sites within the same large, global pharmaceutical company. The third case study presents technical challenges of a Contract Development and Manufacturing Organization (CDMO) as they develop and manufacture an API for a large, global pharmaceutical company with a business plan to market the API as a virtual company.

10.2 Main Challenges

Supply chains for the manufacture of APIs are constantly requiring technology transfers to be performed in the pursuit of lower cost options, process improvements, market requirements, and continuity of supply. In recent years, many of the API technology transfers have been from domestic to global manufacturing sites, often at an external company. Development of API processes has also moved to centralized, global development centers that do not have large scale manufacturing capabilities; this requires a technology transfer to another site for commercial production. The movement to global supply chains for API manufacture requires the technology transfer to understand the differences between the sending and receiving units, specifically the differences in their QMSs, risk tolerance levels, and operating/compliance standards. These differences need to be identified early in the technology transfer process so they can be addressed and factored into the overall transfer plan.

10.3 Technology Transfer Process

As shown in Chapter 1, ICH Q10 [1] defines the product lifecycle elements. Chapter 3 then defines the technology transfer activities as six distinct phases. In the first two case studies, a modified technology transfer process (five stage process as shown in Figure 10.1) was implemented to facilitate the cross-functional framework of the transfer as well as regular review of the transfer risks. In the example process depicted in Figure 10.1, defined checkpoints with a steering committee are used to evaluate risk and provide approval prior to moving to the next stage. The typical functions required in each stage of this transfer model are also shown. For the first two case studies, adjustments to this technology transfer process are discussed to address unique requirements of each of the transfers.

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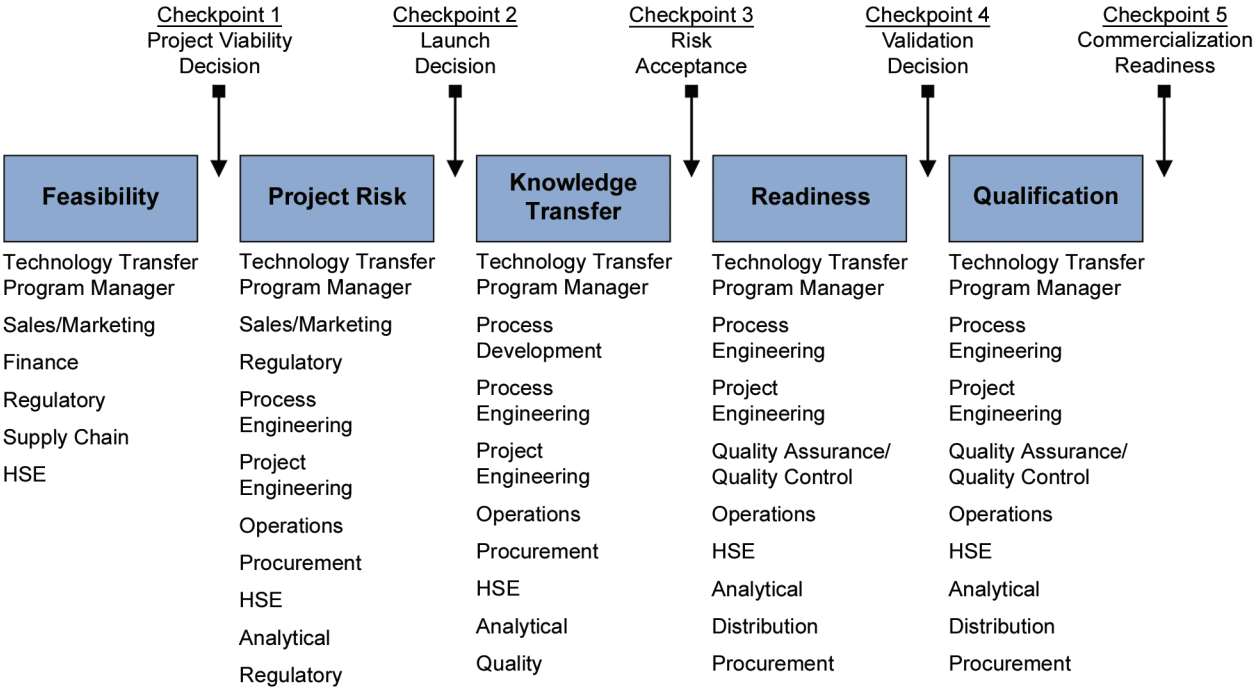
Figure 10.1: Example Five Stage Technology Transfer Process



In this example technology transfer process, early feasibility is defined as the first stage, incorporating key technical functions, to help assess high level risk which is then used to develop the technology transfer charter. An example of a high level risk evaluation for a technology transfer is provided in Table 10.1.

Table 10.1: Example High Level Technology Transfer Risk Evaluation for Feasibility Stage

Sending Unit	Receiving Unit	New Product	Same/ Single Market	New/ Multiple Market(s)	Existing Production Line	New Production Line	Equipment Change	Process Change	> 10x Scale-up
Site 1	Site 1	Low	Low	Low	N/A	Med	Med/High	Med/High	High
Site 1	Site 1	High	High	High	Low/Med	Med	High	N/A	High
Site 1	Site 2	Low	Low	Low	Low	Med	Med/High	Med/High	High
Site 1	Site 2	High	High	High	Med	Med	High	N/A	High

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The feasibility stage is often used for site selection, which is discussed in more detail in Chapter 9.

Detailed risk assessment is then performed in the second stage using a broader cross-functional team to identify all elements of the knowledge transfer needed to perform a successful technology transfer. Changes to the technology transfer risk assessment and risk management during the transfer process may require the team to revisit earlier checkpoints to ensure full visibility and communication.

The third case study is from a CDMO (based in India) that performs both API process development and API manufacturing. Figure 10.2 shows an example CDMO technology transfer model. This model is used to incorporate detailed elements related to the process development and the information required from the ongoing process development for the knowledge transfer. The CDMO completes the development phase by performing laboratory scale reproducibility batches to mimic plant conditions and hold times.

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Figure 10.2: Example Contract Development and Manufacturing Organization (CDMO) Technology Transfer Process Model
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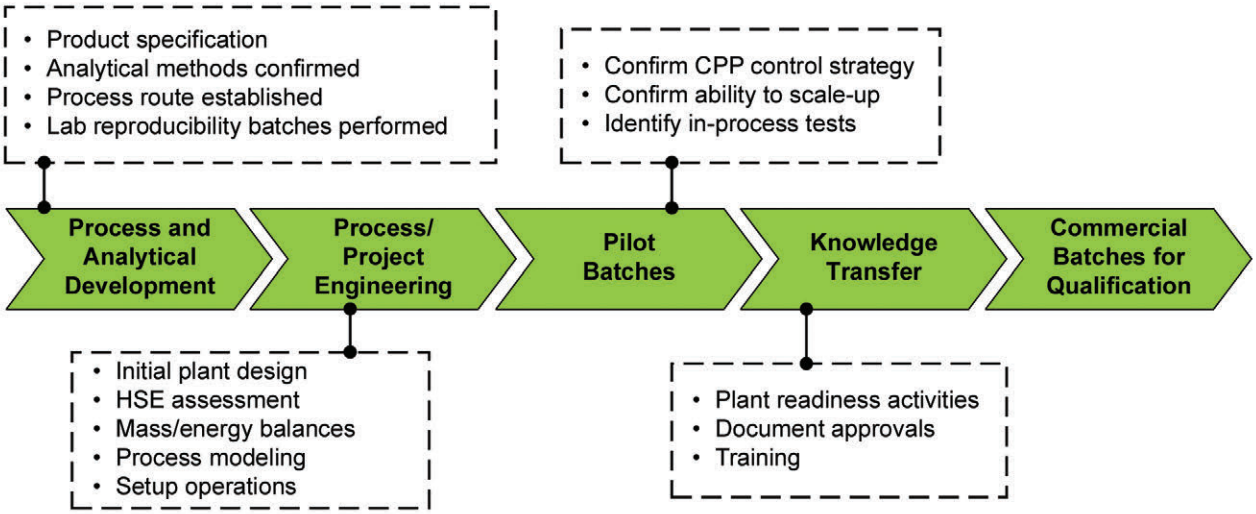

When process and/or method development is required to provide the information required for a technology transfer package, steps in a technology transfer process are often performed in parallel to facilitate the knowledge transfer and overall timing of the transfer. In this third case study, intermediate scale pilot batches were performed to ensure the ability to scale-up the process and to prove the control strategy for the CPPs.

10.4 Technology Transfer Case Studies

The first two case studies address the challenges associated with the potential technical, cultural, and organizational gaps between the sending and receiving units. The first case study identified technology gaps driven by global differences in interpretation of CPPs and how they are controlled, while the second case study discusses differences in the interpretation of the global QMS. Each of these small molecule API technology transfers were for a large, global pharmaceutical company where the sending and receiving units were internal to the same company, demonstrating that these challenges are not unique to external manufacture.

10.4.1 Small Molecule Case Study #1: Supply Chain Driven Technology Transfer

This case study is for a commercial-commercial transfer of an API that was already being commercially manufactured in Europe and imported for sale to the US market. Changes in importation regulations on the API and a portion of its raw materials were going to impact the ability to serve the US market. These changes created the business need to transfer commercial manufacture to the US within 3-6 months in order to maintain continuity of supply. It was assumed the timing required for the technology transfer could be achieved for the following reasons:

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- API was produced commercially at a site internal to the company with well-established process criticality and control strategy
- API had regulatory approval in the US
- Initial plant scale would be smaller than the current equipment
- The US-based development team was already working on a next generation process and as such, analytical methods were already being used.

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At the kickoff meeting for the transfer, the full cross-functional team was provided the technology transfer charter with the goals and boundaries of the transfer. Priority was given to the team and the site to ensure the required timing for filing the API manufacture at the US site. Clear roles and responsibilities for each member of the technology transfer team were communicated as well. Due to the condensed timeframe for the transfer, a full risk assessment could not be performed until after the knowledge transfer had begun. The risk assessment identified key challenges in the initial assumptions as summarized in Table 10.2.

Table 10.2: Small Molecule Case Study #1 – Risk Assessment Challenges and Resolutions

Challenge	Resolution
Clear understanding of process CPPs and control strategy between the sending and receiving units was challenging due to local operational knowledge and a language barrier	Face-to-face meetings between process and analytical team members to facilitate communication of actual practice. Involvement of operations personnel from the sending unit during process validation at the receiving unit.
Key analytical methods for in-process tests were not yet validated/qualified as the focus had been on the next generation process	Gaps in method qualifications were identified early in the transfer which allow the receiving unit to initiate a co-validation between the sending and receiving units in time for the process validation.
New equipment to be used was for a multi-purpose small volume production line that required design modifications specific to this API process	As the process was well-defined, engineering provided design modifications specific to the API process with long lead items ordered shortly after the kickoff meeting. Equipment qualification remained on schedule.
Not all equipment between the sending and receiving units was equivalent, impacting the regulatory strategy to obtain site approval for the API manufacture	Additional development studies were performed to provide data supporting that the product was equivalent with available equipment.
Installation of a new milling suite to achieve particle size distribution requirements of the API; despite milling equipment being identical to the sending unit, problems in obtaining particle size distribution specification	Plant scale milling trials had to be performed to identify milling process parameters that could achieve the particle size distribution specification.

Challenge

This section describes the challenge involved with knowledge transfer of the CPPs and control strategy for this case study. Although the technology transfer was between two internal company sites, access to operations information and data was limited. Review by the receiving unit of the historical manufacturing batch records, validation reports, and CPPs with their ranges provided as part of the knowledge transfer identified significant gaps in understanding for implementation at the receiving unit. These gaps included:

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- Accurate translation of the batch records to English, which impacted the ability to understand common or local operational practices at the sending unit
- Philosophy in the identification and control of CPPs
- Process data availability

For example, pH was to be controlled tightly in a specified range as measured using an in-process control sample. However, information such as the location of the sample, how the measurement was obtained and validated, and accuracy of the data was not well understood based on the technology transfer information provided. With the batch records being in another language, there was a language barrier to further understanding of the actual operations practice.

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Another example was interpretation of the allowed flowrate of a reactant into the reactor. The documented criticality justification stated that the flowrate was a critical parameter to be maintained between 0.5-1.0 liters/min. However, the batch records appeared to measure only the total time rather than a rate across the entire reactant charge. This was a common practice to manage flowrate at the sending unit, though it differed slightly from the criticality justification.

Learning

Face-to-face meetings with the sending unit cross-functional team responsible for manufacture of the API allowed the receiving unit to obtain the required process information needed for successful implementation of the process. A crucial element was the involvement of the technicians with expertise performing the actual batch operations. These face-to-face meetings were essential to allowing the receiving unit to develop an understanding of the process specifics implemented at the sending unit. This interaction also developed a stronger relationship between the receiving and sending units, facilitating remote cooperation throughout the technology transfer. Additionally, technicians/operators and engineers from the sending unit participated at the receiving unit during the manufacture of the process validation batches.

Outcome

Successful process validation was performed at the receiving unit three months after the start of the technology transfer, allowing the receiving unit to be filed as a manufacturer with no gap in API supply for the US market.

10.4.2 Small Molecule Case Study #2: Alignment of Quality Management Requirements for Method Transfer

This case study is for a development-commercial transfer of a set of related APIs that were being developed in Europe for commercial manufacture in the US. While these APIs were new to the company's portfolio of products, they were already commercially available in the US but were being produced using a different process route. For this reason, each API already had CQAs developed and filed with the drug product manufacturers.

The development and technology transfer teams were internal to a large, global pharmaceutical company, which meant they were following the same global QMS guidelines. As the sending unit, the development team was responsible for designing all aspects of the process including the analytical method development. Program management of the technology transfer was performed at the receiving unit with cross-functional members of the receiving unit having oversight in the ongoing development activities.

In this case study, the focus is on the impact of the analytical method requirements on the overall project resources and schedule (see Chapter 4 for detailed information about integration of methods for a technology transfer). For this set of APIs, coordination of the methods was more complicated due to the number of APIs, isolated intermediates, and raw materials. The technology transfer of each API was to be performed sequentially under tight timelines to achieve contract deadlines with the drug product manufacturer. However, the development of each API process was being performed in parallel with analytical methods that significantly overlapped the APIs. With the analytical development being performed in Europe, importation restrictions on the APIs required the team to determine ways to streamline the analytical requirements of the technology transfers. Because of the importation restrictions and the large number of analytical method transfers and validations needed between Europe and the US, steps were initiated very early in the technology transfer to ensure alignment with the local QMS requirements of the sending and receiving units for the full set of APIs.

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Challenge

While both development and manufacturing were being handled at internal locations, interpretation of the global QMS for analytical method transfer, qualification, and validation requirements differed locally between Europe and the US; this was going to cause excessive redundant analysis and documentation activities. The quality departments in these two locations believed that these different interpretations of the global QMS within the same company resulted from supporting information provided to associated regulatory authorities during an audit. Therefore, alignment on the quality philosophy and approach to the analytical method requirements was essential to ensuring project success.

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For compendial methods, the global policy required the method to be confirmed by running the method on a reference sample. The sending unit did not specify the type of reference sample required and often used material generated in the development laboratory batches. The receiving unit required a method qualification in the laboratory that involved testing of the method using a qualified reference standard. As in any development program, availability of qualified reference materials can be a challenge. Supply of R&D (sample) material was also limited due to US importation restrictions. In summary, global policy required validation of non-compendial methods but local differences were found in the testing requirements for the sending and receiving laboratories performing the validation and in the documentation requirements.

Learning

As these were existing APIs already in the market but made from a different process, the project team worked with the drug product manufacturer under contract, the internal business team, and quality/regulatory to establish boundaries for the method development work. The boundaries for the analytical methods included:

- CQAs and the limits already filed by the drug product manufacturer were to align where possible
- Drug product manufacturer analytical methods would be used where applicable and if not proprietary
- New impurities formed from the new process would be identified if greater than 0.10%

The development and technology transfer project managers facilitated negotiations over a six-month period between the local quality groups at the sending and receiving units. A blend of approaches (shown in Tables 10.3 and 10.4) was implemented for the project that would satisfy the key desired QMS elements yet reduce the required amount of sample materials, testing time, and number of resources. For instance, the receiving laboratory acted as the second laboratory verification required for completion of the method validation with both laboratories using the same sample; this approach reduced the time for method validation in half. Prior to any analytical work being performed for the technology transfer, an agreement was signed between the sending and receiving units which established criteria for all analytical work associated with these technology transfers. These boundaries established the workflow for the analysts developing and transferring the methods.

Outcome

The negotiated approach to the analytical method transfer was formally documented between the sending and receiving units. Alignment of the QMS between the sending and receiving units for analytical method transfer, qualification, and validation significantly reduced costs; this also allowed for proper definition of required resources, which would keep the overall project timeline intact. This blended approach allowed for 21 methods to become verifications/qualifications rather than validations, which resulted in a time savings of over a year for one analyst. In total, 80 method verifications/qualifications and 8 full method validations were included in the overall technology transfer plan. Additionally, a system was agreed upon that allowed multiple verifications and products to be combined into fewer documents. This system was adopted internally by the company to be used for future technology transfers.

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Table 10.3: Small Molecule Case Study #2: Non-Compendial Method Transfers

Attribute	Method	Difficulty Level	Number of Compounds Impacted	Verification Requirements
Assay	HPLC	3	2 APIs	Sending unit performs full validation per the receiving unit site SOP. Receiving unit performs three sample preparations at each level to assess intermediate precision, accuracy, linearity, and specificity. System suitability is to be used to assess system precision.
Impurities	HPLC	3	2 APIs	
Chiral Assay	HPLC	3	2 APIs 1 Intermediate	
Detection of Critical Raw Material X	HPLC	3	2 Intermediate	

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Table 10.4: Small Molecule Case Study #2: Compendial Method Qualifications

Attribute	USP Test	Difficulty Level*	Number of Compounds Impacted	Current Use of USP Test by Receiving Unit	Verification Requirements
Appearance	Visual	1	4 APIs 2 Intermediates 5 Raw Materials	Daily	• 1 analysis on compendial material • 1 analysis on internal material (except raw materials) • Both sending and receiving units to meet USP criteria
Identification	<197M>	2	4 APIs 2 Intermediates 5 Raw Materials	Weekly	
Melting Point	<741>	2	2 APIs	Weekly	
Elemental Impurities	ICH Q3D	2	4 APIs	Outsourced	
Testing for Iron	<241>	1	1 Raw Material	Never	
Optical Rotation	<781>	2	4 APIs 1 Intermediate 3 Raw Materials	Weekly	
pH	<791>	2	4 APIs	Daily	
Water Content	<921>	2	2 APIs	Daily	
Loss on Drying	<731>	1	4 APIs 2 Intermediates 5 Raw Materials	Daily	
Residue on Ignition	<281>	1	4 APIs 1 Raw Material	Weekly	
Residual Solvents	<467>	2	4 APIs	Weekly	In addition to the requirements above: • Intermediate precision with comparison between both sending and receiving units • Verification of limit of detection/limit of quantitation
Assay	USP x	3	2 APIs 2 Intermediates	Daily	
Impurities	USP x	3	2 APIs	Daily	
*Difficulty Level: 1 = No special equipment or software required 2 = Specialized equipment, simple software 3 = Specialized equipment; complex software					

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10.4.3 Small Molecule Case Study #3: Development to Commercial at a CDMO

Contributions from: Dr. M. Damodharan (Sr. Vice President, Global Quality and Regulatory Affairs), Tuneer Ghosh (Sr. Vice President, Process Engineering and New Product Manufacturing), Suresh Naik Korra (Assistant General Manager, Process Engineering), and D.V.S. Varma (Associate Vice President, Quality Assurance), Sai Life Sciences Ltd., India

In this case study, the business plan for a global pharmaceutical company supported all development and manufacture of an existing antibiotic API to be performed externally at a CDMO in India. The API was used in a new drug product which was undergoing clinical phase 3 trials. Since the manufacturing process for this API was not novel, the CDMO leveraged platform technologies to begin the development.

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The CDMO followed the model shown in Figure 10.3 to develop, assess, and demonstrate the process required to make the API. In this process, the focus of the work centered around safety and scalability, identifying CPPs, and developing the process engineering requirements for a successful plant design.

Figure 10.3: Small Molecule Case Study #3 – CDMO Process Development/Technology Transfer Model
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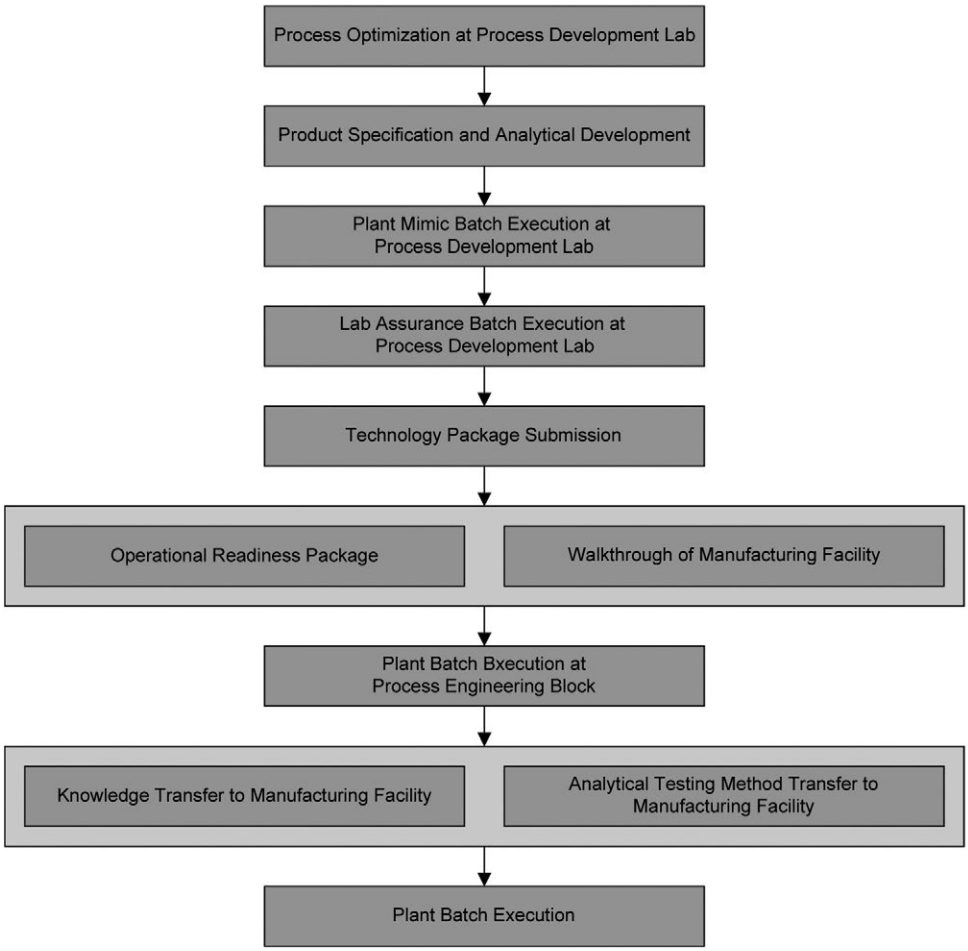

Challenge

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This case study focuses on the technical challenges associated with knowledge transfer between development and commercial manufacture for a greater than 10 times scale-up of the process.

Learning

The API process consisted of five stages to achieve five chemical transformations and another three stages to insert the required side chains. The initial scale of 100 g (3.3 L) was used for all development work, including execution of three laboratory scale batches to demonstrate reproducibility with the successful scale-up to 90 kg (3,000 L). The CPPs that were identified with the control strategy were incorporated into the commercial scale plant design, as summarized in Table 10.5.

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Table 10.5: Small Molecule Case Study #3: Process Criticality for Plant Design

Process Requirement	Process Design
Oxygen free environment for safety reasons	Calculations determined three inertization cycles were needed to achieve the required 4 ppm oxygen level in the system.
Main reaction was highly exothermic—simulation of the reaction kinetics helped determine the heat of reaction, adiabatic temperature rise, and rate required for the limiting reagent	• Mixing required for heat removal was 80 rpm with a dual impeller agitator consisting of a propeller for one impeller and a hydrofoil for the second • Cooling system was designed to maintain -5°C on reactor throughout the addition of the limiting reagent • Flowrate of the limiting reagent was controlled using an orifice
Filtration and cake wash were critical for impurity removal	Filtration and drying studies identified that an agitated filter/dryer combination was the appropriate design for this unit operation

Once the equipment design was established, the technology transfer team was able to coordinate operations activities in parallel with the equipment installation and qualification. These operations activities included drafting the safety assessment, drafting the master batch record, preparation for process validation, and definition of cleaning requirements.

Upon start-up of the commercial scale process, several other critical parameters were identified that were not seen during development despite having performed three reproducibility batches in the laboratory. These technical challenges and their resolutions are shown in Table 10.6.

Table 10.6: Small Molecule Case Study #3: Technical Challenges upon Scale-up and Resolutions

Challenge	Resolution
Longer hold times required for a critical raw material in the drum prior to and during charging resulted in precipitate that settled to the bottom of the drum; precipitation rendered the raw material not suitable for the reaction as precipitate caused the reaction to stall	The raw material drum was positioned such that it did not need to be moved. A vacuum dip tube was used to remove clear solution from the drum with an in-line sight glass to determine solids breakthrough.
An emulsion was formed during the extraction step due to overmixing	The centrifugal pump used to transfer the liquid to the extraction vessel was replaced with a pressure transfer. Additionally, mixing in the extraction vessel was reduced 65%.
An unknown impurity was observed at a percent level that was too high	After extraction, the organic layer required acid neutralization. Mixing was too low, allowing localized areas of high acid concentration. Increase in the mixing speed removed the impurity.

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Outcome

A demonstration batch at commercial scale needs to be factored into the project plan, considering that new critical parameters may be seen for process parameters (such as mixing) that are difficult to mimic in the developmental laboratory environment or even at pilot scale. Extended hold times should be incorporated into the laboratory development work to mimic the expected times for commercial manufacture. Safety considerations are also a key factor for small molecule plant design that needs to be assessed during development and can, at times, present unique challenges to achieving the needed process criticality for product quality. Reaction kinetic modeling is a useful tool in establishing process parameters ranges needed to achieve both a safe environment and product quality.

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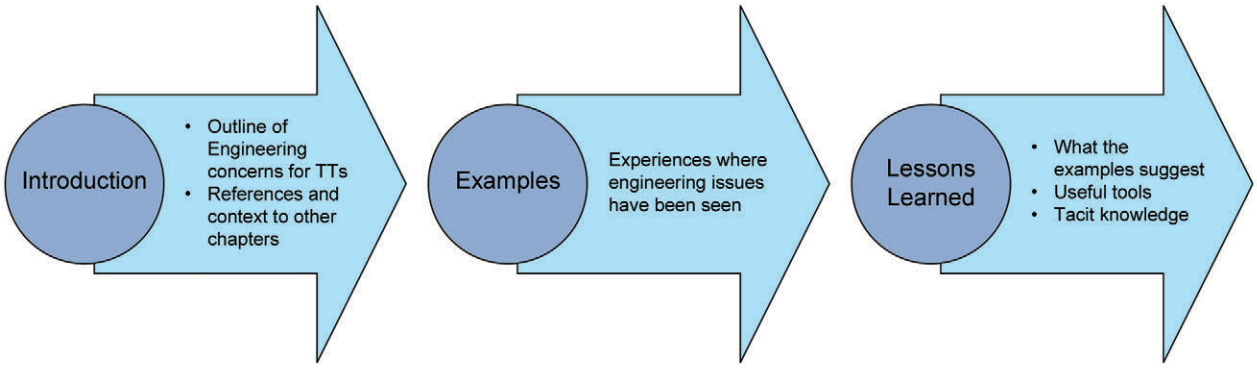
11 Appendix 4 – Engineering Considerations for Technology Transfer

Figure 11.1 provides an overview of the sections presented in this chapter.

Figure 11.1: Chapter Overview



11.1 Introduction

Transfer of engineering knowledge can be an important consideration for transfers, whether for single or multiple sites. Importantly, it should be an ongoing two-way exchange of information as the intricacies and experience gained may enhance process understanding and product quality.

In the context of modern biopharmaceutical manufacturing, engineering knowledge can cover a wide range of information, including activities and information related to facilities/equipment design and maintenance through to chemical and process engineering knowledge and decisions related to manufacturing activities.

This chapter focuses on those activities that do not fall within the scope of process development or product realization as such activities should be captured within the development activities and control strategy for the process.

Consideration to the engineering aspects of a technology transfer should already be planned as part of the technical diligence and detailed risk assessment described as part of development of a technology transfer plan in Chapter 5 (for drug substance) and Chapter 6 (for drug product).

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Overall, it is the success of the transferred process and delivery of product quality (drug substance or drug product meeting the requisite CQAs) that determines whether a transfer is successful. Well developed products and processes will identify critical aspects of processes that support achievement of consistent and reliable manufacturing; where these are of significance, they will form part of the control strategy.

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Engineering controls are critical to maintenance of CPPs and can be key to detection and/or management of potential process influence from material attributes. Additionally, engineering requirements and specifications are important risk mitigations in processes—for example, where materials compatibility and cleanability is required to be assured by control of materials of construction, welding, surface finishes etc; or where instrument suitability is managed by instrument selection and calibration. There are numerous other examples where engineering controls are critical to assurance of CQAs and CPPs.

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It is important that technology transfer planning clearly differentiates the critical engineering aspects of processes and that these decisions are clear to engineering groups at the receiving unit as well as their suppliers. Earlier chapters in this Guide discuss the establishment of CQAs and CPPs. Engineering teams need to be informed of these requirements as part of the technology transfer plan. If engineering teams and suppliers are disconnected from this information, this situation can lead to lack of focus and inefficiency or, worse, may result in failure to achieve the required manufacturing outcomes or delays, which lead to potential shortages in critical products.

Engineering information can also play an important role in the tacit understanding of the process at a level underpinning the critical variables and controls. This knowledge often exists within an organization in systems and locations outside of the mainstream production control systems as Good Engineering Practice (GEP), engineering trials, or inside of maintenance and plant management systems. This information is often assumed to exist within qualification and validation files; however, the outputs of the engineering knowledge are frequently the only portion that is visible, not necessarily the process to get to that point.

Engineering information may include:

- Details and information on the design, fabrication, or procurement of process equipment or supporting utilities, for example:
  - Equipment drawings and specifications, including materials of construction compatible with process solutions and geometry
  - Details of critical measuring instruments, including range/capabilities
  - Physical layout or arrangement and integration of equipment
- Details on the types of equipment to be used for running or supporting a process
- Arrangement for equipment transfer, such as equipment transfer plans and commissioning/qualification methodology
- Control programming/logic and configuration information
- Maintenance and setup instructions
- Equipment service history
- Cleaning process development information
- Details of engineering studies that are of importance to determining process benchmarks for equipment or processes

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Proper technical assessment needs to be carried out across GxP and GEP systems to ensure that key knowledge in these areas is captured to avoid delay or complications with technology transfers. Problems in validation can often be avoided with a greater appreciation of engineering knowledge gaps that existed at the time of transfer.

The major area of focus for this chapter is to the equipment and its use. The process is often presumed to be like-for-like in this respect, but subtle differences can occur between the sending and receiving units in terms of the physical elements, controls, configuration (setup), and use.

There are three possible broad scenarios in terms of knowledge transfer:

- Sending unit transfers the process and like-for-like equipment as part of the project. In this case, knowledge around use and behavior of the equipment related to the product and process under transfer is the focus for knowledge required for transfer.

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- Sending and receiving units use the same or similar equipment (either new or existing) but that exists separately on both sites. In this case, experience of operation, recommended configuration, and details of setup are required.

- Sending and receiving units use similar but not identical equipment, where the receiving unit may need to procure or operate equipment to achieve the process. In this case, the product and process outcomes are important. The means of getting there (operationally) may differ.

Sometimes the receiving unit is at a disadvantage because some technical knowledge or known behavior (tacit knowledge) does not get captured; this information may not form part of the formal control strategy. It is also possible that experience from a receiving unit, which does not fit into the formal validation or knowledge framework used for transfer, does not get fed back to the sending or other units.

In situations where equipment transfers form part of the technology transfer, particular emphasis on engineering information becomes a key success factor to the transfer. Plans need to be made for the sharing of experience from operation of the equipment. The transfer planning process should provide for access by engineering SMEs from the receiving unit to the equipment in use at the sending unit, particularly where the equipment will require mechanical and/or electrical decommissioning and reassembly at the receiving unit. In these situations, detailed engineering plans and documentation are required to ensure the equipment can be recommissioned and qualified in a like-for-like state.

The scope of a technology transfer often includes the use of plant automation and/or some level of enterprise level information technology (such as manufacturing execution systems). Specialist engineering resources should be identified and directed to planning for replication or implementation of suitable automation and/or network infrastructure; particular attention should be given to risks to data integrity and cybersecurity. The automation/network infrastructure or governance often differs between sending and receiving units; therefore, careful planning and assessment early in the technology transfer project is warranted. In addition, specific, engineering reviews may be warranted as part of the project close-out to ensure all risks are addressed.

This chapter presents examples of actual situations where engineering knowledge was not captured originally and/or should have been, highlighting the importance of building a robust engineering assessment and review process into the technology transfer plan.

Poor planning for these aspects can lead to inadequate time allocated to commissioning and start-up activities, resulting in delays in commercial manufacture or poor performance or deviations during manufacturing.

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11.2 Examples

Table 11.1: Engineering Considerations Example #1

Scenario	A technology transfer project required transfer of manufacturing for blending, filling, and packing dry powder inhalation products to three sites in different regions around the world. Identical equipment was centrally procured from a single reputable international equipment vendor who commissioned the lines on each site. Equipment included a filling machine, which needed to be set up carefully to control fill weight of the dose.
Issue	During qualification and initial operation, individual differences in setup and operation of the various machines masked the fact that there was a design feature in the filler where a mechanical part would wear over time and gradually introduce increased variation in fill weight. This was discovered at one site and over time they developed a modified procedure. The procedure consisted of monitoring the fill weight performance and then, at a trigger point, removing and maintaining the parts concerned. Additional lubrication steps resulted in more consistent filling. Other sites had observed the deterioration but attributed it to different causes.
Outcome	The true root cause was the mechanical issue. The site that discovered this over time started to supply unpacked products to the other two sites while their investigations continued, leading to reduced capacity and increase cost overall until the knowledge was shared.

Table 11.2: Engineering Considerations Example #2

Scenario	A technology transfer project required procurement of equipment for three sites to perform an API addition and mixing step under controlled temperature. The plan was to use like-for-like equipment; however, due to site technical preferences, two sites (site 1 and site 2) specified one instrument type and a third site (site 3) used an alternative instrument type.
Issue	A technical audit of the sites, performed later, concluded that the technical preference for site 3 was due to prior experience with repeated drift and recalibration requirements with the brand of the original instrument. Sites 1 and 2 had set standard calibration frequencies and limits; they had seen evidence of the drift but had not identified it as an issue yet. The problem was subsequently confirmed as present at both sites.
Outcome	The API distribution within the product was known to migrate due to heat distribution and the error was significant enough to have potentially caused a manufacturing deviation that could have resulted in a costly investigation across all sites. Sites 1 and 2 replaced the instruments. Validations were required to be repeated at site 2 (a second source of supply to site 3's market) due to variation in product assay, which was in specification but significant from a market authorization filing perspective.

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Table 11.3: Engineering Considerations Example #3

Scenario	A technology transfer project required transfer of a product to a second site by a sending unit that has experience in manufacture and packing of this product. Qualification protocols were prepared based on the original risk assessments and qualification practices used by the sending unit. Device criticality and, therefore, maintenance priority and frequency were specified in the protocols; however, the receiving unit established their own device prioritization and frequency for this equipment at their site.
Issue	Corporate quality considered the sending unit to be the technical lead on the project, given that the process was developed using science and risk-based methodologies; however, the receiving unit had their own history based on historical use and the intricacies of their equipment. Some critical tasks frequencies were decreased at the receiving unit and other tasks that were not critical in the protocol were considered critical at the receiving unit.
Outcome	Corporate quality required a validation deficiency at the sending site to investigate the need for the critical tasks from the receiving unit, resulting in delayed completion of PPQ batches.

Table 11.4: Engineering Considerations Example #4

Scenario	A technology transfer project required transfer of a process for a monoclonal antibody (mAb) API within a multinational company. The transfer was such that the process needed to be scaled up upon transfer. Therefore, the receiving unit was undertaking a large expansion on their site, similar to the sending unit. However, the geometry was not the same due to building layout and local requirements. For the purpose of this example, only the transfer of the cleaning process is discussed, although other processes were transferred from the sending unit to the receiving unit. Note that the product was on an expedited approval timeline.
Issue	Cleaning development studies were conducted at laboratory scale. The cleaning validation reports were available but were based on many weeks of engineering trials at scale. Transfer of the engineering data was initially not planned since it was viewed as impractical given that the strategy had been to meet acceptance criteria. Issues were discovered by the receiving unit that questioned the performance variables and required investigation.
Outcome	The process of reviewing the cleaning validation reports took many weeks to complete because the sending engineering trials were not cohesively summarized and the validation contractors used were no longer available to assist. Validation studies for both sites were delayed due to having to ensure data and limits to be filed were consistent between the sites.

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Table 11.5: Engineering Considerations Example #5

Scenario	A technology transfer project required a process to be transferred from a sending unit in Europe to a receiving unit, a CMO, located in Vietnam. The transfer requires the CMO to purchase and commission new equipment for solution preparation and aseptic fill/finish using blow-fill-seal technology. The equipment is designed and built by well-known European suppliers with operating offices/service presence in Singapore. The engineering team at the receiving unit was not as experienced as the sending unit. Additionally, this is one of only two installations of this type for the supplier in the Asia Pacific region; the other installation was in China for a multinational company.
Issue	The risk to the success of the transfer was considerably increased given these factors, not just for engineering, but in all areas. Even if the technology was well known and supported and the engineering assessment and information was well collated and assembled, it was still possible that considerable effort would be required to ensure the engineering information was delivered in an understandable manner.
Outcome	The company responsible for the transfer considered providing the support of one of their own engineering staff to the receiving unit to assist with project; however, the receiving unit would not fund the role and expenses. Following commissioning, numerous technical issues were encountered. A point of concern was reached when damage occurred to the equipment. Representatives of both the sending unit and European supplier were required to visit the site within short notice. The project was delayed by three months as a consequence of these issues. Commercial tensions developed between the parties and the need to invoke supply penalty clauses were considered.

Table 11.6: Engineering Considerations Example #6

Scenario	A technology transfer project required transfer of an mAb process from a sending site in the US to a receiving site in the EU. No new equipment needed to be purchased in the EU site since the current equipment were to be configured to accommodate the new process. Since the transfer did not involve the start-up of new equipment, no commissioning activities were planned as part of the technology transfer.
Issue	During the first batch at scale, there were problems during execution of the concentration phase during ultrafiltration/diafiltration. Further investigation of the problem revealed that the automation recipe was not properly configured.
Outcome	The project was unnecessarily delayed as the batch was implicated and it was determined as part of the CAPA that the batch needed to be repeated. Engineering felt the configuration items in error were not critical; however, quality and validation felt the issue warranted wider investigation of the adequacy of configuration in other systems.

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11.3 Lessons Learned

Transfer of process related information in the form of specifications, quality documents, qualification and validation documents, production data, work instructions, and the many written technical resources within the GxP frameworks of a biopharmaceutical organization is expected under a robust technology transfer plan. Operations, quality, and other direct staff can be trained in the operation of the equipment and, to some extent, engineers can be trained in the planned maintenance, normal operation, and troubleshooting around a process; however, GxP systems may be insufficient for capturing/transfer of tacit engineering knowledge.

The examples in Section 11.2 highlight the types of issues that may be encountered during technology transfer projects. The lessons learned from these examples are summarized in Table 11.7.

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Table 11.7: Engineering Considerations Example: Lessons Learned

Engineering Considerations Example (from Section 11.2)	Lessons Learned
Example #1	Include feedback loops in technology transfer plans, in anticipation of new knowledge obtained at the receiving unit that can collectively benefit the organization.
Example #2	Solicit prior engineering knowledge for equipment selection from the receiving unit(s) during planning for technology transfer projects.
Example #3	Where similar capabilities exist, conduct gap assessments of GxP engineering practices at sending and receiving units. Understand the potential impact these gaps may have to qualification programs.
Example #4	In addition to reviewing validation activities and outcomes, review the engineering work underpinning these or implied in their completion.
Example #5	In addition to the quality of the information, consider the capability of the receiving unit to receive, understand, and act upon technical information.
Example #6	Review configuration information within control systems to ensure the most current information is captured. Where acceptable ranges are permitted, validation information may not reflect the current configuration.

11.3.1 Useful Techniques for Capturing Engineering Information

Engineering SMEs should plan for knowledge and information to be transferred and shared as part of a technology transfer project. This is particularly relevant where the transfer involves changes to the engineering team in addition to changes in the site. In this case, lessons learned from the procurement and commissioning activities can assist to expedite or reduce risk at the receiving unit.

The following is a list of useful tools.

- **Planning for Technology Transfer During Procurement**

  It is often known or expected that a process will be transferred. For many products, a technology transfer is likely to occur at some point during the product lifecycle. As part of the project planning process, it is recommended to have a structured program comprising areas and associated content such as the example shown in Table 11.8.

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Table 11.8: Examples of Content for Project Planning Process

Area	Examples of Content
Validation	Planning and support for qualification and validation
Commissioning	Inspection and test plans, commissioning, handover
Integration with GxP Systems	Maintenance/calibration to computerized maintenance management system, SOP development
Construction	Project reports, meeting minutes
Process Automation	Automation deliverables, specifications, drawings
Engineering/Design	Specifications, drawings
Document Control/Coordination	Project procedures, training plans, records
Cost and Time Control	Budgets, budget reports, cost and time change reports
Planning	Schedules, work breakdown
Project Management	Key project documents, charter, project execution plan

Deliverables for the project charter should include a process to capture useful engineering information/decisions that would assist any future execution of a similar project. The mechanism for achieving this information capture includes staged review specifically to collect such information, an awareness of the need to collect the information throughout the project, and assigning responsibility for this task to a user representative during the project. It is easier to capture such information during the process rather than with hindsight.

- **Structured Reviews of Engineering Databases for Existing Processes**

  Engineering databases may include computerized maintenance management systems or maintenance schedules, calibration management systems, configuration management systems, equipment history files or logs, setup instructions or check-sheets, training/familiarization materials, and other rich sources of data.

  Engineering SMEs with knowledge of these systems and experience as a sending or receiving unit engineer should review these information resources in a formal manner for information. The degree of rigor in these reviews will reflect the complexity and risk of the project. Data collected should be captured and collated to allow versioning as GEP documentation.

  Companies that conduct frequent transfers often develop checklists for these reviews. Typical data to be considered may include:

  - Maintenance management data
    - > Maintenance and calibration procedures and schedules (current practice)
    - > Routine setup procedures and operational checks
    - > Maintenance and calibration history (or selected history)
    - > Information from engineering files/manuals relating to knowledge gained post-qualification
    - > Checklists or training resources associated with key engineering activities
    - > Log book reviews

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- Configuration data
  - > Control/automation configurable settings lists
  - > Copies or images of automation or configuration records
  - > Program version number information and history of upgrades
  - > Equipment/process setting checklists
  - > Tools/aids or techniques developed to assist setup or checking configuration
- Engineering data supporting qualification (considered to be useful benchmark data for the receiving unit)
  - > FMEAs, Hazard and Operability, and other risk assessments conducted for operation
  - > FMEA analysis to anticipate the need of specific commissioning or wet testing during start-up
  - > Commissioning reports or data
  - > Details of engineering studies conducted
  - > Changes/lessons learned from start-up or initial operation
- Operational excellence data
  - > OEE information
  - > Value stream mapping exercises, Lean/Six Sigma projects (Define, Measure, Analyze, Improve, and Control improvements)
  - > Engineering KPIs
  - > Known troublesome areas for operations arising out of breakdowns, ineffective maintenance, difficulty in setup as reported from defect/deviation records
  - > Reliability analysis, mean time between failures, mean time to failure, or similar data

- **Responsibilities (RACI) Matrix**

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SMEs from the engineering functions of sending and receiving units should meet, ideally face-to-face, to review the proposed transfer after receiving details of the project scope and proposed process. The purpose of this meeting is to review the requirements from each party. A result of this meeting should include a list of required deliverables and a responsibilities matrix. The sending unit should prepare this responsibilities matrix and take into account the capability and maturity of the receiving unit in relation to the technology transfer project. Matters to consider include:

- Role of the engineering SMEs in the sending and receiving units
  - > Involvement in planning of items/knowledge to be shared
  - > Strength and weaknesses in the necessary technical areas at the receiving unit
  - > Identification of skill matrix for required support for the project

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> Future site visits and training needs for both the sending and receiving units

> Specific areas where collaboration which may be required between SMEs from different backgrounds (engineering, scientists and operations specialists) at sending and receiving units

- Role of engineering suppliers and vendors

> Areas where potential exists to leverage existing vendor/supplier networks from either site

> Capability analysis of supplier/vendor networks at the receiving unit to deliver the expected outcomes for the project and areas where this may need to be supported

> The need for ongoing supplier/vendor engagement and support

> The scope and responsibility for preparation of contracts and service agreements as necessary

> Information that will need to be shared with third parties and the applicable confidentiality provisions

> The criticality of third party support packages and the potential need for guarantees and incentivization

- **Engineering Equivalence Assessments (Gap Analysis)**

The level of existing capabilities for key processes or equipment may not be identical at the sending and receiving units. Even where processing and equipment are considered identical (same make, model of equipment, starting materials, specifications, etc.), these systems may be performing differently. Equivalence assessments involve identification and comparison of aspects including:

- Design and specifications of equipment and utility services

- Performance data under similar input and operating conditions

- Engineering data

- **Ongoing Review and Benchmarking**

While the techniques discussed above are useful to identify information which may assist in the planning and execution of a technology transfer project, such projects also present a valuable benchmarking opportunity. It is good practice to verify the outcome, particularly where future transfers may occur. Valuable insights may be gained from the receiving unit and the transfer information provides the context to recognize these. The following activities are useful to undertake and may highlight opportunities to improve the process at both the sending and receiving units and for future sites:

- Review of the original information on one or more occasions during the receiving unit's implementation, to identify areas where information was unnecessary and where gaps exist

- Planning for data sharing and monitoring of performance and engineering metrics, if there is interest from the sending and receiving units—the planning could include:

> Ongoing monitoring and reporting metrics, reporting strategies (see below)

> One-time performance/comparability reporting, Pareto analysis

> Targeted engineering reviews

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> Reliability centered monitoring

> Incident reporting and learning from investigations

> Risk reviews to verify existing risk assessments and revisit these based on actual performance

Data to consider for ongoing monitoring includes:

- Breakdown or incident classification and/or downtime monitored by engineering or technical groups in systems outside GxP scope

- Automation and/or manufacturing execution system data reviews, configuration management activities and lessons learned

- Reliability metrics such as:

  > Ratio of planned preventative interventions versus unplanned interventions for the process

  > The number and nature of changes to configuration information and maintenance instructions per month, or other defined period

  > Number of times per visit, or in percentage terms, that additional activity is undertaken during planned interventions in the process

  > Frequency and nature of vendor/supplier attendance to site

- Capability data and investigations to shifts in performance

- Equipment robustness monitoring—ability for the process to cope with changes to inputs over time

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12 Appendix 5 – Example of a Failure Mode and Effect Analysis (FMEA) for a Non-Sterile Drug Product

Item/ Process Step	Potential Failure Mode	Potential Failure Effects	SEV	Potential Causes	OCC	Current Controls	DET	RPN	Actions Recommended	Responsible	Actions Taken
Tableting and IPC controls	Failure in disintegration time IPC control	Reinforced control and possible rejection of the amount of production not conform	5	Hardness control borderline.	1	IPC every 4 hours	8	40	Use of SPC during tableting. RFT project for optimization of tableting parameters.		
Tableting and IPC controls	Failure in friability IPC control	Reinforced control and possible rejection of the amount of production not conform	5	The instrument is dirty.	1	Visual inspection	8	40	No action required.		
Tableting and IPC controls	Failure in appearance IPC control	Rejection of the amount of production not conform	10	Broken punches. The piece of the punch is not found. The compression force is higher than the maximum admitted by the punches.	2	Metal detector at the discharging of the tablets	8	160 •	Revision of all the recipes to check that the maximum load admitted on punches is aligned with limits.		
Tableting and IPC controls	Failure in appearance IPC control	Rejection of the amount of production not conform	4	Broken punches. The piece of the punch is not found. The compression force is higher than the maximum admitted by the punches.	2	Metal detector at the discharging of the tablets	8	64	Revision of all the recipes to check that the maximum load admitted on punches is aligned with limits.		
Tableting and IPC controls	Failure in appearance IPC control	Rejection of the amount of production not conform	10	Presence of black spot on the tablet surface.	2	IPC control every hour and cleaning of punches	8	160 •	Cleaning of punches after every batch.		
Tableting and IPC controls	Failure in appearance IPC control	Rejection of the amount of production not conform	4	Sticking problems, non-suitable granulate.	2	IPC control every hour	8	64	See action on LOD and tapped density.		
Tableting and IPC controls	Failure in appearance IPC control	Rejection of the amount of production not conform	4	Sticking problems, dirty punches.	2	IPC control every hour	8	64	Clean the punches immediately before their use.		
Note: IPC, In-Process Controls; SPC, Statistical Process Controls; RFT, Right First Time											

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13 Appendix 6 – Example of Information that May be Included in Technology Transfer Report

Document	Comments
Development Information	
General Information	
Product Development History	
Drug Substance	
Development History Overview and References	
Formulation	
Formulation Development Report	
Excipient Compatibility Report	
Manufacturing Process	
Batch Manufacturing History	
Process Comparison Summary (incl. rationale for changes)	
Comparison: Transfer, Demo, ICH, Scale-up, Batches	
Process Understanding/Risk Assessment Reports	
Analytical Methods Development: Specification History/Rationale	
Justification of Specifications	
Summary Memo on Specification History and Rationale	
Methods Overview: Development/Validation/Transfer	
Drug Product Method Transfer Reports	
Analytical Method Transfer Exercise Overview	
Development History and References	
Technology Transfer Documentation	
Packaging	
Summary of Stability Data	
Stability Program Summary	
Container Closure Summary	
Miscellaneous	
In-Process Specifications	
Batch Analysis	

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Document	Comments
Production Information	
General Information	
List of all Lots Manufactured	
ICH Stability Batch Manufacturing Protocol	
ICH Stability Batch Manufacturing Report	
Scale-up Batch Protocol	
Scale-up Batch Report	
Clinical Batch Protocol	
Clinical Batch Report	
Qualification Batch Protocol	
Qualification Batch Report	
Demo Batches Placebo Report + Protocol	
Demo Batches Active Report + Protocol	
Bills of Material	
List of Lots Rejected	
Overview Registration Stability Packaging	
Production Facility	
List of Master Batch Record (MBR) Versions	
Clinical Batches MBR	
Pilot Batches MBR	
ICH Stability Batches MBR	
Demo Batches MBR	
Overview/Rationale for Formulation Changes from pivotal to current product	
Packaging	
List of Packaged ICH Batches	
List of Packaging Batches (reg. samples, stability)	
Packaging Records ICH/Registration Lots	
Overview/Rationale for formulation changes from pivotal to current product	
Bulk	
Shipping Test	
Hold Time Studies	
Purified Water: Quality Statement	
List of Raw Materials	
Material Specifications	

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Document	Comments
Production Information (continued)	
Bulk (continued)	
List of Raw Materials	
Material Specifications	
List of Packaging Material ICH	
List of Packaging Material (registration and stability samples)	
Packaging Material Specifications	
Supplier History and Vendor Audits (packaging material)	
Stability Summary	
Stability Data	
Shipping Tests in Blister/Bottles	
Process Equipment	
List of Equipment	
Calibration Records	
Preventative Maintenance Records	
Qualification Reports	
Packaging Equipment	
List of Equipment	
Qualification Reports	
Analytical Equipment	
List of Equipment	
Qualification Reports	
List of PAT Equipment	
PAT Development History	
PAT Methodology	
Overview Validation	
Validation Plan	
Cleaning Validation	
Overview Cleaning Validation Documents	
Methods for Cleaning Validation	
Cleaning Validation Method Validation	
Raw Data related to Cleaning Validation	

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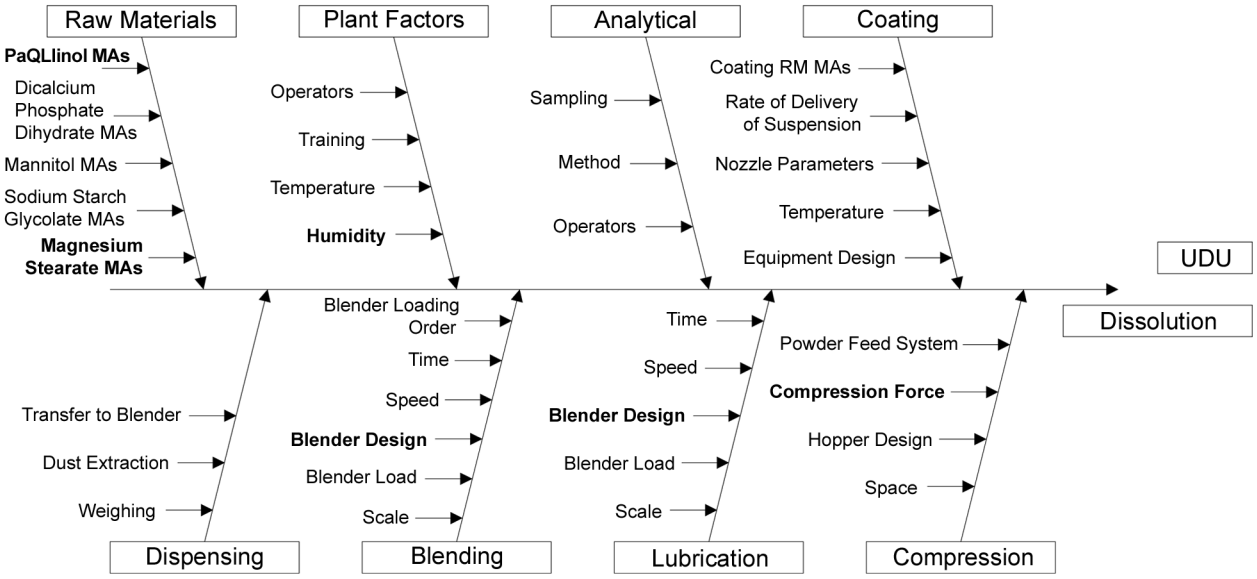
Figure 13.1: Example Ishikawa for Tablet Manufacturing Operations – Identification of Significant Factors for Uniformity of Dose Units and Dissolution



MA = Material Attribute

Note: This figure was adapted from original Figure 2.14 in the *ISPE Guide Series: Product Quality Lifecycle Implementation (PQLI®) from Concept to Continual Improvement, Part 2 – Product Realization using Quality by Design (QbD): Illustrative Example* [44]. In this illustrative example for the manufacture of PaQLInol tablets, factors in **bold** were determined to be significant to the following CQAs: Uniformity of Dose Units (UDU) and Dissolution. Material attributes were studied using DoE techniques and found to be significant. Plant, blending, and compression factors were found to be significant in engineering studies conducted during development. Highlighting these factors in the Ishikawa diagram and providing explanations such as this to the receiving unit assists in the transfer of important process knowledge.

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15 Appendix 8 – Glossary

15.1 Acronyms and Abbreviations

AEX	Anion Exchange (Chromatography)
API	Active Pharmaceutical Ingredient
ASTM	American Society for Testing and Materials
ATP	Analytical Target Profile
BFD	Block Flow Diagram
CAPA	Corrective Action and Preventive Action
CDMO	Contract Development and Manufacturing Organization
CMC	Chemistry, Manufacturing, and Controls
CMO	Contract Manufacturing Organization
COGM	Cost of Goods Manufactured
CPP	Critical Process Parameter
CPV	Continued Process Verification
CQA	Critical Quality Attribute
CQV	Continued Quality Verification
EMA	European Medicines Agency (EU)
FDA	Food and Drug Administration (US)
FMEA	Failure Mode and Effect Analysis
FRS	Functional Requirement Specification
FTE	Full Time Equivalent
GEP	Good Engineering Practice
GMP/CGMP	Good Manufacturing Practice/Current Good Manufacturing Practice
HPLC	High Pressure Liquid Chromatography
HSE	Health, Safety, and Environment
ICH	International Council for Harmonisation
LOD	Limits of Detection
MBR	Master Batch Record
MCB	Master Cell Bank
NIHS	National Institute of Health Sciences (Japan)
OEE	Overall Equipment Efficiency
P&ID	Piping and Instrumentation Diagram
PAT	Process Analytical Technology
PFD	Process Flow Diagram
PPQ	Process Performance Qualification

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PQS	Pharmaceutical Quality System
QA	Quality Assurance
QbD	Quality by Design
QC	Quality Control
QMS	Quality Management System
QRM	Quality Risk Management
QTPP	Quality Target Product Profile
R&D	Research and Development
RACI	Responsible, Accountable, Consulted, Informed
RCB	Research Cell Bank
RFP	Request for Proposal
RPN	Risk Priority Number
SME	Subject Matter Expert
SOP	Standard Operating Procedure
TBD	To Be Determined
URS	User Requirements Specification
UV	Ultraviolet
USP	United States Pharmacopeial
WHO	World Health Organization

15.2 Definitions

Control Strategy (ICH Q10 [1])

A planned set of controls, derived from current product and process understanding, that assures process performance and product quality. The controls can include parameters and attributes related to drug substance and drug product materials and components, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring and control.

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Critical Process Parameter (CPP) (ICH Q8(R2) [3])

A process parameter whose variability has an impact on a critical quality attribute and therefore should be monitored or controlled to ensure the process produces desired quality.

Critical Quality Attribute (CQA) (ICH Q8(R2) [3])

A physical, chemical, biological or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality.

Functional Requirement Specification (FRS)

Specification document, which builds on the URS (User Requirement Specification) and provides a basic narrative on what functions the process and its control system are expected to perform.

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Lifecycle (ICH Q8(R2) [3])

All phases in the life of a product from the initial development through marketing until the product's discontinuation.

Process Analytical Technology (PAT) (ICH Q8(R2) [3])

A system for designing, analyzing, and controlling manufacturing through timely measurements (i.e., during processing) of critical quality and performance attributes of raw and in-process materials and processes with the goal of ensuring final product quality.

Process Validation

The collection and evaluation of data, from the process design stage through commercial production, which establishes scientific evidence that a process is capable of consistently delivering quality product. (FDA Process Validation Guidance [6])

The documented evidence that the process, operated within established parameters, can perform effectively and reproducibly to produce a medicinal product meeting its predetermined specifications and quality attributes. (EMA Process Validation Guideline [7])

A systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control, based on sound science and quality risk management. (ICH Q8(R2) [3]).

Quality Attribute

A physical, chemical, or microbiological property or characteristic that directly or indirectly relates to predefined product quality (safety, identity, strength, purity, and marketability of the product).

Quality by Design (QbD) (ICH Q8(R2) [3])

A systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control, based on sound science and quality risk management.

Quality Management System (QMS)

Management system to direct and control an organization with regard to quality (e.g., ISO). This is equivalent to Quality System as defined in ICH Q9 [4].

Quality Risk Management (QRM) (ICH Q9 [4])

A systematic process for the assessment, control, communication and review of risks to the quality of the drug (medicinal) product across the product lifecycle.

Quality Target Product Profile (QTPP) (ICH Q8(R2) [3])

A prospective summary of the quality characteristics of a drug product that ideally will be achieved to ensure the desired quality, taking into account safety and efficacy of the drug product.

Receiving Unit

An organization, which includes core and auxiliary functions, where a designated product, process, or method is expected to be transferred to.

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Sending Unit

An organization, which includes core and auxiliary functions, where a designated product, process, or method is expected to be transferred from.

Stakeholder (ICH Q9 [4])

Any individual, group, or organization that can affect, be affected by, or perceive itself to be affected by a risk. Decision makers might also be stakeholders. For the purposes of this guideline (ICH Q9 [4]), the primary stakeholders are the patient, healthcare professional, regulatory authority, and industry.

Subject Matter Expert (SME) (ASTM E2500-13 [43])

Individual with specific expertise and responsibility in a particular area or field (for example, quality unit, engineering, automation, development, operations, and so forth).

Technology Transfer Charter

The technology transfer charter captures the strategic intent and defines the overall scope of the technology transfer project. It should document the team members and their roles and responsibilities, explain the effort and time required for the technology transfer project, and identify significant assumptions and risks. The charter helps to ensure that management and team members from both sending and receiving units understand the project and agree upon deliverables.

Technology Transfer Package

The technology transfer package is the collection of all (including local and process related) knowledge required to run the process and analyze the product.

Technology Transfer Plan

The technology transfer plan is based on the technology transfer proposal. It provides a more detailed description of the elements of the project (both technical and project management) that are to be completed by the team to achieve the overall goals.

Technology Transfer Proposal

The technology transfer proposal defines the overall scope of the project, documents the team members and their roles and responsibilities, estimates the effort and time required for the project, establishes high level success criteria, and identifies significant assumptions and risks.

User Requirement Specification (URS)

A description of the requirements of the facility in terms of product to be manufactured, required throughput and conditions in which the product should be made.

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